

Read questions carefully, and answer ~~all~~ ^{all} values in right ^{question properly!}

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

Name: Parthiv Patel

1. (100 pts) A saturated liquid feed at 100 kmol/min consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of propane is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

Butane.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5		5
C3, propane	2.4		10
C4, butane	0.85		30
C5, pentane	0.28		35
C6, hexane	0.1		20
Total in feed	-	-	100

77/100

- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

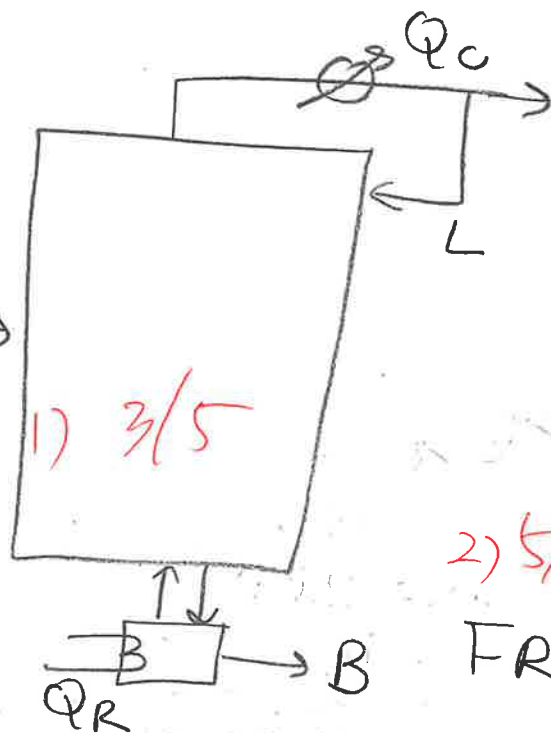
$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,bot} = (BX_{i,bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

① ②



$$F = 100 \frac{\text{kmol}}{\text{min}}$$

$$x_{C_2} = 0.05$$

$$x_{C_3} = 0.10$$

$$x_{C_4} = 0.30$$

$$x_{C_5} = 0.35$$

$$x_{C_6} = 0.20$$

Overall M.B.

component M.B. (?)

③ Distillate

$$D x_{C_2} = 100 \times 0.05 = 5 \frac{\text{kmol}}{\text{hr}} \quad FR_{C_2, D} = 1$$

$$D x_{C_3} = 100 \times 0.10 = 10 \frac{\text{kmol}}{\text{hr}} \quad FR_{C_3, D} = 1$$

$$D x_{C_4} = 0.98 \times 100 \times 0.30 = 29.4 \frac{\text{kmol}}{\text{hr}} \quad FR_{C_4, D} = 0.98$$

$$D x_{C_5} = (1 - 0.98) \times 100 \times 0.35 = 0.7 \frac{\text{kmol}}{\text{hr}} \quad FR_{C_5, D} = 0.02$$

$$D x_{C_6} = 0$$

$$x_{C_2, D} = 0.1092$$

$$x_{C_5, D} = 0.03057$$

$$D = 45.8 \frac{\text{kmol}}{\text{hr}}$$

$$x_{C_3, D} = 0.2183$$

$$x_{C_6, D} = 0$$

$$x_{C_4, D} = 0.6419$$

validate mole fraction sum! = 1

Parth Patel

655437221

② ethane C_2 - LNK

propane C_3 - LNK

A - butane C_4 - LK

B - pentane C_5 - HHK

n-hexane C_6 - HNK

2) 5/5

$$FR_{C_4, D} = 0.98$$

$$FR_{C_5, B} = 0.96$$

$$FR_{C_2, B} = 0, FR_{C_6, D} = 0$$

$$FR_{C_2, D} = 1, FR_{C_6, B} = 1$$

$$N_{F, \min} = \ln \left\{ \frac{x_{A, \text{dist}}}{x_{B, \text{dist}}} \times \frac{z_B}{z_A} \right\}$$

$$= \frac{\ln \alpha_{AB}}{\ln \left\{ \frac{0.6419}{0.03057} \times \frac{0.35}{0.30} \right\}}$$

$$N_{F, \min} = \frac{3.19856}{1.11054} = 2.880 \approx \boxed{3}$$

$$x_{A, \text{dist}} = 0.6419$$

$$x_{B, \text{dist}} = 0.03057$$

$$z_A = 0.30$$

$$z_B = 0.35$$

$$\alpha_{AB} = 3.036$$

$$\text{Q6. } \Delta V_{\text{feed}} = \frac{\sum_{i=1}^C \alpha_{i-\text{ref}} F z_i}{\alpha_{i-\text{ref}} - \phi}$$

x_{ref} is B HK.
 $\downarrow$
 C_5

10/15

$$\frac{\alpha_{C_2 C_5} F z_{C_2}}{\alpha_{C_2 C_5} - \phi} + \frac{\alpha_{C_3 C_5} F z_{C_3}}{\alpha_{C_3 C_5} - \phi} + \frac{\alpha_{C_4 C_5} F z_{C_4}}{\alpha_{C_4 C_5} - \phi}$$

$$+ \frac{\alpha_{C_5 C_5} F z_{C_5}}{\alpha_{C_5 C_5} - \phi} + \frac{\alpha_{C_6 C_5} F z_{C_6}}{\alpha_{C_6 C_5} - \phi}$$

Draw M.B around the feed stage!



$$\Delta V_{\text{feed}} = V_{\text{in}} - V_{\text{out}} = 0$$

for (2)

$$\alpha_{C_2 C_5} = 26.78$$

$$\alpha_{C_3 C_5} = 8.57$$

$$\alpha_{C_4 C_5} = 3.036$$

$$\alpha_{C_5 C_5} = 1$$

$$\alpha_{C_6 C_5} = 0.35714$$

$$\left(\frac{26.78 \times 100 \times 0.05}{26.78 - \phi} \right) + \left(\frac{8.57 \times 100 \times 0.1}{8.57 - \phi} \right) + \left(\frac{3.036 \times 100 \times 0.3}{3.036 - \phi} \right) + \left(\frac{1 \times 100 \times 0.35}{1 - \phi} \right) + \left(\frac{0.35714 \times 100 \times 0.2}{0.35714 - \phi} \right) = 0$$

Q_{urr} V_{feed} = 0 because feed is saturated liquid.

$$0 = \left(\frac{133.9}{26.78 - \phi} \right) + \left(\frac{85.7}{8.57 - \phi} \right) + \left(\frac{91.08}{3.036 - \phi} \right) + \left(\frac{35}{1 - \phi} \right) + \left(\frac{7.1428}{0.35714 - \phi} \right)$$

ϕ should be 1.498 look ideal based on above expression.

$$\textcircled{7.1} V_{\min} = \sum_{i=1}^C \frac{\alpha_{i, \text{ref}} (Dx_{i,D})}{\alpha_{i, \text{ref}} - \phi}$$

$$\frac{\alpha_{C_2 C_S} Dx_{2,D}}{\alpha_{C_2 C_S} - \phi} + \frac{\alpha_{C_3 C_S} Dx_{3,D}}{\alpha_{C_3 C_S} - \phi} + \frac{\alpha_{C_4 C_S} Dx_{4,D}}{\alpha_{C_4 C_S} - \phi} + \frac{\alpha_{C_5 C_S} Dx_{5,D}}{\alpha_{C_5 C_S} - \phi} + \frac{\alpha_{C_6 C_S} Dx_{6,D}}{\alpha_{C_6 C_S} - \phi}$$

$$\Rightarrow \frac{26.78 \times 5}{26.78 - 1.498} + \frac{8.57 \times 10}{8.57 - 1.498} + \frac{3.036 \times 29.4}{3.036 - 1.498} + \frac{1.4}{1 - 1.498} =$$

$$5.29 + 12.12 + 58.035 + 2.811 = \boxed{72.6 \frac{\text{kmol}}{\text{hr}}}$$

9.

$$\frac{N_{F, \min}}{N_{\min}} = \frac{N_F}{N}$$

9) 5/5

$$\frac{2.880}{6.366} \times 17.415 = N_F = \boxed{7.878}$$

7.20

11.

$$FR_{NK, \text{dist}} = (\alpha_{AB})^{N_{\min}}$$

$$\frac{FR_{HK, \text{bot}}}{1 - FR_{HK, \text{bot}}} + (\alpha_{AB})^{N_{\min}}$$

$$= \frac{(3.036)^{6.366}}{0.96 / 0.04} + (3.036)^{6.366}$$

$$= \frac{0.96}{0.04} + (3.036)^{6.366}$$

$$= 1175.8$$

$$\approx \boxed{0.98}$$

$$24 + 1175.8 = 1200$$

$$0.97999$$

5/10

11)

explain or justify
how this value can be
used for question!

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

71/100
 Name: Jeremy Wang

1. (100 pts) A saturated liquid feed at 100 kmol/min^{hr} consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of propane is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5	26.79	5
C3, propane	2.4	8.57	10
C4, butane	0.85	3.036	30
C5, pentane	0.28	1	35
C6, hexane	0.1	0.357	20
Total in feed	-	-	100

- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,Bott} = (BX_{i,Bot})/Fz_i$

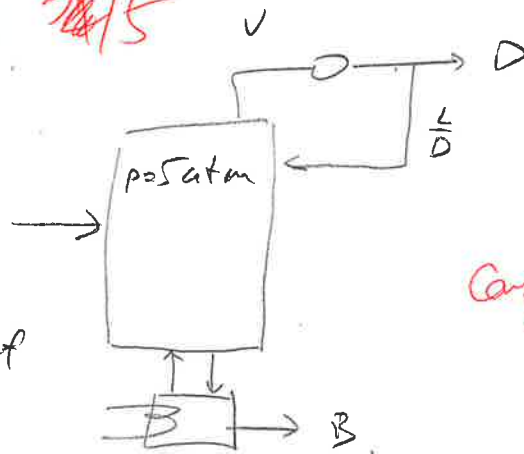
- (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

LK = 2
 HK = 5

①

$$F = 100 \frac{\text{kmol}}{\text{hr}}$$

17 3/5



$$F_{C4D} = 0.98$$

$$F_{C5D} = 1 - 0.98 = 0.02$$

Cap. H.B (?)

→

Sat lig $\xi = \frac{L}{F} = 1 \quad L = F$
 $f = \frac{V}{F} = 0 \quad V = 0$

$$F_{C5B} = 0.96$$

$$F_{C4B} = 1 - 0.96 = 0.04$$

②

$$\alpha_{C2C5} = \frac{7.5}{0.28} = 26.786$$

$$\alpha_{C3C5} = \frac{2.4}{0.28} = 8.57$$

$$\alpha_{C3C5} = \frac{2.4}{0.28} = 8.57$$

$$\alpha_{C4C5} = \frac{0.85}{0.28} = 3.036$$

2) 5/5

$$\alpha_{C5C5} = \frac{0.28}{0.28} = 1$$

$$\alpha_{C6C5} = \frac{0.1}{0.28} = 0.357$$

LNK C₂LK C₄HNK C₆LNK C₃HNK C₅

$$③ \quad D_{C4D} = 100 \times 0.98 \times 0.3 = 29.4$$

$$\frac{\text{kmol}}{\text{hr}} \quad \checkmark$$

$$D_{C5D} = 100 \times (1 - 0.98) \times 0.35 = 1.4$$

$$\frac{\text{kmol}}{\text{hr}} \quad \checkmark$$

$$D_{C2D} = 100 \times 0.05 = 5$$

$$\frac{\text{kmol}}{\text{hr}} \quad \checkmark$$

$$D_{C3D} = 100 \times 0.1 = 10$$

$$\frac{\text{kmol}}{\text{hr}} \quad \checkmark$$

$$D_{C6} = 100 \times 0 = 0$$

$$\frac{\text{kmol}}{\text{hr}} \quad \checkmark$$

D_{tot}

$$\text{Sum} = 45.8 \quad \frac{\text{kmol}}{\text{hr}} \quad \checkmark$$

③ $X_{C4D} = \frac{23.4}{45.8} = 0.51$

$X_{C5D} = \frac{1.4}{45.8} = 0.03$

$X_{C2D} = \frac{5}{45.8} = 0.109$

$X_{C3D} = \frac{10}{45.8} = 0.218$

$X_{C6D} = 0 = 0$

14/15

validation $\sum x_i = 1.0$

-

④ $F = B + D$ $B = F - D = 100 - 45.8 = 54.2 \frac{\text{kmol}}{\text{hr}}$

$B_{C4D} = 100 \times 0.3 \times (1 - 0.98) = 0.6 \frac{\text{kmol}}{\text{hr}}$

$B_{C5D} = 100 \times 0.35 \times 2.96 = 33.6 \frac{\text{kmol}}{\text{hr}}$

$B_{C2D} = 100 \times 0 = 0 \frac{\text{kmol}}{\text{hr}}$

$B_{C3D} = 100 \times 0 = 0 \frac{\text{kmol}}{\text{hr}}$

$D_{C6} = 100 \times 0.2 = 20 \frac{\text{kmol}}{\text{hr}}$

$X_{C4B} = \frac{0.6}{54.2} = 0.011$

$X_{C5B} = \frac{33.6}{54.2} = 0.6199$

$X_{C2B} = 0 = 0$

$X_{C3B} = 0 = 0$

$X_{C6B} = \frac{20}{54.2} = 0.369$

14/15

validation -

⑤

$$N_{min} = \frac{\ln \left[\left(\frac{0.98}{1-0.98} \right)_D \left(\frac{0.86}{1-0.86} \right)_R \right]}{\ln(3.036)}$$

$$\alpha_{C4,C5} = 3.036$$

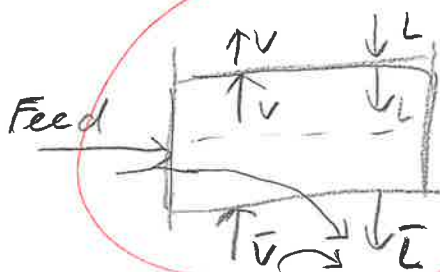
$$= \underline{6.366 \text{ stages}}$$

$$N_{Pm2} = \frac{\ln \left(\left(\frac{0.64}{0.03} \right)_D \times \left(\frac{0.35}{0.3} \right)_F \right)}{\ln(3.036)}$$

5/5

$$= 2.88 \text{ stages}$$

⑥



$$\Delta U = F(\xi - 1)$$

✓

15/15

$$\Delta U = F(\xi - 1) = \sum \frac{\alpha_{i,ref} F z_i}{\alpha_{i,ref} - \phi}$$

$$\alpha_{LHK} < \boxed{\phi} < \alpha_{HK}$$

$$1 < 1.498 < 3.036$$

$$0 = \frac{\alpha_{C2,C5} F z_{C2}}{\alpha_{C2,C5} - \phi} + \frac{\alpha_{C3,C5} F z_{C3}}{\alpha_{C3,C5} - \phi} + \frac{\alpha_{C4,C5} F z_{C4}}{\alpha_{C4,C5} - \phi} + \frac{\alpha_{C5,C5} F z_{C5}}{\alpha_{C5,C5} - \phi} + \frac{\alpha_{C6,C5} F z_{C6}}{\alpha_{C6,C5} - \phi}$$

$$0 = \frac{26.79 \times 100 \times 0.05}{26.79 - \phi} + \frac{8.57 \times 100 \times 0.1}{8.57 - \phi} + \frac{3.036 \times 100 \times 0.3}{3.036 - \phi} + \frac{1 \times 100 \times 0.35}{1 - \phi}$$

⑦

$$L_{min} = V_{min} \rightarrow$$

2/10

$$V_{min} = \frac{26.9 \times 5}{26.9 - 1.488} + \frac{8.57 \times \cancel{0.1}^{10}}{8.57 - 1.488} + \frac{3.036 \times \cancel{0.1}^{29.4}}{3.036 - 1.488} + \frac{1 \times 1.4}{1 - 1.488} +$$

+ 0

$$\geq 2.8$$

$$V_{min} = 72.64 \text{ kcal/h}$$

$$L_{min} = 26.84 "$$

$$(L/D)_{min} = 0.59$$

$$L_{min} = 2.8 - 45.8 = -42.997$$

$$(L/D)_{min} = 0.8388$$

⑧

$$\frac{B}{A_{min}} = 1.2 \quad B = 1.2 \text{ min}$$

2/5

$$\frac{L}{D} = \left(\frac{L}{D}\right)_{min} 1.2$$

$$\frac{L}{D} = -0.94 \times 1.2 = \underline{-1.128}$$

$$X = 0.07$$

$$Y = 0.56$$

$$X = 0.4$$

$$Y = 0.31$$

$$0.31 = \frac{N - N_{min}}{N + 1}$$

$$N = \cancel{9.57}$$

15.91

⑨

$$N \cdot \frac{N_{Fmin}}{N_{min}} \Rightarrow 9.57 \frac{2.87}{6.3} = \underline{4.39} = N_F$$

Method is ok

$$N_F = 7.15$$

~~compare it with~~

4/5

⑩

2/5

$$\frac{4.39}{9.57 - 4.39} = \frac{54.2}{45.8} \left(\frac{0.35}{0.3} \right) \left(\frac{0.11}{0.03} \right)^2 \Rightarrow NF = 7.15$$

Compare it with (9)

⑪

$$FR_{\text{mod}} = \frac{3.036}{\left(\frac{0.96}{1 - 0.96} \right) + 3.036} \quad \begin{matrix} 6.366 \\ 6.366 \end{matrix}$$

$$= 0.979 \approx 0.98 = FR_{\text{UKD}}$$

Justify! 5/10

Find Relative Volatilities using α values

40/100

CHE 313 Transport Phenomena III Exam #2 (03/17/2026)

Name: Tina Muccell

$$\alpha = \frac{K_A}{K_B}$$

1. (100 pts) A saturated liquid feed at 100 kmol/min consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of ~~propane~~ butane is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

	Component	K (y/x) value	α (relative volatility) value	kmol/hr F
LNK →	C2, ethane	7.5	260.79	5
LHK →	C3, propane	2.4	80.57	10
LK →	C4, butane	0.85	30.04	30 D
HK →	C5, pentane	0.28	1	35 B
HNK →	C6, hexane	0.1	0.36	20
	Total in feed	-	-	100

$$1 < \alpha < 30.04$$

- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table. $\leftarrow$ Denom
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}} \quad \left\{ \frac{FR_{LK,D}}{FR_{LK,B}} \times \frac{FR_{HK,B}}{FR_{HK,D}} \right.$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}} \quad \left\{ \frac{0.003}{0.03} \times \frac{0.35}{0.3} \right.$$

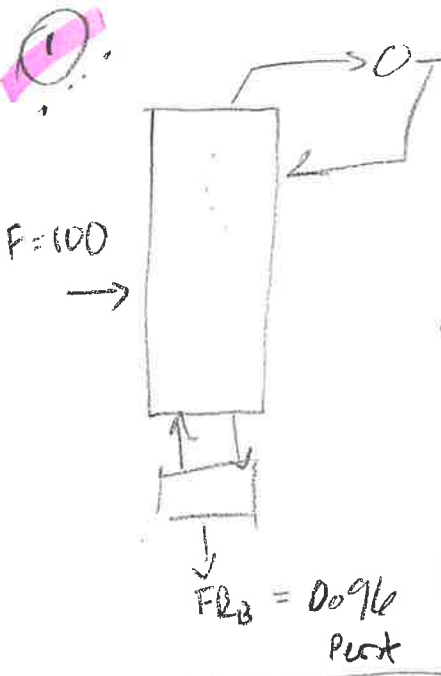
$$= \ln(30.04)$$

where A=LK and B=HK; $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,bot} = (BX_{i,bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a mass balance envelope around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

Don't solve yet
expand the eqn

$$\phi_{HK} < \phi < \phi_{LK} \quad 1 < \phi < 30.04$$



$$FR_B = 0.98 \text{ but}$$

Bal $\Rightarrow F = B + D$ $\propto A/B$
 $Fz_i = x_{B,i}B + x_{D,i}D$

$$FR_{D,i} = x_{D,i}D / F \cdot z_i$$

17 5/5

$$C_2 = \text{ethane} = z = 0.05$$

$$C_3 = \text{Propane} = z = 0.1$$

$$C_4 = \text{butane} = z = 0.3 \leftarrow \text{LK so full } 30 \frac{\text{mol}}{\text{hr}} \text{ to D}$$

$$C_5 = \text{Pentane} = z = 0.25 \leftarrow \text{HK so full } 35 \frac{\text{mol}}{\text{hr}} \text{ to B}$$

$$C_6 = \text{n-hexane} = z = 0.2$$

Relative volatilities

$$\alpha_{C_2} = \frac{7.05}{0.28} = 26.79$$

2) $\alpha_{AB} = K_A/K_B \rightarrow K_2/K_{HK}$

344

$$Fz_i = x_{B,i}B + x_{D,i}D$$

$C_{5,D} = 1.41 \frac{\text{mol}}{\text{hr}}$
 $C_{6,D} = 0$

$$\alpha_{C_3} = \frac{2.04}{0.28} = 8.07$$

$$\alpha_{C_4} = \frac{0.85}{0.28} = 3.04$$

$$\alpha_{C_5} = \frac{0.28}{0.28} = 1 \quad 2) 5/5$$

$$\alpha_{C_6} = \frac{0.0}{0.28} = 0.36$$

Distil Flow Dx_i, dist

$$C_2 = x_{D,i} = 100 (0.05) (0.98) = 4.9 \frac{\text{kmol}}{\text{hr}} \text{ in Dist}$$

$$C_3 = 100 \frac{\text{kmol}}{\text{hr}} (0.1) (0.98) = 9.8 \frac{\text{kmol}}{\text{hr}} \text{ in Dist}$$

$$C_4 = F \cdot z_i = 100 \frac{\text{kmol}}{\text{hr}} (0.3) (0.98) = 29.4 \frac{\text{kmol}}{\text{hr}} \text{ dist}$$

Bottom Flow

$$C_5 = F (FR \cdot z) = 100 \frac{\text{kmol}}{\text{hr}} (0.96) (0.25) = 33.6 \text{ in Bott}$$

$$C_6 = 100 \frac{\text{kmol}}{\text{hr}} (0.96) (0.2) = 19.2 \frac{\text{kmol}}{\text{hr}} \text{ in Bott}$$

Bott Flow

$$C_{4 \text{ Bott}} = 30 - 29.4 \frac{\text{kmol}}{\text{hr}} = 0.6 \frac{\text{kmol}}{\text{hr}} \text{ in bott}$$

$$C_{2 \text{ Bott}} = 5 - 4.9 = 0.1 \frac{\text{kmol}}{\text{hr}} \text{ in Bott}$$

$$C_{3 \text{ Bott}} = 10 - 9.8 = 0.2 \frac{\text{kmol}}{\text{hr}} \text{ in Bott}$$

Distillate Flow

$$C_5' = 35 - 33.6 = 1.4 \frac{\text{kmol}}{\text{hr}} \text{ in Distil}$$

$$C_6 = 20 - 19.2 = 0.8 \frac{\text{kmol}}{\text{hr}} \text{ in Dist}$$

$$\text{Total Dis Flow} = 46.3 \frac{\text{kmol}}{\text{hr}}$$

$$\text{Total Bott Flow} = 100 - 46.3 = 53.7 \frac{\text{kmol}}{\text{hr}}$$

$$B = F - D$$

x_{bottoms}

$$C_2 = 0.1 / 53.7 = 0.0019$$

$$C_3 = 0.2 / 53.7 = 0.0037$$

$$C_4 = 0.6 / 53.7 = 0.011$$

$$C_5 = 33.6 / 53.7 = 0.63$$

$$C_6 = 19.2 / 53.7 = 0.36$$

= 1 so good

$x_{\text{distillation}}$

$$C_2 = 4.9 / 46.3 = 0.11$$

$$C_3 = 9.8 / 46.3 = 0.21$$

$$C_4 = 29.4 / 46.3 = 0.63 \text{ LK}$$

$$C_5 = 1.4 / 46.3 = 0.03 \text{ HK}$$

$$C_6 = 0.8 / 46.3 = 0.017$$

0.997 = 1 so good

3) 10/15

4) 10/15

79/100

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

Name: Mariya Krawczyk

1. (100 pts) A saturated liquid feed at 100 kmol/hr consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of propane is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5	26.78	5
C3, propane	2.4	8.57	10
C4, butane	0.85	3.04	30
C5, pentane	0.28	1.00	35
C6, hexane	0.1	0.38	20
Total in feed	-	-	100

LNK
LNK
LK
HK
HNK

$$\alpha = \frac{K_i}{K_{HK}}$$

- ✓ 1) (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- ✓ 2) (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- ✓ 3) (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- ✓ 4) (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- ✓ 5) (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

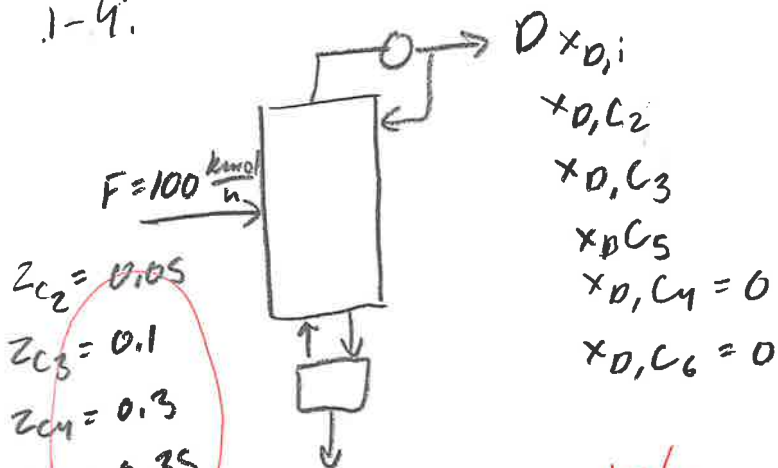
$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (Dx_{i,dist})/Fz_i$ & $FR_{i,bot} = (Bx_{i,bot})/Fz_i$

- 6) (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

1-4.



$$z_{C2} = 0.05$$

$$z_{C3} = 0.1$$

$$z_{C4} = 0.3$$

$$z_{C5} = 0.35$$

$$z_{C6} = 0.2$$

$$B x_{B,i}$$

$$x_{B,C3}$$

$$x_{B,C5}$$

$$x_{B,C6}$$

$$x_{B,C2} = 0$$

$$x_{B,C4} = 0$$

$$1) 5/5$$

$$2) 5/5$$

$$3) 15/15$$

$$4) 15/15$$

$$F(\sum z_i) = D(\sum x_{D,i}) + B(\sum x_{B,i})$$

$$\sum z_i = 1$$

$$\sum x_{D,i} = 1$$

$$\sum x_{B,i} = 1$$

$$D = \sum D x_{D,i} = 5 + 10 + 29.4 + 1.4 = 45.8 \frac{\text{kmol}}{\text{h}}$$

$$\left. \begin{aligned} C_2: x_{D,C2} &= 5/45.8 = 0.11 \\ C_3: x_{D,C3} &= 10/45.8 = 0.22 \\ C_4: x_{D,C4} &= 29.4/45.8 = 0.64 \\ C_5: x_{D,C5} &= 1.4/45.8 = 0.03 \\ C_6: x_{D,C6} &= 0 \end{aligned} \right\} 1.00$$

$$F = D + B$$

$$F z_i = D x_{D,i} + B x_{B,i}$$

3) Distillate

$$C_2: D x_{D,C2} = F z_{C2} = 100(0.05) = 5 \frac{\text{kmol}}{\text{h}}$$

$$C_3: D x_{D,C3} = (FR_{D,C3})(F z_{C3}) = (0.98)(35) = 10 \frac{\text{kmol}}{\text{h}}$$

$$C_4: D x_{D,C4} = (FR_{D,C4})(F z_{C4}) = (0.98)(30) = 29.4 \frac{\text{kmol}}{\text{h}}$$

$$C_5: D x_{D,C5} = (1 - FR_{B,C5})(F z_{C5}) = (1 - 0.96)(35) = 1.4 \frac{\text{kmol}}{\text{h}}$$

$$C_6: D x_{D,C6} = 0 \frac{\text{kmol}}{\text{h}}$$

4) Bottoms

$$C_2: B x_{B,C2} = 0 \frac{\text{kmol}}{\text{h}}$$

$$C_3: B x_{B,C3} = (FR_{B,C3})(F z_{C3}) = 0 \frac{\text{kmol}}{\text{h}}$$

$$C_4: B x_{B,C4} = (1 - FR_{D,C4})(F z_{C4}) = (1 - 0.98)(30) = 0.6 \frac{\text{kmol}}{\text{h}}$$

$$C_5: B x_{B,C5} = (FR_{B,C5})(F z_{C5}) = (0.96)(35) = 33.6 \frac{\text{kmol}}{\text{h}}$$

$$C_6: B x_{B,C6} = F z_{C6} = 20 \frac{\text{kmol}}{\text{h}}$$

Next time,
Try to number for
your answer!

$$B = \sum B_{x_{B,i}} = 0.6 + 33.6 + 20 = 54.2 \frac{\text{kmol}}{\text{h}}$$

$$F = 45.8 + 54.2 = 100 \checkmark$$

$$C_2: x_{B,C_2} = 0$$

$$C_3: x_{B,C_3} = 0$$

$$C_4: x_{B,C_4} = 0.6/54.2 = 0.01$$

$$C_5: x_{B,C_5} = 33.6/54.2 = 0.62$$

$$C_6: x_{B,C_6} = 20/54.2 = 0.37$$

1.00 ✓

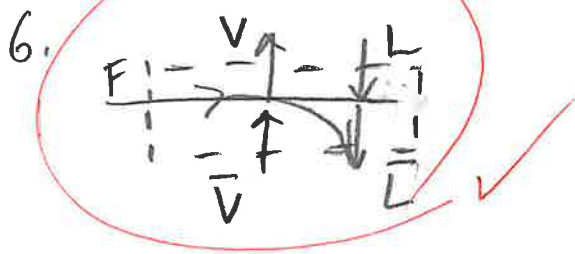
$$5. \quad N_{\min} = \frac{\ln \left\{ \frac{0.98}{(1-0.98)} \cdot \frac{0.96}{(1-0.96)} \right\}}{\ln 3.04}$$

$$\alpha_{AB} = \frac{K_{LK}}{L_{HK}}$$

$$= 6.36 \approx 6 \text{ stages} \checkmark$$

5/5

$$N_{F,\min} = \frac{\ln \left\{ \frac{0.64}{0.03} \cdot \frac{0.35}{0.3} \right\}}{\ln 3.04} = 2.89 \approx 3^{\text{rd}} \text{ stage} \checkmark$$



F - sat liq.

$$F + L = \bar{L}$$

$$\bar{V} = V$$

$$\Delta V_{\text{feed}} = \sum \frac{\alpha_i - \text{ref}}{\alpha_i - \text{ref} - \phi} F z_i$$

$$\Delta V_{\text{feed}} = 0 \checkmark$$

$$0.55 = \frac{N - N_{min}}{N + 1}$$

$$N_{min} = 6.36$$

$$0.45 = \frac{N - 6.36}{N + 1}$$

$$0.45N + 0.45 = N - 6.36$$

$$N - 0.45N = -6.36 + 0.45$$

$$N = 16.31$$

5/5

$$0.55 + 6.36 = N - 0.55N$$

$$N = 15.36 \approx 16 \checkmark$$

9.

$$\frac{N_F}{15.36} = \frac{2.89}{6.36}$$

$$N_F = 7.20$$

1/5

$$10. \left(\frac{N_F - 1}{15.36 - N_F} \right) = \left(\frac{54.2}{45.8} \right) \left(\frac{0.35}{0.3} \right) \left(\frac{0.01}{0.03} \right)^{2 \cdot 0.206}$$

$$N_F - 1 = 0.696 (15.36 - N_F)$$

$$N_F - 1 = 10.44 - 0.696 N_F$$

11.

3/5

25/100

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

Name: Jasmine Torres

- 9=1
1. (100 pts) A saturated liquid feed at 100 kmol/hr consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of propane is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms. Cy butane

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5	26.78	5
C3, propane	2.4	8.57	10
C4, butane	0.85	3.0357	30
C5, pentane	0.28	1	35
C6, hexane	0.1	0.357	20
Total in feed	-	-	100

$$\alpha = \frac{K_i}{K_{HK}}$$

- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,Bott} = (BX_{i,Bot})/Fz_i$

- Draw!
- 6) (15 pts - Underwood equation) After drawing a mass balance envelope around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

Don't
solve

satisfy, choose & explain

1)

$$F = 100 \text{ kmol/h}$$

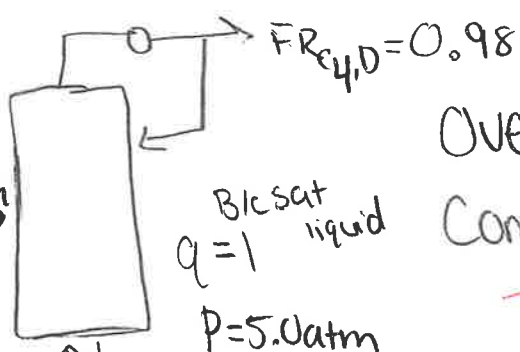
$$C_2 = 0.05$$

$$C_3 = 0.10$$

$$C_4 = 0.30$$

$$C_5 = 0.35$$

$$C_6 = 0.20$$



B/c sat liquid

$$q = 1$$

$$P = 5.0 \text{ atm}$$

$$\text{Overall MB}^\circ F = D + B$$

$$\text{Component MB}^\circ F(z_i) = x_{i,D}(D) + x_{i,B}(B)$$

Jasmine Torres

5/5

2) $C_2 \rightarrow \text{NK}$

$$C_3 \rightarrow \text{LNK}$$

$$C_4 \rightarrow \text{LK}$$

$$C_5 \rightarrow \text{HK}$$

$$C_6 \rightarrow \text{HNK}$$

It's this
is wrong
b/c
D+B=100

$$\alpha_{C_2-C_5} = \frac{7.5}{0.28} = 26.78$$

$$\alpha_{C_3-C_5} = \frac{2.4}{0.28} = 8.57$$

$$\alpha_{C_4-C_5} = \frac{0.85}{0.28} = 3.0357$$

$$\alpha_{C_5-C_5} = 1$$

$$\alpha_{C_6-C_5} = \frac{0.1}{0.28} = 0.357$$

4/5

3) $Dx_{i,\text{LNK},D} = z_{i,C_3} \cdot F = (0.10)(100) = 10 \text{ kmol}$

$$Dx_{i,\text{LK},D} = (0.098)(100)(0.30) = 2.94$$

$$Dx_{i,\text{NK},D} = 0$$

$$Dx_{i,\text{HK},D} = (1 - 0.96)(100)(0.20) = 0.8$$

$$Dx_{i,\text{HNK},D} = 0$$

$$D = \sum (Dx_{i,D}) = 13.74$$

45.8

kmol/h

unit missy

$$Dx_{C_2,D} = 5 \text{ kmol/h}$$

$$Dx_{C_4,D} = 29.4 "$$

$$Dx_{C_5,D} = 1.4 "$$

$$Dx_{C_6,D} = 0$$

3/15

$$4) B_{xi, C_2, B} = 0$$

$$B_{xi, C_4, B} = 100(0.30)(1-0.98) = 30(0.02) = 0.6 \text{ kmol/hr}$$

$$B_{xi, C_3, B} = 0$$

$$B_{xi, C_6, D} = 100(0.20) = 20 \text{ kmol/hr}$$

$$B_{xi, C_5, D} = 100(0.35)(0.96) = 33.6 \text{ kmol/hr}$$

$$B = \Sigma = 54.2 \text{ kmol/h}$$

Mole fractions:

$$B_{xi} = \frac{B_{xi, B} \cdot z_i}{B}$$

10/15

$$F = D + B$$

Supposed to equal 100

5)

$$N_{min} = \ln \left\{ \frac{0.98}{1-0.98} \times \frac{0.96}{1-0.96} \right\}$$

$$\ln \alpha_{AB} = \ln(3.0357)$$

$$N_{min} = \ln \left\{ \frac{0.98}{0.02} \times \frac{0.96}{0.04} \right\}$$

3/5

$$1.11044$$

$$\frac{\ln(49 \times 24)}{1.11044} = \frac{7.06987}{1.11044} = 6.366 \text{ stage}$$

$$N/F_{min} = 2.88$$

$$7) V_{\min} = \sum_{i=1}^c \frac{\alpha_{i-\text{ref}}(D_{x_i, \text{dist}})}{\alpha_{i-\text{ref}} - \phi}$$

87/100

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

Name: Liam Gaza

$q = 1$

1. (100 pts) A saturated liquid feed at 100 kmol/min consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of propane is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKS do not distribute. $\alpha = \frac{K_i}{K_{HK}}$

	Component	K (y/x) value	α (relative volatility) value	kmol/hr
LNK	C2, ethane	7.5	26.79 ✓	5
LNK	C3, propane	2.4	8.57 ✓	10
LK	C4, butane	0.85	3.04 ✓	30
HK	C5, pentane	0.28	1.00 ✓	35
HNK	C6, hexane	0.1	0.357 ✓	20
	Total in feed	-	-	100

- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,Bot} = (BX_{i,Bot})/Fz_i$

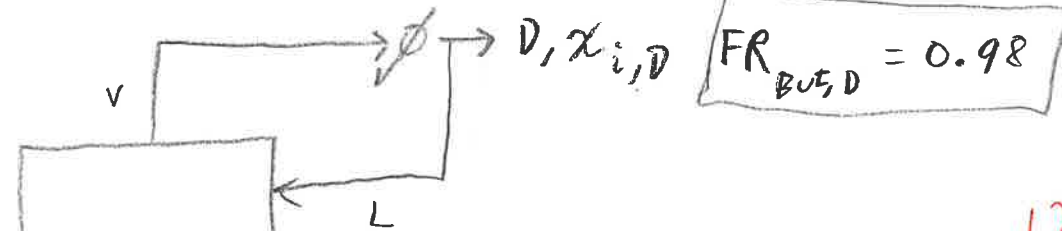
- (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

$\Delta V_{feed} = F(1-q) = (100)(0) = 0$

[Handwritten signature]

1.)

$$P = 5 \text{ atm}$$



$$FR_{But,D} = 0.98$$

1) 5/5

$$\text{Total MB} \quad F = B + D$$

$$\text{Comp MB} \quad F z_i = x_{i,B} B + x_{i,D} D$$

$$z_{Et} = \frac{5}{100} = 0.05 \quad z_{pent} = \frac{35}{100} = 0.35$$

$$z_{proe} = \frac{10}{100} = 0.10 \quad z_{hex} = \frac{20}{100} = 0.20$$

$$z_{but} = \frac{30}{100} = 0.30$$

$$\begin{aligned} 2. \quad & LNK = C_2 = C_3 \\ & LK = C_4 \\ & HK = C_5 \\ & HNK = C_6 \end{aligned}$$

5/5

$$3.) \quad B, x_{i,B} \quad FR_{pent} = 0.96$$

Et)

$$FR_{LNK,D} = 1 = \frac{D x_{Et}}{F z_i} \Rightarrow D x_{Et,D} = 5 \text{ kmol/hr}$$

UNIT MISSING!

$$\text{prop)} \quad 10 = x_{prop,B} B + x_{prop,D} D \Rightarrow 10 = D x_{prop,D}$$

$$\text{But)} \quad 30 = x_{but,B} B + x_{but,D} D \quad 0.98(100)(30) = D x_{but,D} = 29.4$$

$$\text{pent)} \quad 35 = x_{pent,B} B + x_{pent,D} D \quad 0.04(100)(35) = D x_{pent,D} = 1.4$$

$$\text{Hex)} \quad 20 = B x_{hex,B} \quad (FR_{HNK,B} = 1) \quad D x_{hex,D} = 0$$

$$\begin{aligned} & \uparrow \\ & FR_{int,D} = \\ & = 1 - 0.96 = 0.04 \end{aligned}$$

3. continued

$$\sum_i D x_i = D, \quad 5 + 10 + 29.4 + 1.4 + 0 = D = 45.8 \frac{\text{kmol}}{\text{hr}}$$

$$x_{\text{Et}, D} = \frac{5}{45.8} = 0.109$$

$$x_{\text{pro}, D} = \frac{10}{45.8} = 0.218$$

$$x_{\text{but}, D} = \frac{29.4}{45.8} = 0.642$$

$$x_{\text{pent}, D} = \frac{1.4}{45.8} = 0.031$$

$$x_{\text{hex}, D} = 0$$

$$\sum x_{i,D} = 1 \Rightarrow$$

$$\begin{array}{r} 0.109 \\ 0.218 \\ 0.642 \\ 0.031 \\ + 0 \\ \hline 1 \end{array}$$

14/15

5/15

$$4.) F = B + D \Rightarrow F - D = B = 54.2 \frac{\text{kmol}}{\text{hr}}$$

$$x_{\text{Et}, B} = 0$$

$$x_{\text{pro}, B} = 0$$

$$\sum x_{i,B} = 1$$

calculate $Bx_{i, \text{but}}$.

$$Bx_{\text{Et}, B} = Bx_{\text{pro}, B} = 0$$

$$Bx_{\text{but}, B} = 0.6 \text{ kmol/hr}$$

$$Bx_{\text{pent}, B} = 33.6 "$$

$$Bx_{\text{hex}, B} = 20 "$$

$$0 + 0 + 0.011 + 0.620 + 0.369$$

$$= 1 \quad \checkmark$$

$$\frac{30 - 29.4}{54.2} = x_{\text{but}, B} = 0.011$$

(Equations from mass balance)

$$\frac{35 - 1.4}{54.2} = x_{\text{pent}, B} = 0.620$$

$$\frac{20 - 0}{54.2} = x_{\text{hex}, B} = 0.369$$

$$5.) \therefore FR_{A,dist} = 0.98 \quad FR_{B,dist} = 0.04$$

$$\alpha_{AB} = \alpha_{LK-HK} = \frac{3.04}{1}$$

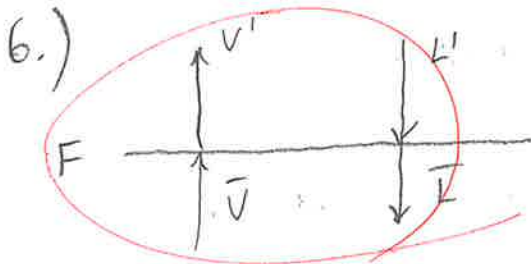
$$FR_{A,Bot} = 0.02 \quad FR_{B,Bot} = 0.96$$

$$N_{min} = \frac{\ln \left(\frac{0.98}{0.02} \cdot \frac{0.96}{0.04} \right)}{\ln(3.04)} = \boxed{6.36 \text{ stages}}$$

$$x_{A,dist} = 0.642 \quad x_{B,dist} = 0.031$$

$$z_A = 0.30 \quad z_B = 0.35$$

$$N_{F,min} = \frac{\ln \left(\frac{0.642}{0.031} \cdot \frac{0.35}{0.30} \right)}{\ln(3.04)} = \boxed{2.86 \text{ stages}}$$



$$F + \bar{V} + L' = \bar{L} + V'$$

$$L_{min} = V_{min} - D$$

$$\Rightarrow V' - \bar{U} = F - \bar{L} + L' = \Delta V_{feed}$$

But since Feed is a saturated liquid, $\Delta V_{feed} = F(1-q) = 0$

$$\Delta V_{feed} = 0 = \frac{26.79(100)(0.05)}{26.79 - \phi} + \frac{8.57 \cdot 10}{8.57 - \phi} + \frac{3.04 \cdot 36}{3.04 - \phi} + \frac{1.35}{1 - \phi} + \frac{0.357 \cdot 20}{0.357 - \phi} = 0$$

15/15

ϕ must sit between the relative volatilities of LK & HK,

$$1 < \phi < 3.04 \quad \therefore \boxed{\phi = 1.498}$$

$$7.) V_{\min} = \frac{26.79(5)}{26.79 - 1.498} + \frac{(8.57)(10)}{8.57 - 1.498} + \frac{3.04(29.4)}{3.04 - 1.498} + \frac{1(1.4)}{1 - 1.498} + 0 = \boxed{72.56 \frac{\text{kmol}}{\text{hr}}}$$

$$L_{\min} = V_{\min} - D = 72.56 - 45.8 = 26.76 \frac{\text{kmol}}{\text{hr}}$$

$$\left(\frac{L}{D} \right)_{\min} = \frac{26.76}{45.8} = 0.584$$

✓ 7) 10/10

$$8.) \frac{(L/D)}{(L/D)_{\min}} = 1.2 \Rightarrow L/D = 1.2 \cdot 0.584 = \underline{0.70}$$

8) 5/5

$$X = \frac{0.7 - 0.584}{0.7 + 1} = \underline{0.0682}$$

From graph
↓

$$N_{\min} = 6.36$$

$$\frac{N - N_{\min}}{N + 1} = 0.58 \Rightarrow N - N_{\min} = 0.58N + 0.58$$

$$\Rightarrow 0.42N = \frac{6.36 + 0.58}{0.42} = \boxed{N = 16.52 \text{ stages}}$$

$$9.) N_F = \frac{N_{F,\min} \cdot N}{N_{\min}} = \frac{2.86 \cdot 16.52}{6.36} \Rightarrow \boxed{N_F = 7.43 \text{ stages}}$$

9) 5/5

29/100

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

Name: Ana Tsikhanovich

$q=1$

F h

1. (100 pts) A saturated liquid feed at 100 kmol/min consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot Z_i$) of propane is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKS do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr $\leftarrow F Z_i$
C2, ethane	7.5	26.79	5
C3, propane	2.4	8.57	10
LK C4, butane	0.85	3.04	30
HK C5, pentane	0.28	1.00	35
C6, hexane	0.1	0.36	20
Total in feed	-	-	100

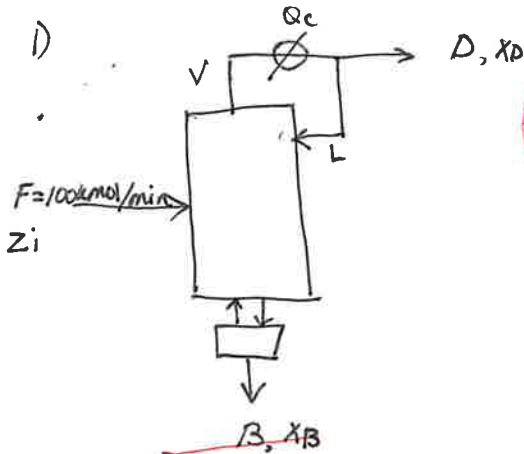
- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (Dx_{i,dist})/FZ_i$ & $FR_{i,Bot} = (Bx_{i,Bot})/FZ_i$

- (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.



$$F = B + D$$

$$F z_i = B x_{B,i} + D x_{D,i}$$

$$\sum z_i = \sum x_{B,i} = \sum x_{D,i} = 1$$

Mole fractions?
 $z_{C2} = 0.05$

5/5

20/20

2) $C2 = \text{LNK}$ $\alpha_i = \frac{K_i}{K_{\text{ref}}} = \frac{K_i}{K_{\text{HK}}}$
 $C3 = \text{LNK}$
 $C4 = \text{LK}$
 $C5 = \text{HK}$
 $C6 = \text{HNK}$

$C2: \alpha_{C2-C5} = 7.5/0.28 = 26.79$

$C3: \alpha_{C3-C5} = 2.4/0.28 = 8.57$

$C4: \alpha_{C4-C5} = 0.85/0.28 = 3.04$

$C5: \alpha_{C5-C5} = 0.28/0.28 = 1.00$

$C6: \alpha_{C6-C5} = 0.1/0.28 = 0.36$

5/5

Distillate

3) $C2: D x_{C2,D} = F R_{C2,D} \cdot F z_{C2} = 1.5 \text{ kmol/hr} = 5 \text{ kmol/hr}$

$C3: D x_{C3,D} = F R_{C3,D} \cdot F z_{C3} = 1 \cdot 10 \text{ kmol/hr} = 10 \text{ kmol/hr}$

$C4: D x_{C4,D} = F R_{C4,D} \cdot F z_{C4} = 0.98 \cdot 30 \text{ kmol/hr} = 29.4 \text{ kmol/hr}$

$C5: D x_{C5,D} = (1 - F R_{C5,B}) F z_{C5} = (1 - 0.96) \cdot 35 \text{ kmol/hr} = 1.4 \text{ kmol/hr}$

$C6: D x_{C6,D} = 0$ b/c HNK

$D = \sum D x_{i,D} = 45.8 \text{ kmol/hr}$

$C2: 5 \text{ kmol/hr} / 45.8 \text{ kmol/hr} = 0.109$

$x_{i,D} = \frac{D x_{i,D}}{D} \Rightarrow$

$C2: 0.218$

$C3: 0.642$

$C5: 0.031$

$C6: 0$

$\sum x_{D,i} = x_{D,C2} + x_{D,C3} + x_{D,C4} + x_{D,C5} + x_{D,C6} = 1 \checkmark$

4) Bottom

$C2: 0$ b/c LNK

$C3: 0$ b/c LNK

$C4: B x_{C4,B} = (1 - F R_{C4,D}) \cdot F z_{C4} = 0.6 \text{ kmol/hr}$

$C5: B x_{C5,B} = F R_{C5,B} \cdot F z_{C5} = 0.96 \cdot 35 \text{ kmol/hr} = 33.6 \text{ kmol/hr}$

$C6: B x_{C6,B} = F R_{C6,B} \cdot F z_{C6} = 1 \cdot 20 \text{ kmol/hr} = 20 \text{ kmol/hr}$

$B = \sum B x_{i,B} = 54.2 \text{ kmol/hr}$

$x_{i,B} = \frac{B x_{i,B}}{B} \Rightarrow$
 $x_{C2,B} = 0, x_{C3,B} = 0$
 $x_{C4,B} = 0.6 \text{ kmol/hr} / 54.2 \text{ kmol/hr} = 0.011$

$x_{C5,B} = 0.620$

$x_{C6,B} = 0.369$

$\sum x_{B,i} = 0.011 + 0.620 + 0.369 = 1 \checkmark$

$F = B + D = 54.2 \text{ kmol/hr} + 45.8 \text{ kmol/hr} = 100 \text{ kmol/hr} \checkmark$

15/15

$$5) N_{min} = \ln \left[\frac{0.98}{0.02} \cdot \frac{0.96}{0.04} \right] = 6.36 \approx 7 \text{ stages}$$

$$\ln(3.04/1.00)$$

$$\alpha_{LK} = \alpha_{C4-C5} = \alpha_{LK-HK}$$

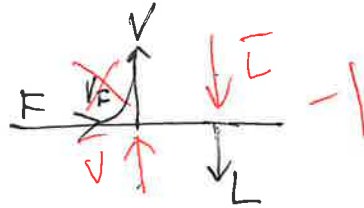
$$N_{F, min} = \ln \left[\frac{0.642}{0.031} \cdot \frac{0.35}{0.30} \right] = 2.86 \approx 3 \text{ stages}$$

$$\ln(3.04)$$

$$6) \alpha_{HK} < \phi < \alpha_{LK}$$

$$1.00 < \phi < 3.04$$

$$\phi = 1.498$$



$$\Delta V_{feed} = F(1-q), \quad q = \frac{L}{F} = 1 \text{ b/c sat. liquid.}$$

$$V = L + D$$

$$F(1-q) = \frac{\alpha_{C2-C5} F_{2C2}}{\alpha_{C2-C5} - \phi} + \frac{\alpha_{C3-C5} F_{2C3}}{\alpha_{C3-C5} - \phi} + \frac{\alpha_{C4-C5} F_{2C4}}{\alpha_{C4-C5} - \phi} + \frac{\alpha_{C5-C5} F_{2C5}}{\alpha_{C5-C5} - \phi} + \frac{\alpha_{C6-C5} F_{2C6}}{\alpha_{C6-C5} - \phi}$$

↓ divide both sides by F and sub $q=1$ on the left side.

$$0 = \frac{26.79 \cdot 0.05}{26.79 - 1.498} + \frac{8.57 \cdot 0.10}{8.57 - 1.498} + \frac{3.04 \cdot 0.30}{3.04 - 1.498} + \frac{1.00 \cdot 0.35}{1.00 - 1.498} + \frac{0.36 \cdot 0.20}{0.36 - 1.498}$$

$$7) V_{min} = L_{min} + \underline{D} \rightarrow L_{min} = V_{min} - D$$

$$V_{min} = \frac{\alpha_{C2-C5} (D x_{C2,D})}{\alpha_{C2-C5} - 1.498} + \frac{\alpha_{C3-C5} (D x_{C3,D})}{\alpha_{C3-C5} - 1.498} + \frac{\alpha_{C4-C5} (D x_{C4,D})}{\alpha_{C4-C5} - 1.498} + \frac{\alpha_{C5-C5} (D x_{C5,D})}{\alpha_{C5-C5} - 1.498}$$

$$+ \frac{\alpha_{C6-C5} (D x_{C6,D})}{\alpha_{C6-C5} - 1.498} \cdot 0$$

$$V_{min} = \frac{26.79(5)}{26.79 - 1.498} + \frac{8.57(10)}{8.57 - 1.498} + \frac{3.04(29.4)}{3.04 - 1.498} + \frac{1.00(1.4)}{1.00 - 1.498} = 5.30 + 12.12 + 57.96 - 2.81$$

$$V_{min} = 72.56 \text{ kmol/hr}$$

$$L_{min} = V_{min} - D = (72.56 - 45.8) \text{ kmol/hr} = 26.76 \text{ kmol/hr}$$

$$(L/D)_{min} = 0.584 = R_{min}$$

10/10

$$8) Y = \frac{N - N_{min}}{N + 1}$$

$$\frac{R}{R_{min}} = 1.2 \rightarrow R = R_{min} \cdot 1.2 = 0.584 \cdot 1.2 = 0.701 = L/D$$

$$X = \frac{R - R_{min}}{R + 1} = \frac{0.701 - 0.584}{0.701 + 1} = 0.069$$

$$Y \approx 0.54 \text{ from graph}$$

$$Y(N+1) = N - N_{min}$$

$$Y \cdot N + Y = N - N_{min}$$

$$YN - N = -N_{min} - Y$$

$$N - YN = N_{min} - Y$$

$$\frac{N}{Y} - N = \frac{N_{min} - Y}{Y}$$

$$\frac{N}{Y} - N = \frac{N_{min}}{Y} - 1$$

$$\frac{N}{0.54} - N = \frac{6.36}{0.54} - 1$$

$$\frac{N}{0.54} - N = 10.778 \rightarrow N \approx 5.82$$

seems calculation error!

$$N \approx 15.91$$

4/5

$$9) N_F = \left(\frac{N_{F,min}}{N_{min}} \right) N = 2.617$$

$$10) \frac{N_F - 1}{N - N_F} = \left(\frac{54.2}{45.8} \right) \left(\frac{0.35}{0.30} \right) \left(\frac{0.011}{0.031} \right) (2)^{0.806} = 0.697$$

$$N_F = 7.15$$

$$(9) \& (10) \Rightarrow N_F \approx 7.0$$

$$11) FR_{NK,D} = \frac{(3.04)^{6.36}}{\frac{0.96}{0.04} + (3.04)^{6.36}} = 0.98$$

$$FR_{NK,D} = 0.98$$

The distillation process is economically reasonable since $FR_{NK,D}$ is high (extra product).

10/10

84/100

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

Name: Nord Weil

1. (100 pts) A saturated liquid feed at 100 kmol/hr ^{hr} consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of propane is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms. ^{C4}

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5	26.78671	5
C3, propane	2.4	8.571429	10
C4, butane	0.85	3.035714	30
C5, pentane	0.28	1	35
C6, hexane	0.1	0.3571429	20
Total in feed	-	-	100

- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table. -
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,Bot} = (BX_{i,Bot})/Fz_i$

- flow mass balance envelope
- (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422**, **1.498** and **5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

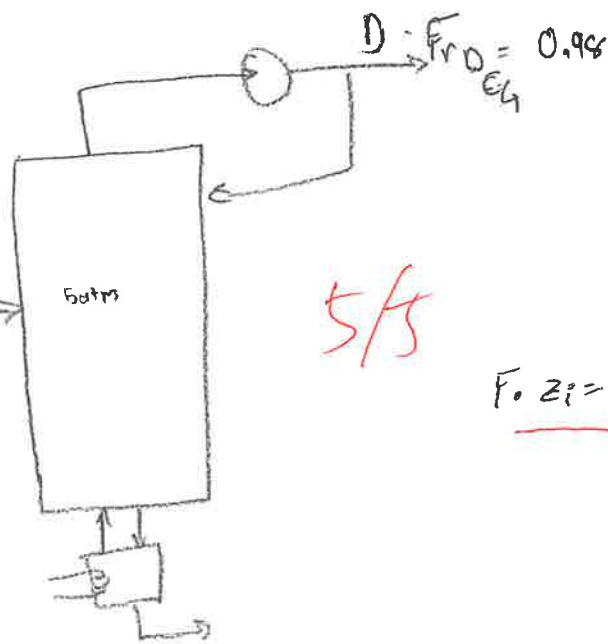
plug all except ϕ

$\alpha_{LK} < \phi < \alpha_{HK}$

①

$$F = 100 \text{ kmol/min}$$

$$\begin{aligned} z_{C_2} &= 0.05 \\ z_{C_3} &= 0.1 \\ z_{C_4} &= 0.3 \\ z_{C_5} &= 0.35 \\ z_{C_6} &= 0.2 \end{aligned}$$



HK ref -
(rel vol)

$$F \cdot z_i = D \cdot x_{Di} + B \cdot x_{Bi}$$

$$B \cdot Fr_{BC_6} = 0.96$$

②

$$C_2 - \text{LNK} = 7.5 / 0.28 = 26.78571$$

$$C_3 - \text{LNK} = 24 / 0.28 = 8.571429$$

$$C_4 - \text{LK} = 0.85 / 0.28 = 3.035714$$

$$C_5 - \text{HK} = 0.28 / 0.28 = 1$$

$$C_6 - \text{HNK} = 0.1 / 0.28 = 0.3571429$$

③

For LNK assume none in bottom so

$$\begin{aligned} C_2 \quad F \cdot z_i = D \cdot x_i &= 5 \text{ kmol/hr} \quad \leftarrow \text{From table} \\ C_3 &= 10 \text{ kmol/hr} \end{aligned}$$

$$C_4 = F \cdot z_i \cdot Fr_{DC_4} = (100)(0.3)(0.98) = 29.4 \text{ kmol/hr}$$

$$C_5 = F \cdot z_i \cdot (1 - Fr_{BC_5}) = (100)(0.35)(1 - 0.96) = 1.4 \text{ kmol/hr}$$

$$C_6 = 0$$

- All HK in bottom no Dist

$$D = 5 + 10 + 29.4 + 1.4 = 45.8 \text{ kmol/hr}$$

Nxt Pg →

$$X_i = \frac{D_i}{D} =$$

$$C_2 = \frac{5}{45.8} = 0.1091703$$

$$C_3 = \frac{10}{45.8} = 0.2183406$$

$$C_4 = \frac{29.4}{45.8} = 0.6419214$$

$$C_5 = \frac{1.4}{45.8} = 0.0305677$$

$$C_6 = 0$$

14/15

validation?

$$\sum X_i = 1.0$$

$$(4) F = B + D$$

$$B = F - D = 54.2 \text{ kmol/hr}$$

No LNK so $C_2 = C_3 = 0$

$$C_2 = 0$$

$$C_3 = 0$$

$$C_4 = (F \cdot Z_i) (1 - F_{rD}) = 100 \cdot (0.3) (1 - 0.98) = 0.6 \text{ kmol/hr}$$

$$C_5 = F \cdot Z_i (F_{rB}) = 100 \cdot (0.35) (0.96) = 33.6 \text{ kmol/hr}$$

$$C_6 = F \cdot Z_i = 20 \text{ kmol/hr}$$

↑
Table

$$B = 0.6 + 33.6 + 20 = 54.2 \text{ kmol/hr}$$

← don't forget the unit!

$$X_{i \text{ out}} = \frac{B \cdot X_i}{B} \quad C_2 = C_3 = 0$$

$$C_4 = \frac{0.6}{54.2} = 0.0110701$$

$$C_5 = \frac{33.6}{54.2} = 0.6199262$$

$$C_6 = \frac{20}{54.2} = 0.3690037$$

validation?

$$\sum X_i = 1.0$$

$$⑤ \quad N_{min} = \frac{\ln \left(\frac{F_{R \text{ dist}}}{F_{R \text{ bot}}} \cdot \frac{F_{R \text{ bot}}}{F_{R \text{ dis}}} \right)}{\ln \alpha_{AB}}$$

$$\frac{\ln \left(\frac{0.98}{1-0.98} \cdot \frac{0.96}{1-0.96} \right)}{\ln (3.035714)}$$

$$N_{min} = \underline{6.366694} =$$

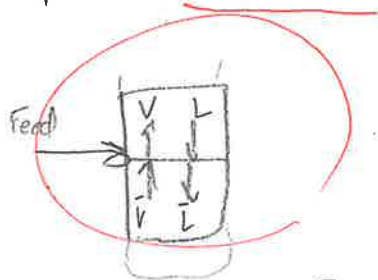
5/5

$$N_{Fmin} = \frac{\ln \left(\frac{x_A \text{ dist}}{x_B \text{ dist}} \times \frac{z_B}{z_A} \right)}{\ln \alpha_{AB}}$$

$$\frac{\ln \left(\frac{0.6419214}{0.0305677} \times \frac{0.35}{0.3} \right)}{\ln (3.035714)}$$

$$N_{Fmin} = \underline{2.880528}$$

⑥



12/15

$$\Delta V_{Feed} = V_{min} - \bar{V}_{min} = 0 \quad \text{as } V_{mm} = \bar{V}_{mm}, f=0.$$

$$\Delta V_{feed} = \sum_{i=1}^c \frac{\alpha_{i-ref} \cdot F_{zi}}{\alpha_{i-ref} - \phi} = \frac{26.78571 \cdot 5}{26.78571 - \phi} + \frac{8.571429 \cdot 10}{8.571429 - \phi} + \frac{3.035714 \cdot 30}{3.035714 - \phi}$$

$$+ \frac{1.35}{1 - \phi} + \frac{0.3571429 \cdot 20}{0.3571429 - \phi}$$

$$\underline{\phi = 1.498}$$

in between α_{LK} and α_{HK}



$$(7) \quad V_{min} = \sum_{i=1}^C \frac{\alpha_{i-ref} (D_{x,dist})}{\alpha_{i-ref} - \phi} = \frac{26.78571 \cdot 5}{26.78571 - 1.498} + \frac{8.571429 \cdot 10}{8.571429 - 1.498} +$$

$$\frac{3.035714 \cdot 29.4}{3.035714 - 1.498} + \frac{1 \cdot 1.4}{1 - 1.498} + \underbrace{\frac{0.3571429 \cdot 0}{0.3571429 - 1.498}}_0 = V_{min}$$

$$V_{min} = 72.64343 \text{ kmol/hr}$$

$$L_{min} = V_{min} + D$$

$$= 72.64343 + 45.8 = L_{min} = 118.4434 \text{ kmol/hr}$$

$$(8) \quad \frac{118.4434}{45.8} = \left(\frac{L}{D} \right)_{min} = 2.586101$$

$$\frac{\frac{L}{D}}{\left(\frac{L}{D} \right)_{min}} = 1.2 = \frac{L}{D} = \frac{L}{D_{min}} (2.586101) = 3.103321$$

$$x = \frac{3.103321 - 2.586101}{4.103321} = 0.1260492$$

$$\text{From graph } y = 0.5$$

$$0.5 = \frac{N - N_{min}}{N + 1}$$

$$\begin{aligned} 0.5N + 0.5 - N &= -N_{min} \\ -0.5N - 0.5 + N &= N_{min} \end{aligned}$$

$$\begin{aligned} 0.5N - 0.5 &= 6.366694 \\ N &= 13.733388 \end{aligned}$$

6/10

4/5

Method is o/c!

$$(9) \quad N_F = N \cdot \frac{N_{Fmin}}{N_{max}}$$

$$\frac{2.880528}{6.366694} \cdot 13.73388 = \frac{6.213716}{7.20} = N_F$$

4/5

(10)

$$\frac{N_F - 1}{13.73388 - N_F} = \left[\frac{54.2}{45.8} \left(\frac{0.35}{0.3} \right) \left[\frac{0.0110701}{0.0305677} \right]^2 \right]^{0.206}$$

$$\frac{N_F - 1}{13.73388 - N_F} = 0.7032646$$



5/5

$$N_F = \frac{6.257719}{7.15}$$

Pretty close. about 0.7% error

(11)

$$F_{RNK-Dist} = \frac{(3.035714)^{6.366694}}{\frac{0.96}{0.04} + (3.035714)^{6.366694}}$$

$$F_{RNK-Dist} = 0.98$$

yes it is reasonably economical
most of the NH stays in Distillate.

10/10

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

A3/100

Name: Iman Abdelrazek

1. (100 pts) A saturated liquid feed at 100 kmol/hr consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of propane is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5		5 C2
C3, propane	2.4		10 C3
C4, butane	0.85		30 C4
C5, pentane	0.28		35 C5
C6, hexane	0.1		20 C6
Total in feed	-	-	100

98 % ←
 96 % ←

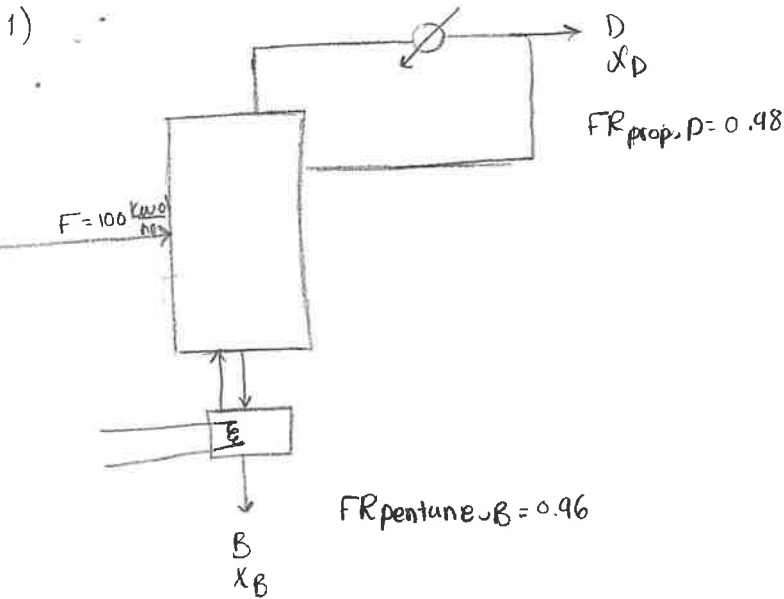
- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,Bott} = (BX_{i,Bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.



3/5

Mole fractions in feed?

e.g. $z_{C2} = 0.05$

⋮

Overall: M.B: $F = B + D$

Comp M.B: $F z_i = B x_{B,i} + D x_{D,i}$

2)

$C_2 \rightarrow \text{LNK}$
 $C_3 \rightarrow \text{LNK}$
 $C_4 \rightarrow \text{LK}$
 $C_5 \rightarrow \text{HK}$
 $C_6 \rightarrow \text{HNK}$

$\alpha_{C2-C5} = \frac{K_{C2}}{K_{C5}} = \frac{7.5}{0.28} = 26.78$ ✓
 $\alpha_{C3-C5} = \frac{K_{C3}}{K_{C5}} = \frac{2.4}{0.28} = 8.57$ ✓
 $\alpha_{C4-C5} = \frac{K_{C4}}{K_{C5}} = \frac{0.85}{0.28} = 3.0357$ ✓
 $\alpha_{C5-C5} = 1$ ✓
 $\alpha_{C6-C5} = \frac{K_{C6}}{K_{C5}} = \frac{1}{0.28} = 3.57$ ✓

2) 5/5

3) Distillate:

$C_2: D x_{C2,D} = F z_{C2} = 5 \text{ kmol/hr}$

$C_3: D x_{C3,D} = F z_{C3} = 10 \text{ kmol/hr}$

$C_4: D x_{C4,D} = F z_{C4} \times FR_{C4,D} = 30 \frac{\text{kmol}}{\text{hr}} (0.98) = 29.4 \text{ kmol/hr}$

$C_5: D x_{C5,D} = F z_{C5} \times (1 - FR_{C5,B}) = 35 \frac{\text{kmol}}{\text{hr}} \times (1 - 0.96) = 1.4 \text{ kmol/hr}$

$C_6: D x_{C6,D} = 0$

$D = 5 + 10 + 29.4 + 1.4 = 45.8 \text{ kmol/hr}$

mole fractions in the distillate:

$C_2: \frac{D x_{2,D}}{D} = \frac{5 \text{ kmol/hr}}{45.8 \text{ kmol/hr}} = 0.1092$

$C_3: \frac{D x_{3,D}}{D} = \frac{10 \text{ kmol/hr}}{45.8 \text{ kmol/hr}} = 0.218$

3) 5/5

7) V

Iman Abdelrazek (4)

$$V_{min} = \frac{\alpha_{c2-c5} D X_{c2,D}}{\alpha_{c2-c5} - \phi} \oplus \frac{\alpha_{c3-c5} D X_{c3,D}}{\alpha_{c3-c5} - \phi} \oplus \frac{\alpha_{c4-c5} D X_{c4,D}}{\alpha_{c4-c5} - \phi} \oplus \frac{\alpha_{c5-c5} D X_{c5,D}}{\alpha_{c5-c5} - \phi} \oplus \frac{\alpha_{c6-c5} D X_{c6,D}}{\alpha_{c6-c5} - \phi}$$

$$= \frac{(26.78)(45.8)(0.104)}{(26.78) - (1.498)} \oplus \frac{(8.57)(45.8)(0.218)}{(8.57) - (1.498)} \oplus \frac{(3.035)(45.8)(0.642)}{(3.035) - (1.498)} \oplus \frac{1(45.8)(0.0304)}{1 - 1.498} \oplus \frac{(3.57)(45.8)(0)}{3.57 - 1.498}$$

$$V_{min} = 3.33778 + 6.67556 + 19.626 + -2.811 + 0 = 72.64 \text{ kN/hL}$$

$$L_{min} = V_{min} - D = 72.64 - 45.8 = 26.84 \text{ kN/hL}$$

$$R_{min} = \frac{L_{min}}{D} = 0.586$$

↑ These values are not correct. I had no time I plugged the full eqn into my calculator and got 72.64.

7) 9/10

$$8) X = \frac{R - R_{min}}{R + 1}$$

$$y = \frac{N - N_{min}}{N + 1}$$

$$\frac{R}{R_{min}} = 1.2$$

$$X = \frac{\frac{R}{R_{min}} - \frac{R_{min}}{R_{min}}}{\frac{R}{R_{min}} + \frac{1}{R_{min}}} = \frac{1.2 - 1}{1.2 + \frac{1}{0.586}} = \frac{1.2 - 1}{1.2 + 1.706} = 0.0688$$

$$\text{at } X = 0.0688$$

$$y = 0.57 \rightarrow \text{From the graph}$$

↑ calculated before.

$$y = \frac{N - N_{min}}{N + 1}$$

A) 5/5

$$yN + y = N - N_{min}$$

$$(N) - yN = y + N_{min}$$

$$N = \frac{y + N_{min}}{1 - y} = \frac{0.57 + 6.4}{1 - 0.57} = 16.2$$

$$= 16.2 \approx 17$$

$$9) \underline{NF} = \frac{NF_{min} \times N}{N_{min}} = \frac{2.879 \times 16.2}{6.4} = 7.9 \approx 8$$

9) 5/5

$$10) \frac{(NF-1)}{N-NF} = \left[\left(\frac{54.2}{45.8} \right) \left(\frac{0.33}{0.30} \right) \left(\frac{0.01}{0.0306} \right)^2 \right]^{0.206}$$

$$\frac{NF-1}{N-NF} = 0.7011$$

NF should be around 7.

10) 2/5

$$NF-1 = 0.7011(N) - 0.7011 NF$$

$$0.2989 NF = 0.7011(N) + 1 \quad \rightarrow 16.2$$

$$NF = \frac{0.7011 N + 1}{0.2989} = 41.35 \quad \times = 42.$$

$$11) FRNK_{ID} = \frac{(3.0357)^{16.2}}{\frac{0.96}{1-0.96} \oplus (3.0357)^{16.2}} = 0.9999$$

11) 4/10

Justify!

e.g.) It is economical

as LNK in deflate

is close 1.0 which

is consistent with assumption

(LNK $\approx$ 1.0 in deflate)

A3/100

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

Name: Bushra Alkarmi

1. (100 pts) A saturated liquid feed at 100 kmol/min consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of propane is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms. (44)

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5	26.786	5
C3, propane	2.4	8.571	10
C4, butane	0.85	3.030	30
C5, pentane	0.28	1.00	35
C6, hexane	0.1	0.357	20
Total in feed		-	100

- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

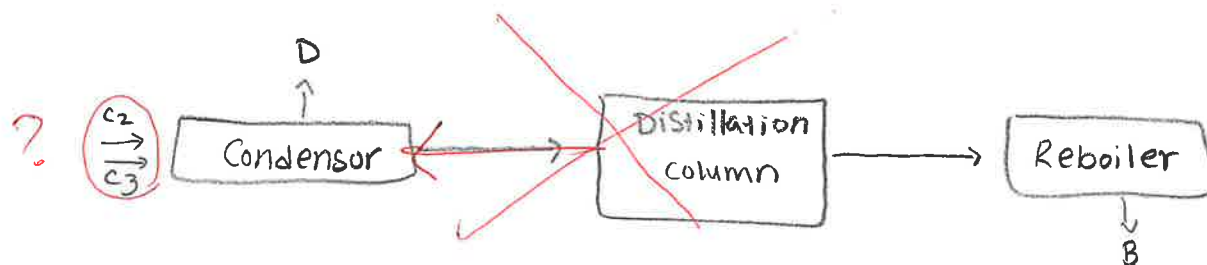
$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (Dx_{i,dist})/Fz_i$ & $FR_{i,Bott} = (Bx_{i,Bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

①



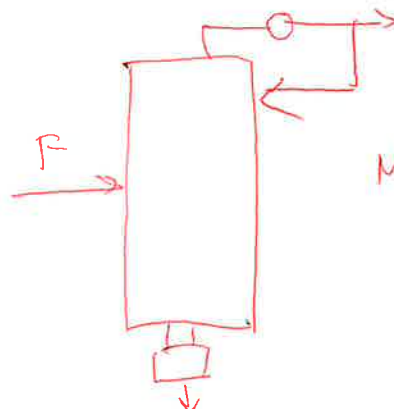
• overall mass balance:

$$F = D + B$$

• Component mass balance:

$$Fz_i = Dx_{i,D} + Bx_{i,B}$$

3/5



Mole fractions?
e.g. $z_{C2} = 0.05$
...

②

$$C_2 = LNK$$

$$C_3 = LNK - 1$$

$$C_4 = LK$$

$$C_5 = HK$$

$$C_6 = HNK$$

$$a_i = \frac{k_i}{k_{HK}} = \frac{k_i}{0.28}$$

4/5

$$a_{C2} = \frac{7.5}{0.28} = 26.786$$

$$a_{C3} = \frac{2.4}{0.28} = 8.571$$

$$a_{C4} = \frac{0.83}{0.28} = 3.036$$

$$a_{C5} = \frac{0.28}{0.28} = 1.00$$

$$a_{C6} = \frac{0.1}{0.28} = 0.357$$

④ $B_i = F_i - D_i$

$B_{C2} = 5 - 5 = 0$ kmol/hr

$B_{C3} = 10 - 10 = 0$

$B_{C4} = 30 - 29.4 = 0.6$

$B_{C5} = 35 - 1.4 = 33.6$

$B_{C6} = 20 - 0 = 20$

unit missing!

-1

$B = 0 + 0 + 0.6 + 33.6 + 20$

$B = \boxed{54.2}$ kmol/hr

$X_{C2, B} = 0$

$X_{C3, B} = 0$

$X_{C4, B} = \frac{0.6}{54.2} = 0.01107$

$X_{C5, B} = \frac{33.6}{54.2} = 0.6199$

$X_{C6, B} = \frac{20}{54.2} = 0.3690$

13/15

validation!

$\sum X_i = 1.0$

• overall balance

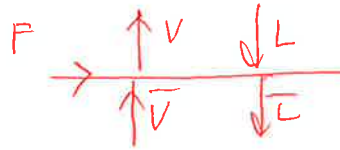
$F = D + B = 45 + 54.2 = 100$

⑥

$$\Delta V_{\text{feed}} = V_{\text{min}} - \bar{V}_{\text{min}} = 0$$

Draw M.B envelope!

$$0 = \sum_{i=2}^6 \frac{a_i F_{zi}}{a_i - \phi}$$



-3

$$0 = \frac{26.786(5)}{26.786 - \phi} + \frac{8.571(10)}{8.571 - \phi} + \frac{3.036(30)}{3.036 - \phi} + \frac{1.0(35)}{1 - \phi} + \frac{0.357(20)}{0.357 - \phi}$$

$$a_{HK} < \phi < a_{LK}$$

$$1.0 < \phi < 3.036$$

the root is

$$1.0 < 1.498 < 3.036$$

12/15

$$\phi = 1.498$$

$$\textcircled{8} \quad \frac{R}{R_{\min}} = 1.2$$

$$R = \frac{L}{D} = 1.2 R_{\min}$$

$$R = 1.2(0.586) = 0.703$$

$$X = \frac{R - R_{\min}}{R + 1}$$

$$X = \frac{0.703 - 0.586}{0.703 + 1} = 0.0688$$

$$Y = \frac{N - N_{\min}}{N + 1} = 0.586$$

$$0.586 = \frac{N - 6.37}{N + 1}$$

$$0.586 \cdot N + 0.586 = N - 6.37$$

$$0.413N = 6.957$$

$$N = 16.82 \approx 17 \text{ stages}$$

5/5

⑪

$$FR_{NK, dist} = \frac{(8.571)^{6.37}}{\frac{0.96}{0.04} + (8.571)^{6.37}}$$

$$= 0.99997 = \underline{99.97\%}$$

✓

10/10

Yes the separation is reasonably economical, because it stays
entirely in the distillate ✓

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

Name: _____

43/100

Andrew Kauth

668817735

1. (100 pts) A saturated liquid feed at 100 kmol/min consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of ~~propane~~ **butane** is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5	26.79	5
C3, propane	2.4	8.57	10
C4, butane	0.85	3.04	30
C5, pentane	0.28	1	35
C6, hexane	0.1	0.36	20
Total in feed	-	-	100

- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

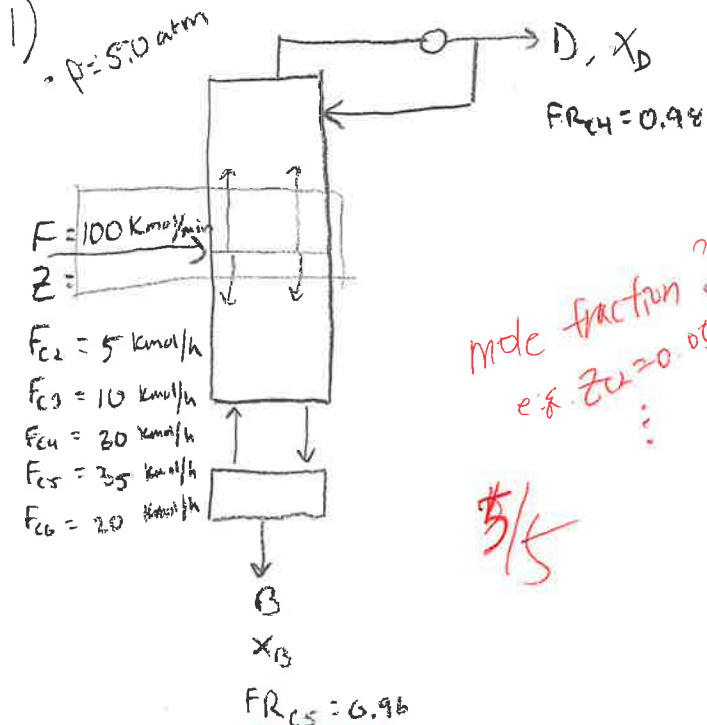
$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,Bott} = (BX_{i,Bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

1) $P = 5.0 \text{ atm}$



mode fraction?
e.g. $z_{C2} = 0.05$

5/5

$$F_i z_i = X_{i,D} D + X_{i,B} B$$

Where $F_i z_i$ is equal to the feeds outlined in the table expression in kmol/hr.

$$F_{C2} = X_{C2,D} \cdot D + X_{C2,B} \cdot B$$

$$F_{C3} = X_{C3,D} \cdot D + X_{C3,B} \cdot B$$

$$F_{C4} = X_{C4,D} \cdot D + X_{C4,B} \cdot B$$

$$F_{C5} = X_{C5,D} \cdot D + X_{C5,B} \cdot B$$

$$F_{C6} = X_{C6,D} \cdot D + X_{C6,B} \cdot B$$

$$X_{C2} = \frac{5}{45.8} = 0.11$$

$$X_{C3} = \frac{10}{45.8} = 0.21$$

$$X_{C4} = \frac{29.4}{45.8} = 0.64$$

$$X_{C5} = \frac{1.4}{45.8} = 0.03$$

$$X_{C6} = \frac{0}{45.8} = 0$$

Validation!

$$\sum X_i = 1.0$$

2)

$$C2 = \text{LNK}$$

$$C3 = \text{LNK}$$

$$C4 = \text{LK}$$

$$C5 = \text{HK}$$

$$C6 = \text{HNK}$$

$$\alpha_{C2} = \frac{K_{C2}}{K_{C5}} = \frac{7.5}{0.28} = 26.79$$

$$\alpha_{C3} = \frac{K_{C3}}{K_{C5}} = \frac{2.4}{0.28} = 8.57$$

$$\alpha_{C4} = \frac{K_{C4}}{K_{C5}} = \frac{0.85}{0.28} = 3.04$$

$$\alpha_{C5} = \frac{K_{C5}}{K_{C5}} = \frac{0.28}{0.28} = 1$$

$$\alpha_{C6} = \frac{K_{C6}}{K_{C5}} = \frac{0.1}{0.28} = 0.36$$

5/5

Need to write down the process!

$$3) 0.98 = \frac{X_{C4,D} \cdot D}{30}$$

$$X_{C4,D} \cdot D = 29.4 \text{ kmol/hr}$$

$$29.4 + 5 + 10 + 1.4 = D$$

$$0.04 = \frac{X_{C5,D} \cdot D}{35} = 1.4$$

$$D = 45.8 \text{ kmol/hr}$$

$X_{C2,D} = 0.11$	$X_{C2,D} \cdot D = 5 \text{ kmol/hr}$
$X_{C3,D} = 0.21$	$X_{C3,D} \cdot D = 10$
$X_{C6,D} = 0$	$X_{C6,D} \cdot D = 0$
$X_{C4,D} = 0.64$	$X_{C4,D} \cdot D = 29.4$
$X_{C5,D} = 0.03$	$X_{C5,D} \cdot D = 1.4$

unit messy -

$$4) 0.96 = \frac{X_{C5,B} \cdot B}{35}$$

$$B = 100 - 45.8 = 54.2$$

$$B = 54.2 \text{ kmol/hr}$$

$$X_{C4,B} = \frac{0.6}{54.2} = 0.011$$

$$X_{C5,B} = \frac{33.6}{54.2} = 0.62$$

$$X_{C6,B} = \frac{20}{54.2} = 0.369$$

11/15

$X_{C2,B} = 0$	$X_{C2,B} \cdot B = 0$
$X_{C3,B} = 0$	$X_{C3,B} \cdot B = 0$
$X_{C4,B} = 0.011$	$X_{C4,B} \cdot B = 0.6$
$X_{C5,B} = 0.62$	$X_{C5,B} \cdot B = 33.6$
$X_{C6,B} = 0.369$	$X_{C6,B} \cdot B = 20$

- unit messy

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

74/100

Name: Pasquale Morrone

1. (100 pts) A saturated liquid feed at 100 kmol/min consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot Z_i$) of propane is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

	Component	K (y/x) value	α (relative volatility) value	kmol/hr
LNK	C2, ethane	7.5	26.74	5
LNK	C3, propane	2.4	8.57	10
LK	C4, butane	0.85	3.035	30
HK	C5, pentane	0.28	1	35
HNK	C6, hexane	0.1	0.357	20
	Total in feed	-	-	100

HK reference

- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
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- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

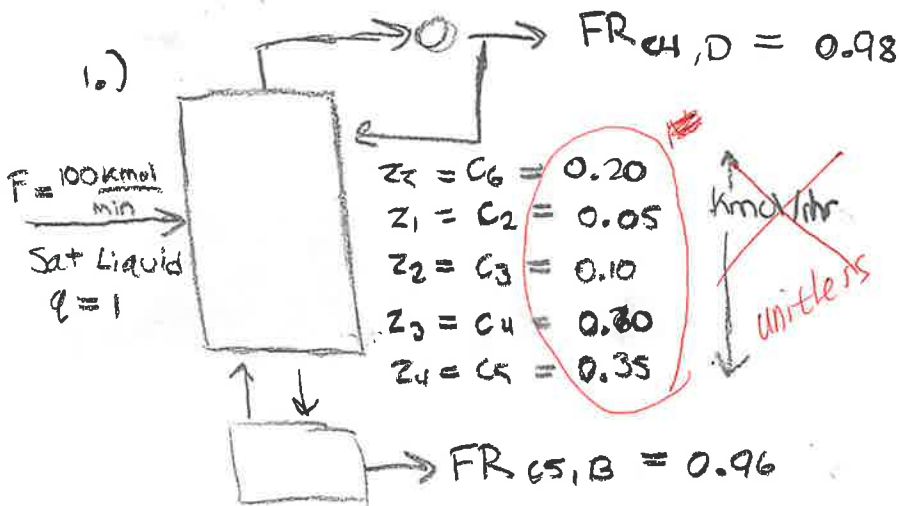
$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (Dx_{i,dist})/Fz_i$ & $FR_{i,Bott} = (Bx_{i,Bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

Pasquale Morrone



Component Mass Balance

$$Fz_i = Dx_{i,D} + Bx_{i,B}$$

2. C_2 : LNK $C_2\alpha = 7.5 / 0.28 = 26.79$

C_3 : LNK $C_3\alpha = 2.4 / 0.28 = 8.57$

C_4 : LK $C_4\alpha = 0.85 / 0.28 = 3.035$

C_5 : HK $C_5\alpha = 0.28 / 0.28 = 1$

C_6 : HNK $C_6\alpha = 0.1 / 0.28 = 0.357$

3.) Find all $Dx_{i,D}$ and D values?

$$Dx_{i,D} = Fz_i \cdot FR_{i,D}$$

$$C_4: 100(0.30) \cdot (0.98) = 29.4 \text{ kmol/hr}$$

$$C_5: 100(0.35) \cdot (1 - 0.96) = 1.4 \text{ kmol/hr}$$

$$C_6: \text{HNK} = 0$$

$$FR_{C_2,D} = \frac{(\alpha_{LK,HK})^{N_{min}}}{\frac{FR_{HK,D}}{1 - FR_{HK,B}} + (\alpha_{LK,HK})^{N_{min}}}$$

$$FR_{LK,D} = \frac{(3.035)^{6.368}}{\frac{0.96}{0.04} + (3.035)^{6.368}} = 0.979$$

$$\alpha_{C_4,C_5} = 3.035 / 1 = 3.035$$

$$N_{min} = \frac{\ln \left[\frac{FR_{C_4,D}}{FR_{C_4,B}} \cdot \frac{FR_{C_5,B}}{FR_{C_5,D}} \right]}{\ln [\alpha_{C_4,C_5}]}$$

$$N_{min} = \frac{\ln \left[\frac{0.98}{0.02} \cdot \frac{0.96}{0.04} \right]}{\ln (3.035)}$$

$$N_{min} = 6.368$$



$$5.) N_{min} = \frac{\ln \left[\frac{0.98}{0.02} \cdot \frac{0.96}{0.04} \right]}{\ln(3.035)}$$

$$\alpha_{CH,CS} = 3.035 / 1 = 3.035$$

= 6.368 or 6 stages

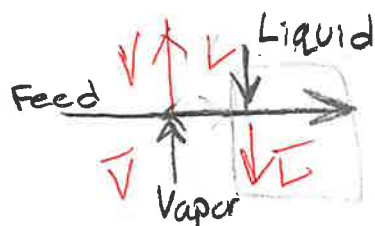
$$N_{F,min} = \frac{\ln \left[\frac{x_{LK,D}}{x_{HK,D}} \cdot \frac{z_{HK}}{z_{LK}} \right]}{\ln[\alpha_{LK,HK}]}$$

$$= \frac{\ln \left[\frac{0.646}{0.03} \cdot \frac{0.35}{0.30} \right]}{\ln(3.035)}$$

$$= \frac{2.088}{0.402}$$

calculator error!

6.) Mass Balance Envelope:



$$\Delta V_{feed} = V_{min} - \bar{V}_{min} = 0 \text{ as } f=0$$

$$\Delta V_{feed} = V_{min} - \bar{V}_{min} = \sum_{i=1}^c \frac{\alpha_{i-ref} F z_i}{\alpha_{i-ref} - \phi}$$

$$\text{Expanded: } = \frac{26.79 \cdot 100(0.05)}{26.79 - \phi} + \frac{8.57(100)(0.1)}{8.57 - \phi} + \frac{3.035(100)(0.3)}{3.035 - \phi} + \frac{1 \cdot 100(0.35)}{1 - \phi} + \frac{0.357(100)(0.2)}{0.357 - \phi}$$

ϕ must meet these conditions: $\alpha_{HK} < \phi < \alpha_{LK}$
1 3.035

ϕ must be 1.498 to meet this condition

$$7.) V_{min} = \sum \frac{\alpha_{i-ref} (D x_{i,D})}{\alpha_{i-ref} - \phi}$$

next page

$$10.) \frac{N_F - 1}{N - N_F} = \left[\frac{B}{D} \left(\frac{Z_{HK}}{Z_{LK}} \right) \left\{ \frac{X_{LK,B}}{X_{HK,D}} \right\}^2 \right]^{0.206}$$

$$= \left[\frac{54.2}{45.485} \left(\frac{0.35}{0.3} \right) \left[\frac{0.011}{0.03} \right]^2 \right]^{0.206}$$

$$\frac{N_F - 1}{N - N_F} = 3.677$$

2/5

$$\frac{N_{Fmin}}{N_{min}} + 1 - \frac{1}{N} + \frac{1}{N_F} = 3.677$$

$$N_F = 0.353$$

Compare your result with (9)

$$11.) FR_{LNK,D} = \frac{(\alpha_{LK,LK})^{N_{min}}}{\frac{FR_{LK,B}}{1 - FR_{LK,B}} + (\alpha_{LK,LK})^{N_{min}}} = 0.979$$

Economically reasonable why?

5/10

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

99/100

Name: vianca Escalante

1. (100 pts) A saturated liquid feed at 100 kmol/^{hr} consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of ^{propane} ~~propane~~ is desired in the distillate. A 96.0% recovery of ^{butane} ~~pentane~~ is desired in the bottoms. HK ref

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

	Component	K (y/x) value	α (relative volatility) value	kmol/hr
E	C2, ethane	7.5	26.8	5
D	C3, propane	2.4	8.57	10
A	C4, butane	0.85	3.04	30
B	C5, pentane	0.28	1	35
C	C6, hexane	0.1	0.36	20
	Total in feed	-	-	100

- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table. pentane HK
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

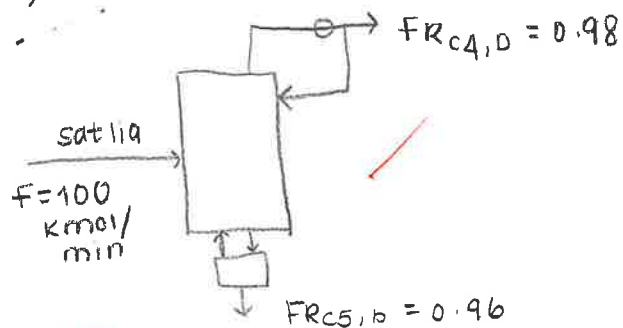
$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,Bott} = (BX_{i,Bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

1.)



$$z_i = \frac{F_i}{F} \quad FR_{i,D} = \frac{x_{i,D} \cdot D}{F \cdot z_i}$$

$$F = B + D$$

$$Fz_i = Bx_{i,B} + Dx_{i,D}$$

$$z_{c2} = 0.05$$

$$z_{c3} = 0.10$$

$$z_{c4} = 0.30$$

$$z_{c5} = 0.35$$

$$z_{c6} = 0.20$$

17 5/5

2.) c₂ ethane LNKc₃ propane LNKc₄ butane LKc₅ pentane HKc₆ hexane HNK

$$\alpha_{AB} = \frac{K_A}{K_B}$$

$$\alpha_{c2-c5} = \frac{7.5}{0.28} = 26.8$$

$$\alpha_{c3-c5} = \frac{2.4}{0.28} = 8.57$$

$$\alpha_{c4-c5} = 3.64$$

$$\alpha_{c5-c5} = 1$$

$$\alpha_{c6-c5} = 0.36$$

2) 5/5

$$3.) FR_{i,D} = \frac{x_{i,D} \cdot D}{F \cdot z_i}$$

Distillate

$$c_2: F \cdot z_{c2} = x_{c2,D} \cdot D \Rightarrow (100)(0.05) = x_{c2,D} \cdot D$$

$$5 = x_{c2,D} \cdot D$$

$$\text{kmol/hr}$$

$$c_3: F \cdot z_{c3} = x_{c3,D} \cdot D \Rightarrow (100)(0.10) = x_{c3,D} \cdot D$$

$$10 = x_{c3,D} \cdot D$$

$$\text{kmol/hr}$$

$$c_4: F \cdot z_{c4} \cdot FR_{c4,D} = x_{c4,D} \cdot D \Rightarrow (100)(0.30)(0.98) = x_{c4,D} \cdot D$$

$$29.4 = x_{c4,D} \cdot D$$

$$\text{kmol/hr}$$

$$c_5: F \cdot z_{c5} \cdot FR_{c5,D} = x_{c5,D} \cdot D \Rightarrow (100)(0.35)(0.04) = x_{c5,D} \cdot D$$

$$1.4 = x_{c5,D} \cdot D$$

$$\text{kmol/hr}$$

$$c_6: 0$$

mole fractions

$$c_2: 5/45.8 = 0.11$$

$$c_3: 10/45.8 = 0.22$$

$$c_4: 29.4/45.8 = 0.64$$

$$c_5: 1.4/45.8 = 0.03$$

$$c_6: 0$$

$$\sum x_i = 1$$

$$D = \sum D x_i = 45.8 \text{ kmol/hr}$$

15/15

4.) Bottom

$$c_2: 0$$

$$c_3: 0$$

$$c_4: F \cdot z_{c4} \cdot FR_{c4,b} = x_{c4,b} \cdot B \Rightarrow (100)(0.30)(0.02) = x_{c4,b} \cdot B$$

$$6.0 = x_{c4,b} \cdot B$$

$$\text{kmol/hr}$$

$$c_5: F \cdot z_{c5} \cdot FR_{c5,b} = x_{c5,b} \cdot B \Rightarrow (100)(0.35)(0.96) = x_{c5,b} \cdot B$$

$$33.6 = x_{c5,b} \cdot B$$

$$\text{kmol/hr}$$

$$c_6: F \cdot z_{c6} = x_{c6,b} \cdot B \Rightarrow (100)(0.20) = x_{c6,b} \cdot B$$

$$20 \text{ kmol/hr} = x_{c6,b} \cdot B$$

$$B = \sum B x_i = 54.2 \text{ kmol/hr}$$

$$c_2: 0$$

$$c_3: 0$$

$$c_4: 6.0/54.2 = 0.11$$

$$c_5: 33.6/54.2 = 0.62$$

$$c_6: 20/54.2 = 0.37$$

$$\sum x_i = 1$$

$$8.) \frac{R}{R_{\min}} = 1.2 \quad R = \frac{L}{D}$$

$$R = 1.2(0.58) = 0.7$$

$$x = \frac{0.7 - 0.58}{0.7 + 1} = 0.071$$

5/5

$$y = 0.57 = \frac{N - N_{\min}}{N + 1} = \frac{N - 6.36}{N + 1}$$

$$0.57N + 0.57 = N - 6.36$$

$$6.93 = 0.43N$$

$$N = 16$$

$$9.) \frac{2.9}{6.36} = \frac{N_F}{16} \quad N_F = 7.3$$

5/5

$$10.) \frac{N_F - 1}{16 - N_F} = \left[\frac{54.2}{45.8} \left(\frac{0.35}{0.30} \right) \left[\frac{0.011}{0.03} \right]^2 \right]^{0.206}$$

$$= \left[1.18 \cdot 1.167 \cdot 0.134 \right]^{0.206} = 0.71$$

$$\frac{N_F - 1}{16 - N_F} = 0.71$$

5/5

$$0.71(16) - 0.71N_F = N_F - 1$$

$$11.36$$

$$12.36 = 1.71N_F$$

$$N_F = 7.23$$

similar value ✓

$$11. \quad FR_{NKid} = \frac{(3.04)^{6.36}}{\left(\frac{0.96}{0.04} \right) + (3.04)^{6.36}} = \frac{1177.8}{24 + 1177.8} = \underline{0.98}$$

It is reasonably economical the FR_{NK} goes almost fully to 1 also is the same value $FR_{LK,0} = 0.98$ ✓

10/10

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

100/100

Name: Nguyen Anh Khoi

1. (100 pts) A saturated liquid feed at 100 kmol/min consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of ~~propane~~ ^{Butane} is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5	26.79	5
C3, propane	2.4	8.57	10
C4, butane	0.85 ^{LK}	3.04	30
C5, pentane	0.28 ^{HK}	1	35
C6, hexane	0.1	0.36	20
Total in feed	-	-	100

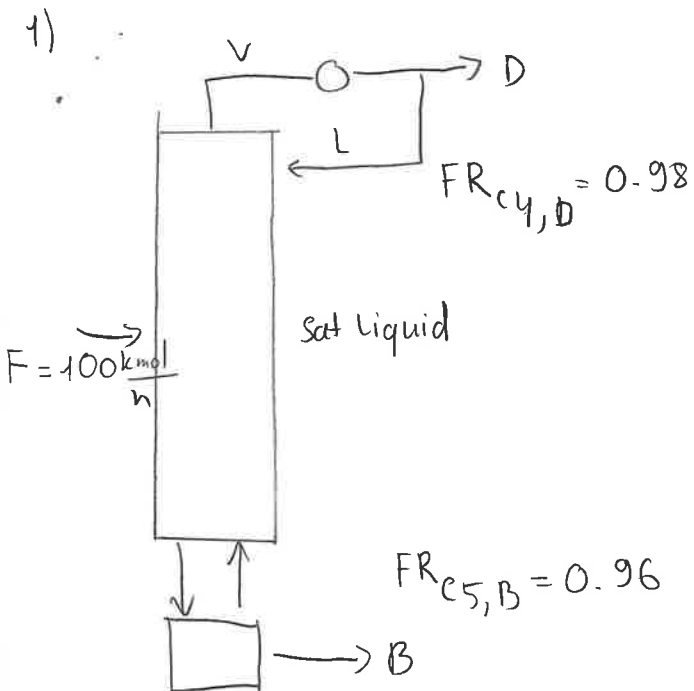
- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,Bott} = (BX_{i,Bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.



$$z_{C2} = \frac{5}{100} = 0.05$$

$$z_{C3} = \frac{10}{100} = 0.10$$

$$z_{C4} = \frac{30}{100} = 0.30$$

$$z_{C5} = \frac{35}{100} = 0.35$$

$$z_{C6} = \frac{20}{100} = 0.20$$

5/5

$$F = D + B \text{ overall mass balance}$$

$$Fz_i = Dx_{i,D} + Bx_{i,B} \rightarrow \text{Component mass balance}$$

2)

$C_2 - \text{LNK}$

Now calculate $\alpha_{i-\text{ref}}$, ref is HK (C_5) $\Rightarrow \alpha_{i-\text{ref}} = \frac{k_i}{k_{\text{ref}}}$

$C_3 - \text{LNK}$

$$\alpha_{C2-C5} = \frac{7.5}{0.28} = 26.79$$

$C_4 - \text{LK}$

$$\alpha_{C3-C5} = \frac{2.4}{0.28} = 8.57$$

$C_5 - \text{HK}$

$$\alpha_{C4-C5} = \frac{0.85}{0.28} = 3.04$$

$C_6 - \text{HNK}$

$$\alpha_{C5-C5} = 1.00$$

5/5

no non key

$$\alpha_{C6-C5} = \frac{0.1}{0.28} = 0.36$$

3)

$$Dx_{C2,D} = Fz_{C2} = 5 \frac{\text{kmol}}{\text{h}}$$

$$Dx_{C6,D} = 0 \frac{\text{kmol}}{\text{h}}$$

$$Dx_{C3,D} = Fz_{C3} = 10 \frac{\text{kmol}}{\text{h}}$$

$$Dx_{C4,D} = FR_{C4,D} Fz_{C4} = 0.98 \times 30 = 29.4 \frac{\text{kmol}}{\text{h}}$$

$$Dx_{C5,D} = (1 - FR_{C5,B}) Fz_{C5} = (1 - 0.96) \times 35 = 1.4 \frac{\text{kmol}}{\text{h}}$$

$$D = 5 + 10 + 29.4 + 1.4 = 45.8 \frac{\text{kmol}}{\text{h}}$$

Rewrite $N_{\min}$

$$N_{\min} = \frac{\ln \left(\frac{FR_{C4,D}}{(1-FR_{C4,D})} \times \frac{FR_{C5,D}}{(1-FR_{C5,D})} \right)}{\ln \alpha_{C4-C5}}$$

Calculate top portion

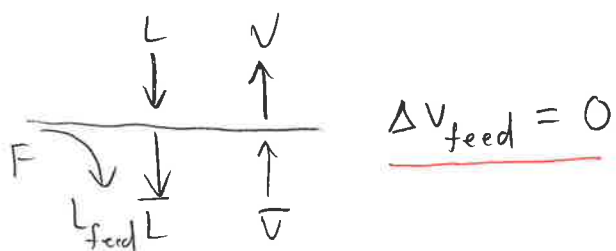
$$\frac{FR_{C4,D}}{(1 - FR_{C4,D})} \times \frac{FR_{C5,B}}{(1 - FR_{C5,B})} = \frac{0.98}{1 - 0.98} \times \frac{0.96}{1 - 0.96} = 1176$$

$$N_{\min} = \frac{\ln 1176}{\ln 3.04} = 6.36 \text{ stages}$$

Now $N_{F, \min}$

$$N_{F, \min} = \frac{\ln \left(\frac{x_{C4, D}}{x_{C5, D}} \times \frac{z_{C5}}{z_{C4}} \right)}{\ln \alpha_{C4-C5}} = \frac{\ln \left(\frac{0.642}{0.031} \times \frac{0.35}{0.30} \right)}{\ln 3.04} = 2.86 \text{ stages}$$

6)



feed is sat liquid

no vapor

$$q = 1$$

$$\Delta V_{\text{feed}} = F(1-q) = F(1-1) = 0$$

$$0 = \frac{\alpha_{c2-c5} F_{zc2}}{\alpha_{c2-c5} - q} + \frac{\alpha_{c3-c5} F_{zc3}}{\alpha_{c3-c5} - q} + \frac{\alpha_{c4-c5} F_{zc4}}{\alpha_{c4-c5} - q} + \frac{\alpha_{c5-c5} F_{zc5}}{\alpha_{c5-c5} - q} + \frac{\alpha_{c6-c5} F_{zc6}}{\alpha_{c6-c5} - q}$$

$$\frac{N - N_{\min}}{N + 1} = 0.572 \rightarrow \text{From graph}$$

$$\frac{N - 6.36}{N + 1} = 0.572$$

$$N = 16.42 \text{ stages}$$

9)

$$N_F = \frac{N_{F,\min}}{N_{\min}} \cdot N = \frac{2.86}{6.36} \times 16.42 = 7.25 \text{ stage}$$

10)

$$\frac{N_F - 1}{16.42 - N_F} = \left[\frac{54.2}{45.8} \left(\frac{0.35}{0.30} \left(\frac{0.044}{0.031} \right)^2 \right)^{0.206}$$

$$N_F = 7.24 \text{ so very close to other value in 9}$$

11)

$$FR_{NK,D} = \frac{(3.04)^{6.36}}{\frac{0.96}{1-0.96} + (3.04)^{6.36}} = 0.98$$

98% of LNK go to the distillate which make it economically viable ~~for~~ economically

10/10

65/100

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

Name: Daniel Brown

1. (100 pts) A saturated liquid feed at 100 kmol/min consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of ~~propane~~ ^{butane} is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5	26.1857	5
C3, propane	2.4	8.571	10
C4, butane	0.85	3.0357	30
C5, pentane	0.28	1.000	35
C6, hexane	0.1	.357	20
Total in feed	-	-	100

- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}} \quad \frac{x_{4,D}}{x_{5,D}} = \frac{.6419}{.0306}$$

where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,Bott} = (BX_{i,Bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

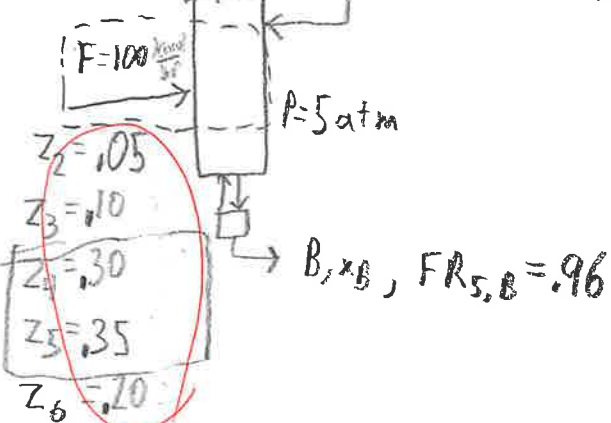
Daniel Brodner

1) $D, x_D, FR_{4,D} = .98$

$$F = D + B$$

$$Fz_i = Dx_{i,D} + Bx_{i,B}$$

5/5



2) $C_2 = \text{LNK}$

$C_3 = \text{LNK}$

$C_4 = \text{LK}$

$C_5 = \text{HK}$

$C_6 = \text{HNK}$

$$\alpha = \frac{K}{K_{HK}}$$

5/5

$$3) Dx_i = \frac{(FR_{i,D})(Fz_i)}{1} \Rightarrow Dx_{4,D} = (.98)(100 \frac{\text{kmol}}{\text{hr}})(.3) = 29.40 \frac{\text{kmol}}{\text{hr}}$$

3) 14/15

$$Dx_2 = Fz_2 = (100 \frac{\text{kmol}}{\text{hr}})(.05) = 5 \frac{\text{kmol}}{\text{hr}}$$

$$Dx_{5,D} = (.04)(100 \frac{\text{kmol}}{\text{hr}})(.35) = 1.40 \frac{\text{kmol}}{\text{hr}}$$

$$Dx_3 = Fz_3 = (100 \frac{\text{kmol}}{\text{hr}})(.10) = 10 \frac{\text{kmol}}{\text{hr}}$$

4) 14/15

$$Dx_6 = 0 \frac{\text{kmol}}{\text{hr}}$$

$$D = \sum Dx_i = 29.40 \frac{\text{kmol}}{\text{hr}} + 5 \frac{\text{kmol}}{\text{hr}} + 10 \frac{\text{kmol}}{\text{hr}} + 1.40 \frac{\text{kmol}}{\text{hr}} = 45.80 \frac{\text{kmol}}{\text{hr}}$$

$$x_4 = \frac{Dx_{4,D}}{D} = 0.6419; x_5 = \frac{Dx_{5,D}}{D} = 0.0306; x_2 = \frac{Dx_{2,D}}{D} = 0.109; x_3 = \frac{Dx_{3,D}}{D} = 0.218$$

$\sum x_i = 1.0$ validate

$$4) Bx_i = (FR_{i,B})(Fz_i) \Rightarrow Bx_{5,B} = (.96)(100 \frac{\text{kmol}}{\text{hr}})(.35) = 33.60 \frac{\text{kmol}}{\text{hr}}$$

$$Bx_6 = Fz_6 = (100 \frac{\text{kmol}}{\text{hr}})(.2) = 20 \frac{\text{kmol}}{\text{hr}}$$

$$Bx_{4,B} = (.02)(100 \frac{\text{kmol}}{\text{hr}})(.3) = .60 \frac{\text{kmol}}{\text{hr}}$$

$$Bx_2 = 0 \frac{\text{kmol}}{\text{hr}}$$

$$B = \sum Bx_i = 33.60 \frac{\text{kmol}}{\text{hr}} + 20 \frac{\text{kmol}}{\text{hr}} + 0.60 \frac{\text{kmol}}{\text{hr}} = 54.20 \frac{\text{kmol}}{\text{hr}}$$

$$Bx_3 = 0 \frac{\text{kmol}}{\text{hr}}$$

$$x_4 = \frac{Bx_{4,B}}{B} = 0.0110; x_5 = \frac{Bx_{5,B}}{B} = 0.6199; x_6 = \frac{Bx_{6,B}}{B} = 0.3690$$

validation
 $\sum x_i = 1.0$

6) 1.498 because ϕ must satisfy $\alpha_{HK} < \phi < \alpha_{LK}$

$$\alpha_{HK} = 1.000 \text{ and } \alpha_{LK} = 3.0357$$

7)

97/100

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

Name: Tyler Eaton

1. (100 pts) A saturated liquid feed at 100 kmol/min consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of propane is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5	$7.5/2.4 = 3.125$	5
C3, propane	2.4	$2.4/0.85 = 2.824$	10
C4, butane	0.85	$0.85/0.28 = 3.036$	30
C5, pentane	0.28		35
C6, hexane	0.1	$0.1/0.1 = 1.0$	20
Total in feed	-	-	100

- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}} = 6.37$$

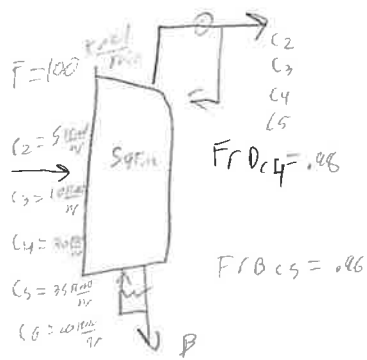
$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}} = 2.9$$

where A=LK and B=HK, $FR_{i,dist} = (Dx_{i,dist})/Fz_i$ & $FR_{i,Bott} = (Bx_{i,Bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

1)

3/5



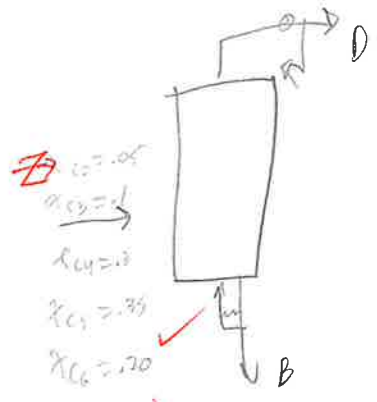
2)

- $C_2 = LMK$ ✓
- $C_3 = LMK$ ✓
- $C_4 = LK$ ✓
- $C_5 = HK$ ✓
- $C_6 = HMK$ ✓

5/5

Tyler Eaton

comp. M.B.?
-2



3)

$$Dx_{C2} = 5 \frac{\text{kmol}}{\text{hr}}$$

$$Dx_{C3} = 10 \cdot 1 = 10 \frac{\text{kmol}}{\text{hr}}$$

$$Dx_{C4} = 30 \cdot 0.98 = 29.4 \frac{\text{kmol}}{\text{hr}}$$

$$Dx_{C5} = 35 \cdot 0.04 = 1.4 \frac{\text{kmol}}{\text{hr}}$$

$$D = 45.8 \frac{\text{kmol}}{\text{hr}}$$

$$x_{C2} = \frac{5}{45.8} = 0.11$$

$$x_{C3} = \frac{10}{45.8} = 0.22$$

$$x_{C4} = \frac{29.4}{45.8} = 0.64$$

$$x_{C5} = \frac{1.4}{45.8} = 0.03$$

$$x_{C6} = 0$$

$$\sum x_i = 1$$

15/15

7)

$$V_{min} = \sum \frac{\alpha_{ref} (D_{k, dist})}{\alpha_{ref} - \phi} \quad | \phi = 1.498$$

$$V_{min} = \frac{26.76 (5)}{26.76 - 1.498} + \frac{8.57 \cdot 10^1}{8.57 - 1.498} + \frac{3.09 \cdot (29.4)}{3.09 - 1.498} + \frac{1 \cdot 1.4}{1 - 1.498} + 0$$

$$5.3 + 12.12 + 57.46 + -2.811 = \boxed{72.57 \frac{\text{kmol}}{\text{hr}}} \quad \checkmark$$
~~$$11.65 + 1.2667 + 1.26 - 1.468 = \boxed{1.752} = V_{min}$$~~

$$V_{min} = L_{min} + D$$

10/10

$$D = 49.8 \frac{\text{kmol}}{\text{hr}}$$

$$V_{min} = 72.57 \frac{\text{kmol}}{\text{hr}}$$

$$L_{min} = (72.57 - 49.8) \frac{\text{kmol}}{\text{hr}} = 26.77 \frac{\text{kmol}}{\text{hr}}$$

$$\left(\frac{L}{D} \right)_{min} = \left(\frac{26.77}{49.8} \right) \frac{\text{kmol}}{\text{hr}} = \boxed{.58} \quad \checkmark$$

8) $R/P_{min} = 1.2$

$$R = 1.2 \cdot .58 = .696 \quad \checkmark$$

5/5

$$\frac{.696 - .58}{.696 + 1} = \boxed{.068} = \text{K-value} \quad \text{so} \quad \frac{N - N_{min}}{N + 1} \approx .6 \quad \checkmark$$

$$N_{min} = 6.37$$

$$\frac{N - N_{min}}{N + 1} = .6$$

$$\frac{N - 6.37}{N + 1} = .6$$

$$N - 6.37 = .6N - .6$$

$$.4N = 6.97$$

$$\boxed{N = 17.425} - 18 \text{ stages} \quad \checkmark$$

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

100/100
 Name: Iliana Perez

1. (100 pts) A saturated liquid feed at 100 kmol/min consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of ^{butane}propane is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.
 Heavy key is ~~propane~~ ^{butane} C5

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5	26.79	5
C3, propane	2.4	8.57	10
C4, butane	0.85	3.04	30
HK C5, pentane	0.28	1	35
C6, hexane	0.1	0.36	20
Total in feed	-	-	100

- ✓ 1) (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- ✓ 2) (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- ✓ 3) (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- ✓ 4) (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- ✓ 5) (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,bot} = (BX_{i,bot})/Fz_i$

- ✓ 6) (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

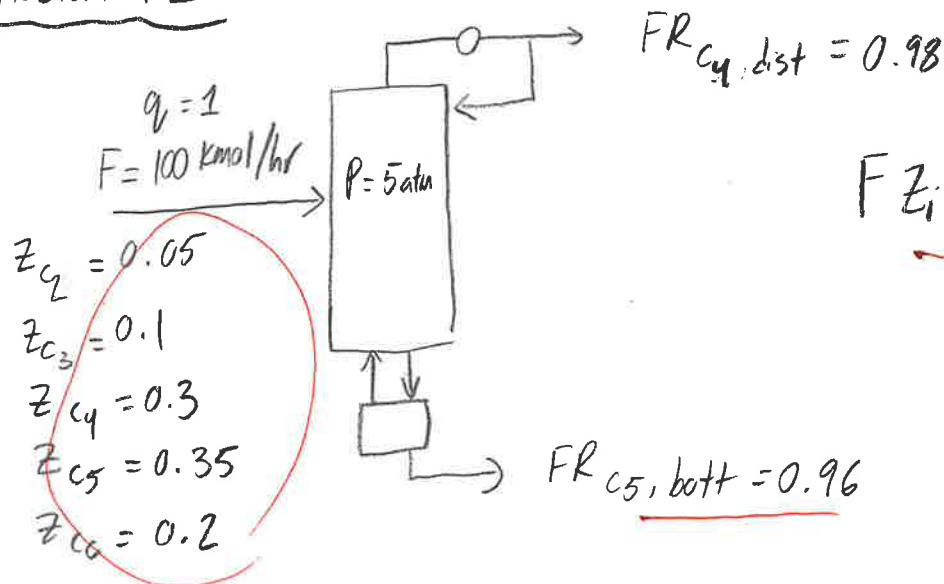
$V_{min} - \bar{U}_{min}$

don't solve expand

HK ϕ LK

Iliana Perez

Problem #1



$$Fz_i = Dx_{i,D} + Bx_{i,B}$$

5/5

Problem #2

C_2 , Ethane = LNK

C_3 , Propane = LNK

C_4 , butane = LK

C_5 , Pentane = HK

C_6 , hexane = HNK

$$\alpha_{C_2-C_5} = \frac{1.5}{0.28} = 26.79$$

$$\alpha_{C_3-C_5} = 2.4/0.28 = 8.57$$

$$\alpha_{C_4-C_5} = 0.85/0.28 = 3.04$$

$$\alpha_{C_5-C_5} = 0.28/0.28 = 1$$

$$\alpha_{C_6-C_5} = 0.1/0.28 = 0.36$$

5/5

Problem #3

$$Dx_{C_2, \text{dist}} = (1)(5) = 5 \frac{\text{kmol}}{\text{hr}}$$

$$Dx_{C_3, \text{dist}} = (1)(10) = 10 \frac{\text{kmol}}{\text{hr}}$$

$$Dx_{C_4, \text{dist}} = (0.98)(30) = 29.4 \frac{\text{kmol}}{\text{hr}}$$

$$Dx_{C_5, \text{dist}} = (0.04)(35) = 1.4 \frac{\text{kmol}}{\text{hr}}$$

$$Dx_{C_6, \text{dist}} = (0)(20) = 0 \frac{\text{kmol}}{\text{hr}}$$

$$\Rightarrow D = \sum Dx_{i,D} = 5 + 10 + 29.4$$

$$+ 1.4 = 45.8 \frac{\text{kmol}}{\text{hr}}$$

15/15

$$x_{C_2, \text{dist}} = 5/45.8 = 0.11$$

$$x_{C_3, \text{dist}} = 10/45.8 = 0.22$$

$$x_{C_4, \text{dist}} = 29.4/45.8 = 0.64$$

$$x_{C_5, \text{dist}} = 1.4/45.8 = 0.03$$

$$x_{C_6, \text{dist}} = 0/45.8 = 0$$

$$\sum x_i = 0.11 + 0.22 + 0.64 + 0.03 + 0 = 1$$

Iliana Perez

Problem # 6 continued

$$\Delta V_{\text{feed}} = V_{\text{min}} - \bar{V}_{\text{min}} = \sum_{i=1}^C \frac{\alpha_{i-\text{ref}} (FZ_i)}{\alpha_{i-\text{ref}} - \phi}$$

$$0 = \frac{\alpha_{C_2-C_5} (FZ_{C_2})}{\alpha_{C_2-C_5} - \phi} + \frac{\alpha_{C_3-C_5} (FZ_{C_3})}{\alpha_{C_3-C_5} - \phi} + \frac{\alpha_{C_4-C_5} (FZ_{C_4})}{\alpha_{C_4-C_5} - \phi} + \frac{\alpha_{C_5-C_5} (FZ_{C_5})}{\alpha_{C_5-C_5} - \phi} + \frac{\alpha_{C_6-C_5} (FZ_{C_6})}{\alpha_{C_6-C_5} - \phi}$$

$$0 = \frac{26.79(5)}{26.79 - \phi} + \frac{8.57(10)}{8.57 - \phi} + \frac{3.04(30)}{3.04 - \phi} + \frac{1(35)}{1 - \phi} + \frac{0.36(20)}{0.36 - \phi}$$

$$0 = \frac{133.95}{26.79 - \phi} + \frac{85.7}{8.57 - \phi} + \frac{91.2}{3.04 - \phi} + \frac{35}{1 - \phi} + \frac{7.2}{0.36 - \phi}$$

15/15

$$\alpha_{\text{HK-HK}} < \phi < \alpha_{\text{LK-HK}} \rightarrow 1 < \phi < 3.04$$

The correct root is $\phi = 1.498$ because that is the root that falls in between the HK relative volatility and LK relative volatility.

Problem # 7

$$V_{\text{min}} = \frac{26.79(5)}{26.79 - 1.498} + \frac{8.57(10)}{8.57 - 1.498} + \frac{3.04(29.4)}{3.04 - 1.498} + \frac{1(1.4)}{1 - 1.498} + \frac{0.36(0)}{0.36 - 1.498}$$

$$V_{\text{min}} = 5.296 + 12.118 + 57.961 + (-2.811) + 0 = 72.56 \frac{\text{kmol}}{\text{hr}}$$

$$L_{\text{min}} = V_{\text{min}} - D = 72.56 - 45.8 = 26.76 \frac{\text{kmol}}{\text{hr}}$$

$$\left(\frac{L}{D}\right)_{\text{min}} = \frac{26.76}{45.8} = 0.58$$

10/10



Iliana Perez

Problem # 11

$$FR_{NK, dist} = \frac{(3.04)^{6.36}}{\frac{0.96}{1-0.96} + (3.04)^{6.36}} = 0.98$$

10/10

This process is reasonably economical because majority of NK is returned in the distillate, which makes sense because the NK's do not distribute and stay together.

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

Name: An The Phan

1. (100 pts) A saturated liquid feed at 100 kmol/min consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of propane is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

	Component	K (y/x) value	α (relative volatility) value	kmol/hr
LNK	C2, ethane	7.5	26.786	5
LNK	C3, propane	2.4	8.571	10
LK	C4, butane	0.85	3.036	30
HK	C5, pentane	0.28	1	35
HNK	C6, hexane	0.1	0.357	20
	Total in feed	-	-	100

- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

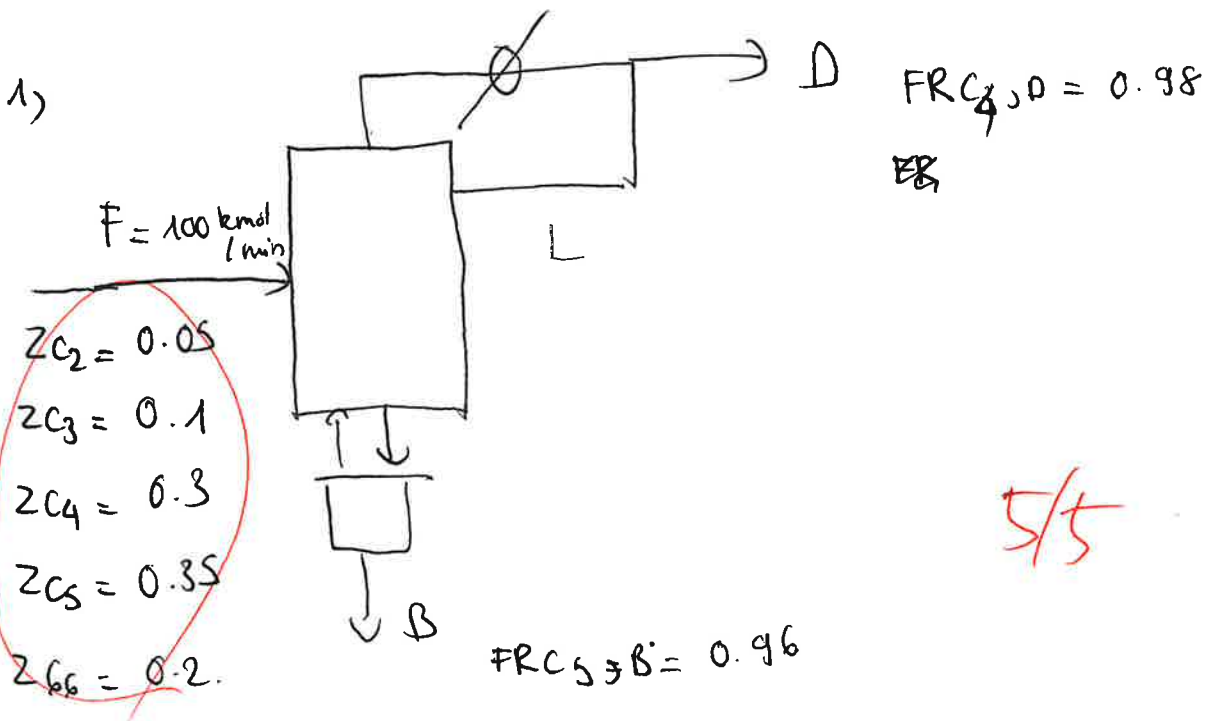
where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,Bott} = (BX_{i,Bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

F(29)

Q1

ex 1)



5/5

$$S = Fz_{C2} = Bx_{C2,B} + Dx_{C2,D}$$

$$10 = Fz_{C3} = Bx_{C3,B} + Dx_{C3,D}$$

$$30 = Fz_{C4} = Bx_{C4,B} + Dx_{C4,D}$$

$$35 = Fz_{C5} = Bx_{C5,B} + Dx_{C5,D}$$

$$20 = Fz_{C6} = Bx_{C6,B} + Dx_{C6,D}$$

Q2

LNK: C_2 LK: C_4 LNK: C_3 HK: C_5 HNK: C_6

I calculate

q in the first penic

kmol/h unit messy! 5/5
-1

$$Q3 \quad FRC_{2,D} = 1 \Rightarrow S = Dx_{C2,D}$$

$$FRC_{3,D} = 0.98 \Rightarrow 10 = Dx_{C3,D}$$

$$FRC_{4,D} = 0.98 \Rightarrow 29.4 = Dx_{C4,D}$$

$$FRC_{5,D} = 1 - 0.96 \Rightarrow 1.4 = Dx_{C5,D}$$

$$= 0.04$$

$$FRC_{6,D} = 0$$

$$\Rightarrow Dx_{C6,D} = 0$$

$$\Rightarrow D = 45.8 \text{ kmol/h}$$

$$x_{C2,D} = 0.1092$$

$$\Rightarrow x_{C3,D} = 0.218$$

$$x_{C4,D} = 0.642$$

$$x_{C5,D} = 0.0306 \Rightarrow \sum x = 0.9992$$

14/15

Q4.

$$FRC_{2,B} = 0 \Rightarrow \overset{B}{X}_{C_{2,B}} = 0$$

15/15

$$FRC_{3,B} = 0 \Rightarrow B \times C_{3,B} = 0$$

$$FRC_{4,B} = 1 - 0.98 = 0.02 \Rightarrow B \times C_{4,B} = 0.6 \text{ kmol/min hr}$$

$$FRC_{5,B} = 0.96 \Rightarrow B \times C_{5,B} = 33.6 \text{ kmol/min hr}$$

$$FRC_{6,B} = 1 \Rightarrow B \times C_{6,B} = 20 \text{ kmol/hr.}$$

$$\Rightarrow B = 54.2 \text{ kmol/min hr}$$

$$\Rightarrow (B + D = 45.8 + 94.2) \approx 100$$

$$\Rightarrow X_{C_{4,B}} = 0.0111$$

$$X_{C_{5,B}} = 0.62 \Rightarrow \sum x = 1.0001 \approx 1$$

$$X_{C_{6,B}} = 0.369$$

validate: $FZC_3 = 0 + 10 = 10 \approx 10$

$$FZC_2 = S = 54.2 \cdot 0 + 45.8 \cdot 0.1092$$

$$\Rightarrow S \approx 5.00136$$

$$FZC_4 = 30 = 54.2 \cdot 0.0111 + 45.8 \cdot 0.642$$

$$30 \approx 30.00522$$

$$FZC_5 = 33.6 + 1.4 = 35 = 35$$

$$FZC_6 = 20 + =$$

Q5.

$$N_{min} = \ln \left\{ \frac{FRC_{4,D}}{1 - FRC_{4,D}} \cdot \frac{FRC_{5,B}}{1 - FRC_{5,B}} \right\}$$

$$= \frac{\ln \left\{ \frac{0.98}{1 - 0.98} \cdot \frac{0.96}{1 - 0.96} \right\}}{\ln 3.036} = 6.3661$$

$$N_{Fmin} = \frac{\ln \left\{ \frac{x_{C4,D}}{x_{C6,D}} \cdot \frac{z_{C45,R}}{C_{4,P}} \right\}}{\ln \alpha_{C4,C6}}$$

$$= \frac{\ln \left\{ \frac{0.642}{0.0306} \cdot \frac{0.35}{0.3} \right\}}{\ln 3.036}$$

$$= 2.879$$

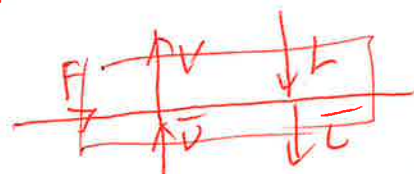
5/5

M.B envelope!

Q6. $F = B + D$

$$F = L + V$$

$$q = \frac{L}{F} \quad \text{as } q = 1 \Rightarrow L = F$$



$$V_{min} = V_{min} - 3$$

$$\Rightarrow F = F + V \Rightarrow F(1 - q) = 0 = V_{feed}$$

$$\Rightarrow F(1 - q) = V$$

$$\Rightarrow 0 = \frac{26.786 \cdot 5}{26.786 - \phi} + \frac{8.571 \cdot 10}{8.571 - \phi} + \frac{3.036 \cdot 30}{3.036 - \phi} + \frac{1 \cdot 35}{1 - \phi}$$

$$+ \frac{0.357 \cdot 20}{0.357 - \phi}$$

$$(1 < \phi < 3.036)$$

↓

⇒ the correct root is 1.498

12/15

Q7. $\Rightarrow V_{min} = \frac{26.786 \cdot 5}{26.786 - 1.498} + \frac{8.571 \cdot 10}{8.571 - 1.498} + \frac{3.036 \cdot 29.4}{3.036 - 1.498} + \frac{1 \cdot 1.4}{1 - 1.498}$

$$= 72.638 \text{ kmol/min hr.}$$

$$\Rightarrow L_{min} = V_{min} - D$$

$$= 72.638 - 45.8$$

$$= 26.838$$

$$\Rightarrow \left[\frac{L}{D} \right]_{min} = R_{min} = \frac{26.838}{45.8} = 0.586$$

10/10

Q8. $R/R_{min} = 1.2$

$\Rightarrow R = 1.2 \cdot 0.586$

$= 0.7032$

$\Rightarrow x = \frac{0.7032 - 0.586}{0.7032 + 1} = \underline{0.0688}$

$\Rightarrow$ According to fit curve: $y = \underline{0.545}$

$\Rightarrow 0.545 = \frac{N - 6.3661}{N + 1}$

$\Rightarrow 0.545N + 0.545 = N - 6.3661$

$\Rightarrow 6.9111 = 0.455N$

$\Rightarrow N = \underline{15.189}$ ✓

5/5

Q9. $\frac{N_F, min}{N_{min}} = \frac{N_F}{N} \Rightarrow \frac{2.879}{6.3661} = \frac{N_F}{15.189}$

5/5

$\Rightarrow N_F = 6.8691$

Q10: $\frac{N_F - 1}{15.189 - N_F} = \left[\frac{54.2}{45.8} \cdot \frac{0.85}{0.3} \cdot \left[\frac{0.0111}{0.0306} \right]^2 \right]^{0.266}$

$\Rightarrow \frac{N_F - 1}{15.189 - N_F} = 0.7037$

$\Rightarrow N_F - 1 = 10.689 - 0.7037 N_F$

5/5

$\Rightarrow 1.7037 N_F = 11.689$ ✓

$\Rightarrow N_F = \underline{6.861} \approx N_F = 6.8691$

(211)

$$\text{FRWK, dist} = \frac{(3.036)^{6.3661}}{\frac{0.96}{1-0.96} + 3.036^{6.3661}}$$

$$= 0.9799989$$

⇒ this distillation process is reasonably economical.

why?

5/10

95/100

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

Name: Patrick Flores

1. (100 pts) A saturated liquid feed at 100 kmol/min consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of ~~propane~~ propane is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5	26.74	5
C3, propane	2.4	8.57	10
C4, butane	0.85	3.04	30
C5, pentane	0.28	1.0	35
C6, hexane	0.1	0.36	20
Total in feed	-	-	100

- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B , D , and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

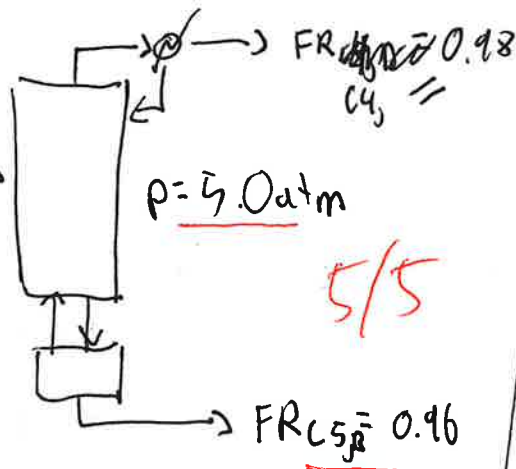
$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,Bot} = (BX_{i,Bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a mass balance envelope around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

$F = 100 \text{ kmol/hr}$
 $z_{C2} = 0.05$
 $z_{C3} = 0.10$
 $z_{C4} = 0.30$
 $z_{C5} = 0.35$
 $z_{C6} = 0.10$



$p = 5.0 \text{ atm}$

5/5

mass balance: $F = D + B \rightarrow 100 = D + B$

$\sum_i F z_i = \sum_i D x_{i,D} + \sum_i B x_{i,B}$

2.) LNK: Propane (C3) LNK: ethane (C2) ~~LNK: butane (C4)~~
 HK: Pentane (C5) HNK: hexane (C6) LK: 1

$\alpha_{C2-C5} = 7.5 / 0.18 = 26.79$
 $\alpha_{C3-C5} = 2.4 / 0.18 = 8.57$
 $\alpha_{C4-C5} = 0.85 / 0.18 = 3.04$
 $\alpha_{C5-C5} = 0.18 / 0.18 = 1$
 $\alpha_{C6-C5} = 0.10 / 0.18 = 0.36$

5/5

14/15

3.) $C2: 1 = \frac{D \cdot x_{C2,D}}{F \cdot z_{C2}} \rightarrow D \cdot x_{C2,D} = 5 \text{ kmol/hr}$
 $C3: 1 = \frac{D \cdot x_{C3,D}}{F \cdot z_{C3}} \rightarrow D \cdot x_{C3,D} = 10 \text{ kmol/hr}$
 $C4: 0.98 = \frac{D \cdot x_{C4,D}}{F \cdot z_{C4}} \rightarrow D \cdot x_{C4,D} = 29.4 \text{ kmol/hr}$
 $C5: 1 - 0.96 = \frac{D \cdot x_{C5,D}}{F \cdot z_{C5}} \rightarrow D \cdot x_{C5,D} = 1.4 \text{ kmol/hr}$
 $C6: 0 = \frac{D \cdot x_{C6,D}}{F \cdot z_{C6}} \rightarrow D \cdot x_{C6,D} = 0 \text{ kmol/hr}$

$x_{C2,D} = 0.109$
 $x_{C3,D} = 0.218$
 $x_{C4,D} = 0.642$
 $x_{C5,D} = 0.031$
 $x_{C6,D} = 0.0$

$\frac{5}{D} + \frac{10}{D} + \frac{29.4}{D} + \frac{1.4}{D} = 1 \rightarrow$

$D = 49.8 \text{ kmol/hr}$

validation $\sum x_i = 1.0$

6. continued

$$\Delta V_{\text{feed}} = \frac{\alpha_{C2-C5} F Z_{C2}}{\alpha_{C2-C5} - \phi} + \frac{\alpha_{C3-C5} F Z_{C3}}{\alpha_{C3-C5} - \phi} + \frac{\alpha_{C4-C5} F Z_{C4}}{\alpha_{C4-C5} - \phi} + \frac{\alpha_{C5-C5} F Z_{C5}}{\alpha_{C5-C5} - \phi} + \frac{\alpha_{C6-C5} F Z_{C6}}{\alpha_{C6-C5} - \phi}$$

7.

$$V_{\min} = \frac{26.79 \cdot 5}{26.79 - 1.498} + \frac{8.57 \cdot 10}{8.57 - 1.498} + \frac{2.04 \cdot 29.4}{3.04 - 1.498} + \frac{1.0 \cdot 1.4}{1.0 - 1.498} + 0$$

$V_{\min} = 72.56 \text{ kmol/hr}$

10/10

$$V_{\min} = L_{\min} + D \rightarrow L_{\min} = 72.56 - 45.8 = 26.76 \text{ kmol/hr}$$

$$\left(\frac{L}{D}\right)_{\min} = \frac{26.76}{45.8} = 0.584$$

8. $\frac{R}{0.584} = 1.2 \rightarrow R = 0.7008$

$$X = \frac{0.7008 - 0.584}{0.7008 + 1} = 0.069 \xrightarrow{\text{Gilliland}} Y = 0.55$$

5/5

$$0.55 = \frac{N - N_{\min}}{N + 1} \rightarrow 0.55N + 0.55 = N - N_{\min} \rightarrow N_{\min} = N - 0.55$$

$$0.45 \cdot N = N_{\min} + 0.55$$

$$N = 15.33 \text{ stages}$$

9. $\frac{2.86}{6.35} = \frac{N_F}{15.33} \rightarrow N_F = 6.90 \text{ stages}$

5/5

11.

$$FR_{NK, dist} = \frac{3.04}{\frac{0.96}{1-0.96} + 3.04} 6.35 = 0.98$$

~~This distillation process is NOT reasonably economical, since potential nonkey components would go almost entirely into the distillate.~~

This distillation process IS economical. Only ~6 to 7 stages give a highly recovery of the target component in the distillate. ✓

10/10

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

95/100

Name: Anaan Chougale

1. (100 pts) A saturated liquid feed at 100 kmol/hr ^{is} ~~consisting~~ consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of ~~propane~~ ^{butane} is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5	26.74	5
C3, propane	2.4	8.57	10
C4, butane	0.85	3.04	30
C5, pentane	0.28	1	35
C6, hexane	0.1	0.36	20
Total in feed	-	-	100

- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

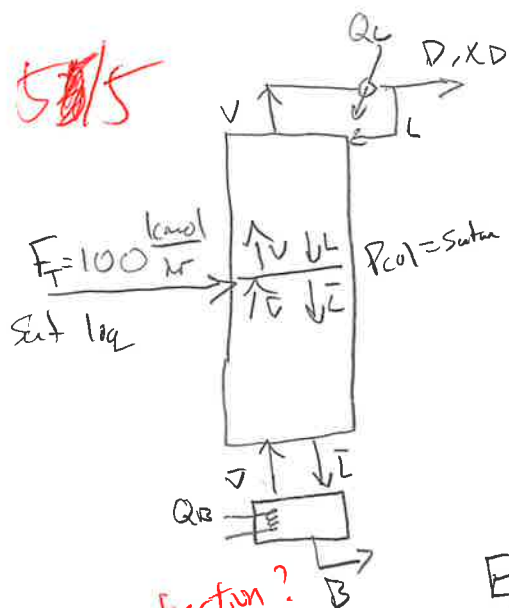
$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,Bott} = (BX_{i,Bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

1.)

5/5



$$FR_{D3} = 0.98$$

$$FR_{B4} = 0.96$$

$$F_1 = 5 \frac{\text{kmol}}{\text{hr}} \quad K_1 = 7.5$$

$$F_2 = 10 \frac{\text{kmol}}{\text{hr}} \quad K_2 = 2.4$$

$$F_3 = 30 \frac{\text{kmol}}{\text{hr}} \quad K_3 = 0.85$$

$$F_4 = 35 \frac{\text{kmol}}{\text{hr}} \quad K_4 = 0.28$$

$$F_5 = 20 \frac{\text{kmol}}{\text{hr}} \quad K_5 = 0.1$$

2.)
ethane $\rightarrow 1$: LNK
Propane $\rightarrow 2$: LNK
butane $\rightarrow 3$: LK
pentane $\rightarrow 4$: HK
n-hexane $\rightarrow 5$: HNK

Note

Feed Mole fraction?
e.g. $z_{C2} = 0.05$
...

External Mass Balances

$$\text{Tot: } F = B + D$$

$$\text{Comp: } F z_i = B x_{Bi} + D x_{Di} \quad \text{where } i \text{ is } 1, 2, 3, 4, \text{ or } 5$$

$$F \left(\frac{F_i}{F} \right) = B \left(\frac{B_i}{B} \right) + D \left(\frac{D_i}{D} \right)$$

$$F_i = B_i + D_i$$

2.)

Using HK as ref

$$\alpha_{14} = \frac{K_1}{K_4} = \frac{7.5}{0.28} = 26.79$$

$$\alpha_{24} = \frac{K_2}{K_4} = \frac{2.4}{0.28} = 8.57$$

$$\alpha_{34} = \frac{K_3}{K_4} = \frac{0.85}{0.28} = 3.04$$

$$\alpha_{44} = \frac{K_4}{K_4} = \frac{0.28}{0.28} = 1$$

$$\alpha_{54} = \frac{K_5}{K_4} = \frac{0.1}{0.28} = 0.36$$

5/5

3.) Distillate

$$DX_{D1} = D_1 = F_1 = 5 \frac{\text{kmol}}{\text{hr}}$$

$$DX_{D2} = D_2 = F_2 = 10 \frac{\text{kmol}}{\text{hr}}$$

$$DX_{D3} = D_3 = FR_{D3} F_3 = 0.98(30) = 29.4 \frac{\text{kmol}}{\text{hr}}$$

$$DX_{D4} = D_4 = (1 - FR_{B4}) F_4 = (1 - 0.96) 35 = 1.4 \frac{\text{kmol}}{\text{hr}}$$

$$DX_{D5} = D_5 = 0 \frac{\text{kmol}}{\text{hr}}$$

(Answer Choukale)

$$D = \sum_i D_i$$

$$D = 5 + 10 + 29.4 + 1.4 + 0 = 45.8 \frac{\text{kmol}}{\text{hr}}$$

15 ~~15~~ / 15

4.) Bottom

$$BX_{B1} = B_1 = 0 \frac{\text{kmol}}{\text{hr}}$$

$$BX_{B2} = B_2 = 0 \frac{\text{kmol}}{\text{hr}}$$

$$BX_{B3} = B_3 = (1 - FR_{D3}) F_3 = (1 - 0.98) 30 = 0.6 \frac{\text{kmol}}{\text{hr}}$$

$$BX_{B4} = B_4 = FR_{B4} F_4 = 0.96(35) = 33.6 \frac{\text{kmol}}{\text{hr}}$$

$$BX_{B5} = B_5 = F_5 = 20 \frac{\text{kmol}}{\text{hr}}$$

$$B = \sum_i B_i$$

$$B = 0 + 0 + 0.6 + 33.6 + 20 = 54.2 \frac{\text{kmol}}{\text{hr}}$$

15 ~~15~~ / 15

mole fractions

3.) Dist

$$x_{D1} = \frac{D_1}{D} = \frac{5}{45.8} = 0.11$$

$$x_{D2} = \frac{D_2}{D} = \frac{10}{45.8} = 0.22$$

$$x_{D3} = \frac{D_3}{D} = \frac{29.4}{45.8} = 0.64$$

$$x_{D4} = \frac{D_4}{D} = \frac{1.4}{45.8} = 0.03$$

$$x_{D5} = 0$$

4.) Bot

$$x_{B1} = 0$$

$$x_{B2} = 0$$

$$x_{B3} = \frac{B_3}{B} = \frac{0.6}{54.2} = 0.01$$

$$x_{B4} = \frac{B_4}{B} = \frac{33.6}{54.2} = 0.62$$

$$x_{B5} = \frac{B_5}{B} = \frac{20}{54.2} = 0.37$$

Validation: $\sum x_i = 1.0$

Validation: $\sum x_i = 1.0$

Validating Answers

$$F = B + D = 45.8 + 54.2 = 100 \frac{\text{kmol}}{\text{hr}} \checkmark$$

$$\sum_{i=1}^5 x_{Di} = 0.11 + 0.22 + 0.64 + 0.03 = 1 \checkmark$$

$$\sum_{i=1}^5 x_{Bi} = 0 + 0 + 0.01 + 0.62 + 0.37 = 1 \checkmark$$

8.)

Anuran Chougale

$$\frac{\frac{L}{D}}{\left(\frac{L}{D}\right)_{\min}} = 1.2 \Rightarrow \frac{L}{D} = 1.2 \left(\frac{L}{D}\right)_{\min} = 1.2(0.58) = 0.696$$

$$X\text{-axis} = \frac{0.696 - 0.58}{0.696 + 1} = 0.068$$

5/5

$$y \approx 0.53 \text{ (from graph)} = \frac{N - N_{\min}}{N + 1} \Rightarrow 0.53N + 0.53 = N - N_{\min}$$

$$0.47N = 0.53 + N_{\min} \Rightarrow N = \frac{0.53 + N_{\min}}{0.47} = \frac{0.53 + 6.36}{0.47} = 14.66 \text{ stages}$$

9.)

$$\frac{N_{F,\min}}{N_{\min}} = \frac{N_F}{N} \Rightarrow N_F = \frac{N_{F,\min}}{N_{\min}} \cdot N = \frac{2.63}{6.36} (14.66) = 6.04 \text{ stages}$$

But OK!

7.50

10.)

$$\left(\frac{N_F - 1}{14.66 - N_F} \right) = \left[\frac{54.2}{45.8} \left(\frac{0.35}{0.3} \right) \left[\frac{0.01}{0.03} \right]^2 \right]^{0.206}$$

5/5

$$\frac{N_F - 1}{14.66 - N_F} = 0.68 \Rightarrow N_F - 1 = 9.97 - 0.68N_F = 1.68N_F = 10.97$$

$$N_F = 6.53 \text{ stages}$$

7.15

The value of the Kirkbride eqn for N_F is larger than the ration eqns. ($6.04 < 6.53$)

slightly larger. But they are close.

$$11.) FR_{D3} = \frac{(\alpha_{34})^{N_{\min}}}{FR_{B4} + (\alpha_{34})^{N_{\min}}} = \frac{(3.04)^{2.63}}{\frac{0.96}{1 - 0.96} + (3.04)^{2.63}} = 0.44$$

~ 1.0

5/10

The fractional recovered of butane in the distillate at the desired fractional recovery of pentane in the bottom (0.6%) is 44%. The distillation process is not reasonably economical due to the low fractional recovery of butane in the distillate

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

653846557 72/100

Name: Peter-James Postlewait

1. (100 pts) A saturated liquid feed at 100 kmol/min consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of ~~propane~~ ^{of propane} is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5	26.79	5
C3, propane	2.4	8.571	10
C4, butane	0.85	3.036	30
C5, pentane	0.28	1	35
C6, hexane	0.1	0.357	20
Total in feed	-	-	100

- 1) (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- 2) (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- 3) (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- 4) (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- 5) (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,Bott} = (BX_{i,Bot})/Fz_i$

- 6) (15 pts - Underwood equation) After drawing a mass balance envelope around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: 0.422, **1.498**, and **5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

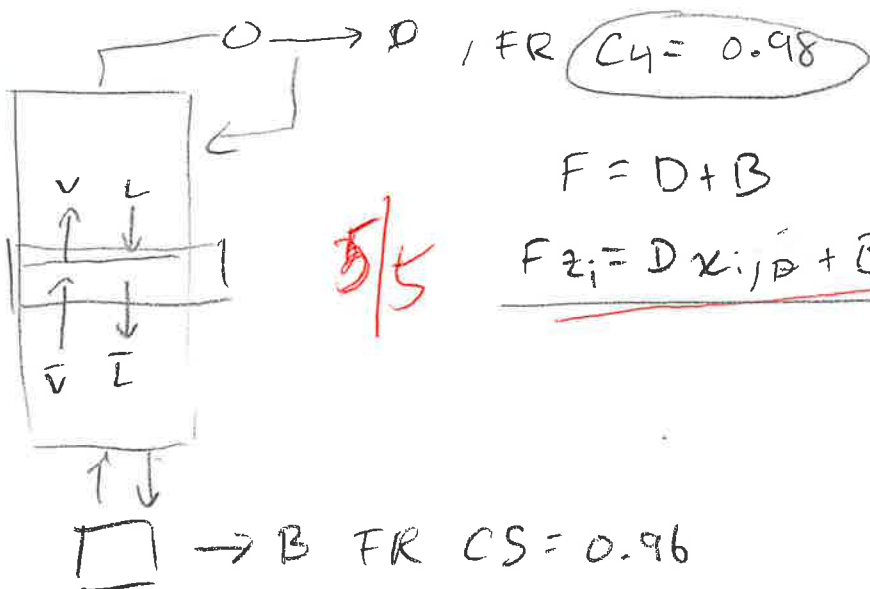
PJ Postlewait

①

C_2 LNK
 C_3 LNK
 C_4 LK
 C_5 HK
 C_6 HNK

 $F = 100 \frac{\text{kmol}}{\text{h}}$

SAT
LIQ



Feed mole fractions?
e.g. $z_2 = 0.05$
⋮

②

 C_2 LNK C_3 LNK C_4 LK C_5 HK C_6 HNK

$$\alpha_i = \frac{k_i}{k_{HK}} = k_{C_5} = 0.28$$

$$C_2 = 26.79 -$$

$$C_3 = 8.571 - \quad 5/5$$

$$C_4 = 3.636 -$$

$$C_5 = 1 -$$

$$C_6 = 0.357 -$$

$$③ \quad F = B + D$$

$$D x_i = 0$$

because

F is SAT LIQ

C_2	5
C_3	10
C_4	36
C_5	35
C_6	20

$\downarrow$
 0
 0
 6
 6
 6

$$D x_{i, \text{dist}} = D x_i \times \text{FR}_{i, \text{dist}}$$

$$C_2 \quad 5 \times 1 = 5 \text{ kmol/hr}$$

$$C_3 \quad 10 \times 1 = 10 \quad "$$

$$C_4 \quad 30 \times 0.98 = 29.4 \quad "$$

$$C_5 \quad 35 \times 0.04 = 1.4 \quad "$$

$$C_6 \quad 0$$

—
unit missing

$$D = 5 + 10 + 29.4 + 1.4 = 45.8 \frac{\text{kmol}}{\text{h}}$$

X_i :

$$D_{xi, dist} = F_{zi} \times FR_{, dist}$$

$$\frac{F_{zi} \times FR_{, dist}}{D} = X_{i, dist}$$

C2	$\frac{5 \times 1}{45.8} = 0.109$ ✓
C3	$\frac{10 \times 1}{45.8} = 0.2183$ ✓
C4	$\frac{30 \times 0.98}{45.8} = 0.642$ ✓
C5	$\frac{35 \times 0.04}{45.8} = 0.0305$ ✓
C6	0

14/15

$$\sum X_{i, dist} = 1$$

$$0.109 + 0.2183 + 0.642 + 0.0305 = 1$$

④ $F_{zi} = D_{xi, D} + B_{xi, B} \rightarrow B_{xi, B} = F_{zi} - D_{xi, D}$

C2	5 - 5 = 0
C3	10 - 10 = 0
C4	30 - 29.4 = 0.6 <i>km/hr</i>
C5	35 - 1.4 = 33.6
C6	20 - 0 = 20

$$B_{xi} = F_{zi} \rightarrow \frac{SAT}{L+Q}$$

C2	5
C3	10
C4	30
C5	35
C6	20

14/15

$$B = 0.6 + 33.6 + 20 = 54.2 \text{ *km/hr*} \rightarrow 100 - 45.8 = 54.2$$

$X_{i, D}$

C2	0
C3	0
C4	$(30 \times 0.02) / 54.2 = 0.01107$
C5	$(35 \times 0.98) / 54.2 = 0.6328$
C6	$(20) / 54.2 = 0.369$

$$F - D = B$$

$$\sum = 1$$

$$N_{min} = \ln \left\{ \frac{(0.99)(0.98)}{(0.04)(0.02)} \right\}$$

$$\frac{k_{LK}}{k_{HK}} = \frac{k_{LU}}{k_{LS}}$$

$$\ln(\alpha_{AB} = 3.036)$$

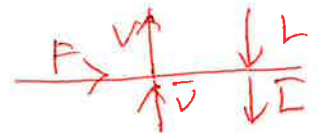
4/5

$$N_{min} = \cancel{7.61}^{6.37} \sim 8 \text{ stages}$$

$$N_{F,min} = \ln \left\{ \frac{0.642}{0.0305} \times \frac{0.35}{0.30} \right\} = 2.89 \sim 3 \text{ stages}$$

$$\textcircled{6} \Delta V_F = V_{min} - \bar{V}_{min} = \sum_{i=2}^6 \frac{\alpha_{i,ref} F z_i}{\alpha_{i,ref} - \phi}$$

M.B envelope -3



$$\begin{aligned} Q = & \frac{\alpha_{C2-CS} z_{C2}}{\alpha_{C2-CS} - \phi} + \frac{\alpha_{C3-CS} z_{C3}}{\alpha_{C3-CS} - \phi} + \frac{\alpha_{C4-CS} z_{C4}}{\alpha_{C4-CS} - \phi} \\ & + \frac{\alpha_{CS-CS} z_S}{\alpha_{CS-CS} - \phi} + \frac{\alpha_{C6-CS} z_6}{\alpha_{C6-CS} - \phi} \end{aligned}$$

$$\Delta V_{feed} = V_{in} - \bar{V}_m = 0 \text{ as } f = 0$$

5/15

$$Q = \frac{26.79 \cdot 0.05}{26.79 - \phi} + \frac{8.571 \cdot 0.1}{8.571 - \phi} + \frac{3.036 \cdot 0.3}{3.036 - \phi} + \frac{0.35}{1 - \phi} + \frac{0.357 \cdot 0.2}{0.357 - \phi}$$

5/10

$$0.28 \leq \phi \leq 0.85, \phi = 0.422$$

$$\textcircled{7} \Delta V_{min} = (26.79)(5) + \frac{(8.571)(10)}{8.571 - 0.422} + \frac{(3.036)(29.4)}{(3.036) - 0.422} + \frac{1.4}{1 - 0.422} = 0$$

set up ok!

$$\Delta V_{min} = 53.06 \text{ km/hr}$$

$$L_{min} = V_{min} - D$$

$$53.06 - 45.8 = 7.26 \text{ km/hr}$$

$$L_{min} = 7.26$$

$$(y_0)_{min} = 0.1586$$

unit m/sy : -1
wrong all values : -4

$$\textcircled{10} \text{FR}_{\text{NR,D}} = \frac{(3.036)^{7.64}}{0.98}$$

8/10

$$\frac{0.98}{0.02} + (3.036)^{7.64} = 0.99$$

This is economical because of the high FR of LNK
 + ~~low~~ $\frac{L}{D}$ ratio

in distillate

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

97/100
 Name: Mujtaba Mezher

1. (100 pts) A saturated liquid feed at 100 kmol/hr consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of propane is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

HK reference

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5	26.79	5
C3, propane	2.4	8.57	10
C4, butane	0.85	3.04	30
C5, pentane	0.28	1	35
C6, hexane	0.1	0.36	20
Total in feed	-	-	100

- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,bot} = (BX_{i,bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

1.) HK is reference

$$\alpha_{C_2-C_5} = \frac{7.5}{.28}$$

$$\alpha_{C_3-C_5} = \frac{2.4}{.28}$$

$$\alpha_{C_4-C_5} = \frac{.85}{.28}$$

$$\alpha_{C_5-C_5} = \frac{.28}{.28}$$

$$\alpha_{C_6-C_5} = \frac{.1}{.28}$$

2.)

C_2 - LNK

C_3 - LNK

C_4 - LK

C_5 - HK

C_6 - HNK

2) 5/5

3.)

Minitube Mezh

Distillate

$$Dx_{C_6, dist} = 0$$

$C_2 = C_3 = 1$ - NK no distribute

$$5 + 10 + (.98)(30) + (.04)(35)$$

$$D = 45.8 \text{ kmol/hr}$$

$$Dx_{C_2, dist} = \frac{5}{45.8} = .11$$

$$"C_3" = \frac{10}{45.8} = .22$$

$$"C_4" = \frac{29.4}{45.8} = .64$$

$$"C_5" = \frac{1.4}{45.8} = .03$$

$$\sum Dx_{i, dist} = 1$$



1) 5/5

4.) Bottom

$$Bx_{C_2} = C_3 = 0$$

$$Bx_{C_6} = 1$$

$$(.02)(30) + (.96)(35) + 20$$

$$B = 54.2 \text{ kmol/hr}$$

15/15

$$F = D + B$$

$$F_{zi} = Dx_{i, dist} + Bx_{i, bot}$$

also

$$F = B + D$$

$$F = 100 = 54.2 + 45.8$$

$Bx_{i, bot}$

$$C_4 = .01$$

$$C_5 = .62$$

$$C_6 = .37$$

$$\sum Bx_{i, bot} = 1$$

7.)

$$V_{min} = \sum_{i=1}^n \frac{\alpha_{i-ref} (D_{ref, dist})}{\alpha_{i-ref} - \psi}$$

$$V_{min} = \frac{(26.79)(5)}{(26.79) - 1.498} + \frac{(8.57)(10)}{(8.57) - 1.498} + \frac{(3.04)(29.4)}{(3.04) - 1.498}$$

$$+ \frac{1(14)}{(1) - 1.498} = \boxed{72.56 \text{ km/hr}} \quad \checkmark \quad 10/10$$

$$L_{min} = V_{min} - D = 72.56 - 45.8 = \boxed{26.76 \text{ km/hr}} \quad \checkmark$$

8.)

$$\frac{R}{R_{min}} = 1.2$$

$$R = 40$$

$$R_{min} = (40)_{min}$$

$$x = \frac{.701 - .584}{.701 + 1} \approx \overset{0.7}{\cancel{.07}}$$

$$y \approx .54$$

8) 5/5

$$R_{min} = \frac{26.76}{45.8} = .584$$

$$R = (1.2)(.584) = .701$$

$$.54 = \frac{N - 6.36}{N + 1}$$

9) 5/5

$$.54N + .54 = N - 6.36$$

$$6.9 = .46N$$

$$\boxed{N = 15} \quad \checkmark$$

$$9.) \frac{3.02}{6.36} = \frac{NF}{15}$$

$$\boxed{NF = 7.12}$$

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

96/100
 Name: Queen Adedire

1. (100 pts) A saturated liquid feed at 100 kmol/min consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of ~~propane~~ ^{butane} is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5	26.79	5
C3, propane	2.4	8.57	10
C4, butane	0.85	3.04	30
C5, pentane	0.28	1	35
C6, hexane	0.1	0.35	20
Total in feed	-	-	100

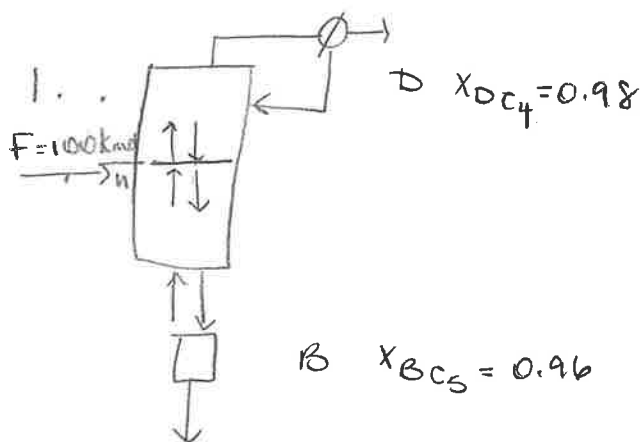
- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (Dx_{i,dist})/Fz_i$ & $FR_{i,Bott} = (Bx_{i,Bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

Overall Mass Balance

$$F = D + B$$

Component Mass Balance

$$C_2 \quad Fz_{C2} = Dx_{DC2} + Bx_{BC2}$$

$$C_3 \quad Fz_{C3} = Dx_{DC3} + Bx_{BC3}$$

$$C_4 \quad Fz_{C4} = Dx_{DC4} + Bx_{BC4}$$

$$C_6 \quad Fz_{C6} = Dx_{DC6} + Bx_{BC6}$$

2) C_2 : ethane: LNK

$$\alpha_{C2-C5} = \frac{K_{C2}}{K_{C5}} = 26.79$$

 C_3 : Propane: LNK

$$\alpha_{C3-C5} = \frac{K_{C3}}{K_{C5}} = 8.51$$

 C_4 : butane: LK

$$\alpha_{C4-C5} = \frac{K_{C4}}{K_{C5}} = 3.04$$

 C_5 : Pentane: HK C_6 : Hexane: HNK

$$\alpha_{C5-C5} = \frac{K_{C5}}{K_{C5}} = 1$$

$$\alpha_{C6-C5} = \frac{K_{C6}}{K_{C5}} = 0.35$$

$$3) \quad Fz_{i,dist} = \frac{Dx_{i,dist}}{Fz_i}$$

$$Dx_{C5} = (1 - 0.96)(35) = 1.4 \frac{\text{kmol}}{h}$$

$$Dx_{C2} = (1)(5) = 5 \frac{\text{kmol}}{h}$$

$$Dx_{C6} = (0)(20) = 0 \frac{\text{kmol}}{h}$$

$$Dx_{C3} = \frac{(1)(10)}{10} = 9.5 \frac{\text{kmol}}{h} ?$$

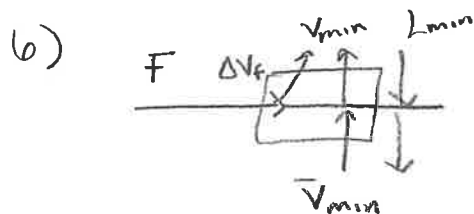
$$D = \sum Dx_i = 5 + 9.5 + 29.4 + 1.4 =$$

$$Dx_{C4} = (0.98)(30) = 29.4 \frac{\text{kmol}}{h}$$

$$\boxed{45.8 \frac{\text{kmol}}{h}}$$

$$5) N_{min} = \frac{\ln \left(\frac{0.98}{1-0.98} \times \frac{0.96}{1-0.96} \right)}{\ln(3.04)} = \boxed{6.35} \quad \checkmark$$

$$N_{F,min} = \frac{\ln \left\{ \frac{0.65}{0.031} \times \frac{0.35}{0.30} \right\}}{\ln(3.04)} = \boxed{2.88} \quad \checkmark$$



$$\Delta V_{feed} = V_{min} - \bar{V}_{min}$$

$$\frac{V_F}{F} = f = 1 - q \rightarrow V_F = F(1 - q)$$

$$q = \frac{L}{F} = 1 \leftarrow \text{Saturated Liquid}$$

$$\Delta V_{feed} = 0 = \sum_{i=2}^C \frac{\alpha_{i-ref} F z_i}{\alpha_{i-ref} - \phi}$$

15/15

$$0 = \frac{(26.79)(5)}{26.79 - \phi} + \frac{(8.57)(10)}{8.57 - \phi} + \frac{(3.04)(30)}{3.04 - \phi} + \frac{(1)(35)}{1 - \phi} + \frac{(0.35)(20)}{0.35 - \phi}$$

$\alpha_{HK} < \phi < \alpha_{LK}$ Only value in between

3.04 and 1 is (1.498)

$$7) V_{min} = \frac{(26.79)(5)}{(26.79) - 1.498} + \frac{(8.57)(10)}{8.57 - 1.498} + \frac{(3.04)(29.4)}{3.04 - 1.498} + \frac{(1)(1.4)}{1 - 1.498} + \frac{(0.35)(10)}{0.35 - 1.498} = 71.95 \frac{\text{kmol}}{n}$$

72.64

10/10

$$V_{min} = D + L_{min}$$

$$L_{min} = V_{min} - D$$

$$L_{min} = (71.95 - 45) \frac{\text{kmol}}{n} = 26.95 \frac{\text{kmol}}{n}$$

$$\left(\frac{L}{D} \right)_{min} = 0.59 = 0.60 \quad \checkmark$$

o/c!

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

99/100
 Name: Aliya Kareemullah

1. (100 pts) A saturated liquid feed at 100 kmol/hr consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of ~~propane~~ ^{butane} is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5	26.79	5
C3, propane	2.4	8.57	10
C4, butane	0.85	3.04	30
C5, pentane	0.28	1	35
C6, hexane	0.1	0.36	20
Total in feed	-	-	100

- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table. $\alpha_i - HK$
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

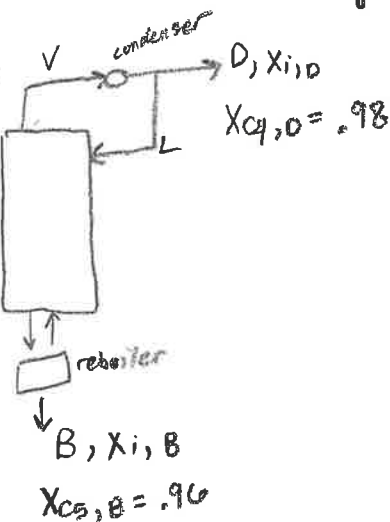
$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,Bott} = (BX_{i,Bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a mass balance envelope around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

Problem 1

1.)

 $F = 100 \frac{\text{kmol}}{\text{min}}$
 Z_i


Mass balance: $F = D + B$

Component: $F Z_i = D X_{i,D} + B X_{i,B}$

~~10/10~~

5/5

2.)	C2	C3	C4	C5	C6
	LNK	LNK	LK	HK	HNK

5/5

3.) $Z_i = \frac{F_i}{F}$

$Z_{C2} = .05 \quad Z_{C3} = .1 \quad Z_{C4} = .3 \quad Z_{C5} = .35 \quad Z_{C6} = .2$

Distillate

$D X_{C2,D} = .05 \times 100 = 5 \frac{\text{kmol}}{\text{hr}}$

$D X_{C3,D} = .1 \times 100 = 10 \frac{\text{kmol}}{\text{h}}$

$D X_{C4,D} = 30 \times .98 = 29.4 \frac{\text{kmol}}{\text{h}}$

$D X_{C5,D} = 35 \times (1 - .96) = 1.4 \frac{\text{kmol}}{\text{h}}$

$D X_{C6,D} = 0 \text{ (HNK)}$

$D = \sum D X_{i,D} = 45.8 \frac{\text{kmol}}{\text{hr}}$

$X_{i,D} = \frac{D X_{i,D}}{D}$

$X_{C2,D} = \frac{5}{45.8} = .109 \quad X_{C3,D} = .218 \quad X_{C4,D} = .642 \quad X_{C5,D} = .031 \quad X_{C6,D} = 0$

Validate: $.109 + .218 + .642 + .031 = 1$ ✓ correct ✓

15/15

4.) Bottom

$B X_{C2,B} = B X_{C3,B} = 0 \text{ (LNK)}$

$B X_{C4,B} = 30 \times (1 - .98) = .6 \frac{\text{kmol}}{\text{h}}$

$B X_{C5,B} = 35 \times .96 = 33.6 \frac{\text{kmol}}{\text{h}}$

$B X_{C6,B} = 20 \frac{\text{kmol}}{\text{hr}} \text{ (HNK)}$

$B = \sum B X_{i,B} = 54.2 \frac{\text{kmol}}{\text{h}}$

$X_{i,B} = \frac{B X_{i,B}}{B}$

$X_{C2,B} = X_{C3,B} = 0 \quad X_{C4,B} = .011 \quad X_{C5,B} = .620 \quad X_{C6,B} = .369$

Validate: $35 = 1.4 + 33.6 = 35$ (from component balance for C5) ✓

validation for mole fraction! -1

14/15

$$10.) \frac{N_F - 1}{14.998 - N_F} = \left[\frac{54.2}{45.8} \times \frac{35}{30} \times \left[\frac{.02}{.04} \right]^2 \right]^{.206} = .803$$

$$N_F - 1 = -.803 N_F + 12.04$$

$$1.803 N_F = 13.04$$

$$N_F = 7.236 \text{ stages}$$

It is higher than the other N_F by about .5, so 7th stage could be seen as the optimal N_F based on these results.

In fact two results are similar each other, ~ 7 stages

5/5

$$11.) FR_{C3,0} = \frac{3.04^{6.359}}{\frac{.96}{.04} + 3.04^{6.359}} = .98$$

10/10

Based on our assumption, $FR_{C3,0}$ should be 1. Based on this information, it might not be economical because of this 2% difference.

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

Name: Caleb Mura

98 +1
(from class) credit
99/100

1. (100 pts) A saturated liquid feed at 100 kmol/hr consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of ~~propane~~ ^{butane} is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5	26.8	5
C3, propane	2.4	8.57	10
C4, butane	0.85	3.04	30
C5, pentane	0.28	1	35
C6, hexane	0.1	0.36	20
Total in feed	-	-	100

- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,Bott} = (BX_{i,Bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

1) 

$q=1$

5/5

$FR_{c5,B} = 0.96$

$K_{c2} = 7.5$

$K_{c3} = 2.4$

$K_{c4} = 0.85$

$K_{c5} = 0.28$

$K_{c6} = 0.1$

$F_{c2} = 5 \frac{\text{kmol}}{\text{h}}$

$F_{c3} = 10 \frac{\text{kmol}}{\text{h}}$

$F_{c4} = 30 \frac{\text{kmol}}{\text{h}}$

$F_{c5} = 35 \frac{\text{kmol}}{\text{h}}$

$F_{c6} = 20 \frac{\text{kmol}}{\text{h}}$

2) $C_2 \rightarrow \text{LNK}$

$C_3 \rightarrow \text{LNK}$

$C_4 \rightarrow \text{LK}$

$C_5 \rightarrow \text{HK}$

$C_6 \rightarrow \text{HNK}$

$\alpha_{25} = \frac{K_2}{K_5} = 26.8$

$\alpha_{35} = \frac{K_3}{K_5} = 8.57$

$\alpha_{45} = \frac{K_4}{K_5} = 3.04$

$\alpha_{55} = 1$

$\alpha_{65} = \frac{K_6}{K_5} = 0.36$

5/5

3) $Dx_2 = FR_{2,D} \cdot F \cdot z_2 = 1 \cdot 5 = 5 \frac{\text{kmol}}{\text{h}}$

$Dx_3 = FR_{3,D} \cdot F \cdot z_3 = 1 \cdot 10 = 10 \frac{\text{kmol}}{\text{h}}$

$Dx_4 = FR_{4,D} \cdot F \cdot z_4 = 0.98 \cdot 30 = 29.4 \frac{\text{kmol}}{\text{h}}$

$Dx_5 = (1 - FR_{5,B}) \cdot F \cdot z_5 = (1 - 0.96) \cdot 35 = 1.4 \frac{\text{kmol}}{\text{h}}$

$Dx_6 = FR_{6,D} \cdot F \cdot z_6 = 0 \cdot 20 = 0 \frac{\text{kmol}}{\text{h}}$

$D = \sum Dx_i = 45.4 \frac{\text{kmol}}{\text{h}}$

45.8

14/15

5 contd....

$$N_{E,min} = \frac{\ln \left(\frac{x_{LK,D}}{x_{HK,D}} \cdot \frac{z_{HK}}{z_{LK}} \right)}{\ln(\alpha_{LK} + 1)}$$

$$z_{HK} = \frac{z_{HK} F}{F} = \frac{35}{100} = 0.35$$

$$z_{LK} = \frac{z_{LK} F}{F} = \frac{30}{100} = 0.30$$

$$\hookrightarrow N_{E,min} = \frac{\ln \left(\frac{0.65}{0.023} \cdot \frac{0.35}{0.30} \right)}{\ln(3.04)} = \frac{2.88}{\ln(3.04)} = \boxed{3.18 \approx 3}$$

4/5

6)



$$\rightarrow V = V' + V_F \rightarrow V - V' = V_F \rightarrow \Delta V_{feed} = V_F$$

$$\& f = \frac{V_F}{F} \& f = 1 - q \rightarrow V_F = F(1 - q) \rightarrow \boxed{\Delta V_{feed} = F(1 - q)}$$

$$\hookrightarrow \Delta V_{feed} = F(1 - q) = \sum \frac{\alpha_{i-HK} F z_i}{\alpha_{i-HK} - q} \quad \& q = 1$$

$$\hookrightarrow 0 = \frac{\alpha_{25} F z_2}{\alpha_{25} - q} + \frac{\alpha_{35} F z_3}{\alpha_{35} - q} + \frac{\alpha_{45} F z_4}{\alpha_{45} - q} + \frac{\alpha_{55} F z_5}{\alpha_{55} - q} + \frac{\alpha_{65} F z_6}{\alpha_{65} - q}$$

$$\hookrightarrow 0 = \frac{26.8 \cdot 5}{26.8 - q} + \frac{8.57 \cdot 10}{8.57 - q} + \frac{3.04 \cdot 30}{3.04 - q} + \frac{1 \cdot 35}{1 - q} + \frac{0.36 \cdot 20}{0.36 - q}$$

$$q = 0.422, 1.498, 5.969 \rightarrow \text{from exam}$$

$$\hookrightarrow \text{we know } \alpha_{HK,HK} < q < \alpha_{LK,HK} \rightarrow 1 < q < 3.04$$

$$\hookrightarrow \text{therefore } \boxed{q = 1.498}$$

15/15

11)

$$FR_{2,D} = \frac{(\alpha_{NKHK})^{N_{min}}}{\frac{FR_{HK,B}}{1 - FR_{HK,B}} + (\alpha_{NKHK})^{N_{min}}} = \frac{(26.8)^{6.36}}{\frac{0.96}{1 - 0.16} + (26.8)^{6.36}} = 0.99999 \approx 1$$

$$FR_{6,D} = \frac{0.56^{6.36}}{\frac{0.96}{1 - 0.96} + 0.56^{6.36}} = 6.3 \times 10^{-5} \approx 0$$

10/10

yes it is viable as NKs do not distribute!

97/100

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

Name: Nanya Majumdar

1. (100 pts) A saturated liquid feed at 100 kmol/min consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of propane is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

	Component	K (y/x) value	α (relative volatility) value	kmol/hr
LNK	C2, ethane	7.5	26.79	5
LNK	C3, propane	2.4	8.571	10
LK	C4, butane	0.85	3.036	30
HK →	C5, pentane	0.28	1.00	35
HNK	C6, hexane	0.1	0.357	20
	Total in feed	-	-	100

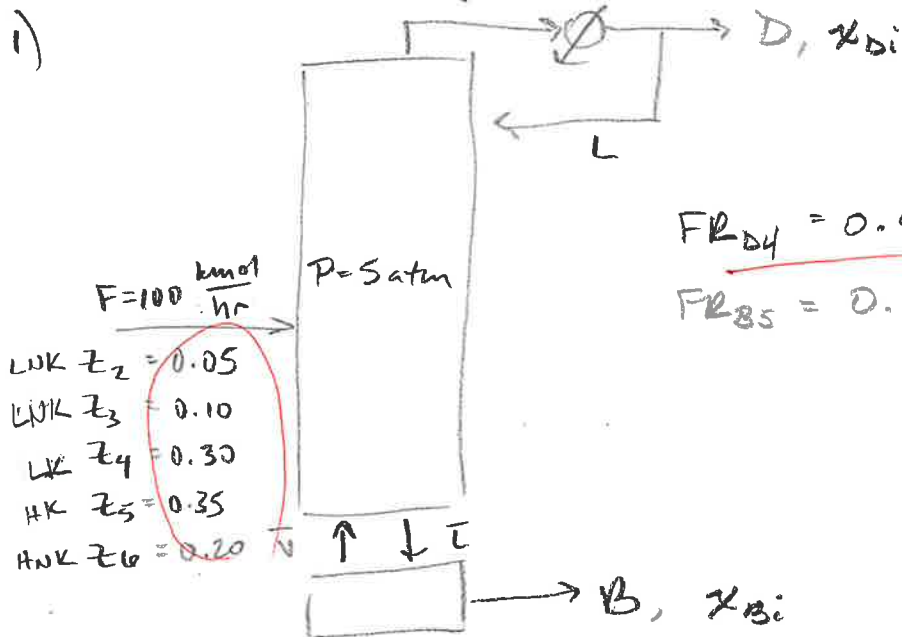
- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,Bot} = (BX_{i,Bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a mass balance envelope around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.



5/5

$$F = B + D, \quad Fz_i = Bx_{Bi} + Dx_{Di}$$

2)

LNK	: C2
LNK	: C3
LK	: C4
HK	: C5
HNK	: C6

$$\alpha_{AB} = \frac{K_A}{K_B}$$

5/5

$$\alpha_{i, HK} = \frac{k_i}{k_{5r}} \quad (HK)$$

3)

$$FR_D = \frac{Dx_{Di}}{Fz_i}$$

15/15

$$C6: 0 \text{ kmol/h} = Dx_{D6}$$

$$C2: Dx_{D2} = (1)(100)(0.05) = 5 \text{ kmol/hr} = Dx_{D2}$$

$$C3: Dx_{D3} = (1)(100)(0.10) = 10 \text{ kmol/hr} = Dx_{D3}$$

$$C4: Dx_{D4} = (0.98)(100)(0.3) = 29.4 \text{ kmol/h} = Dx_{D4}$$

$$C5: Dx_{D5} = (1 - 0.96)(100)(0.35) = 1.4 \text{ kmol/h} = Dx_{D5}$$

$$D = 45.8 \text{ kmol/h}$$

$$x_{D2} = 0.109 \quad x_{D5} = 0.031$$

$$x_{D3} = 0.218 \quad x_{D6} = 0.00$$

$$x_{D4} = 0.642$$

$$x_i = \frac{Dx_{Di}}{D}$$

all sum to 1

6) (cont.) $0.25 N_{Fmin} < \phi < 1.25 N_{Fmin}$

✓ $0.7175 < \phi < 3.3575$

∴ $\boxed{\phi = 1.498}$ → plug into ΔV_{feed}

7)

$$V_{min} = \sum_{i=1}^c \frac{\alpha_{direct} \cdot D \cdot x_{Di}}{\alpha_{direct} - \phi}$$

$$D + L_{min} = V_{min}$$

$$= \frac{26.76(5)}{26.76 - 1.498} + \frac{8.571(10)}{8.571 - 1.498} + \frac{3.036(29.4)}{3.036 - 1.498} + \frac{1(1.4)}{1 - 1.498} + 0$$

$\boxed{= 72.64 \text{ kmol/hr} = V_{min}}$

10/10

$$D + L_{min} = V_{min} = 72.64$$

↳ $L_{min} = 72.64 - 45.8 = 26.84$

$\boxed{\left(\frac{L}{D}\right)_{min} = 0.586}$

8) $\frac{(L/D)}{(L/D)_{min}} = 1.2$ $N_{min} = 6.37$ $N_{Fmin} = 2.87$

$$(L/D) = 1.2(0.586) = 0.703$$

$$\frac{0.703 - 0.586}{0.703 + 1} = 0.069 \sim 0.07$$

5/5

from graph: $x = 0.07$, $y = 0.55$

$$0.55 = \frac{N - 6.37}{N + 1}$$

$$0.55N + 0.55 = N - 6.37$$

✓ $N = 15.38$

or 16 stages

100/100

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

Name: IZA CERNY

1. (100 pts) A saturated liquid feed at 100 kmol/min consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of ~~propane~~ ^{C3} is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr	Σ
C2, ethane	7.5	26.7857	5	0.05
→ C3, propane	2.4	8.5714	10	0.10
C4, butane	0.85	3.0357	30	0.30
C5, pentane	0.28	1.0000	35	0.35
C6, hexane	0.1	0.3571	20	0.20
Total in feed	-	-	100	

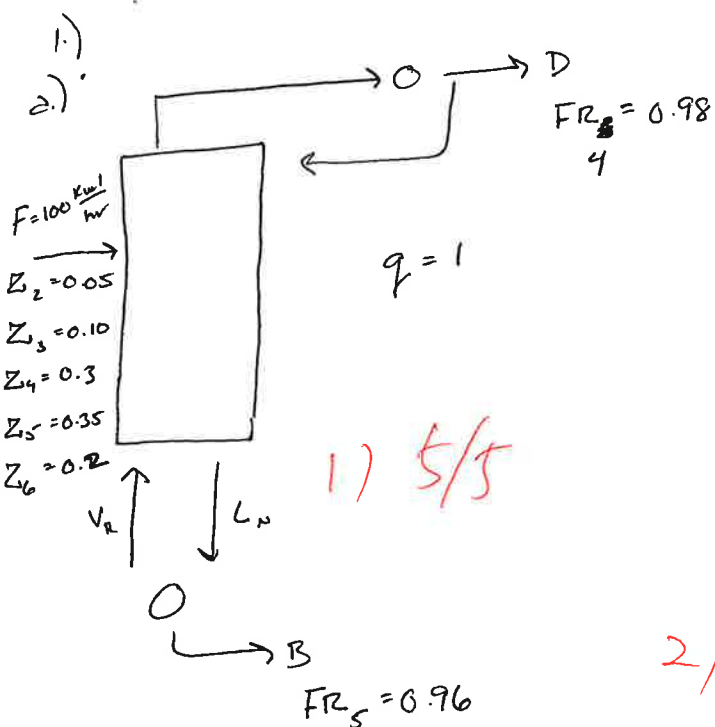
- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,Bott} = (BX_{i,Bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.



b.) Comp

	K	$\alpha_{i,HK}$
LWK C ₂	7.5	26.7857
LHK C ₃	2.4	8.5714
LWK C ₄	0.85	5.0357
LHK C ₅	0.28	1.00
LWK C ₆	0.1	0.3571

c.) $Dx_2 = 0.05(100)$

$Dx_2 = 5 \frac{\text{kmol}}{\text{hr}}$ $x_{2D} = 0.1092$

$Dx_3 = 0.1(100)$
 $= 10 \frac{\text{kmol}}{\text{hr}}$ $x_{3D} = 0.2183$

$Dx_4 = 0.3(100)(0.98)$
 $= 29.4 \frac{\text{kmol}}{\text{hr}}$ $x_{4D} = 0.6419$

$Dx_5 = 0.35(100)(1-0.96)$
 $= 1.4 \frac{\text{kmol}}{\text{hr}}$ $x_{5D} = 0.0306$

$D = 45.8 \frac{\text{kmol}}{\text{hr}}$ $\sum x_{iD} = 1 \checkmark$

d) $Bx_4 = 100(0.3)(1-0.98)$
 $= 0.6 \frac{\text{kmol}}{\text{hr}}$ $x_{4B} = 0.0111$

$Bx_5 = 100(0.35)(0.96)$
 $= 33.6 \frac{\text{kmol}}{\text{hr}}$ $x_{5B} = 0.6199$

$Bx_6 = 100(0.2)$
 $= 20 \frac{\text{kmol}}{\text{hr}}$ $x_{6B} = 0.3690$

$B = 54.2 \frac{\text{kmol}}{\text{hr}}$ $\sum x_{iB} = 1 \checkmark$

d.) $B + D = F$
 $54.2 + 45.8 = 100$
 $100 = 100 \checkmark$

$C_2 \rightarrow 100(0.05) = 45.8(0.1092) + 54.2(0)$
 $\frac{\text{kmol}}{\text{hr}} 5 = 5 \frac{\text{kmol}}{\text{hr}} \checkmark$

$C_3 \rightarrow 100(0.1) = 45.8(0.2183) + 54.2(0)$
 $\frac{\text{kmol}}{\text{hr}} 10 = 10 \frac{\text{kmol}}{\text{hr}} \checkmark$

$C_4 \rightarrow 100(0.3) = 45.8(0.6419) + 54.2(0.0111)$
 $30 \frac{\text{kmol}}{\text{hr}} = 30 \frac{\text{kmol}}{\text{hr}} \checkmark$

$C_5 \rightarrow 100(0.35) = 45.8(0.0306) + 54.2(0.6199)$
 $35 \frac{\text{kmol}}{\text{hr}} = 35 \frac{\text{kmol}}{\text{hr}} \checkmark$

$C_6 \rightarrow 100(0.2) = 45.8(0) + 54.2(0.3690)$
 $20 \frac{\text{kmol}}{\text{hr}} = 20 \frac{\text{kmol}}{\text{hr}} \checkmark$

e.) $N_{min} = \frac{\ln\left[\frac{0.98}{0.02} / \frac{0.04}{0.96}\right]}{\ln(5.0357)}$

$N_{Fmin} = \frac{\ln\left[\frac{0.6419}{0.0306} / \frac{0.3}{0.35}\right]}{\ln(3.0357)}$

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5/5

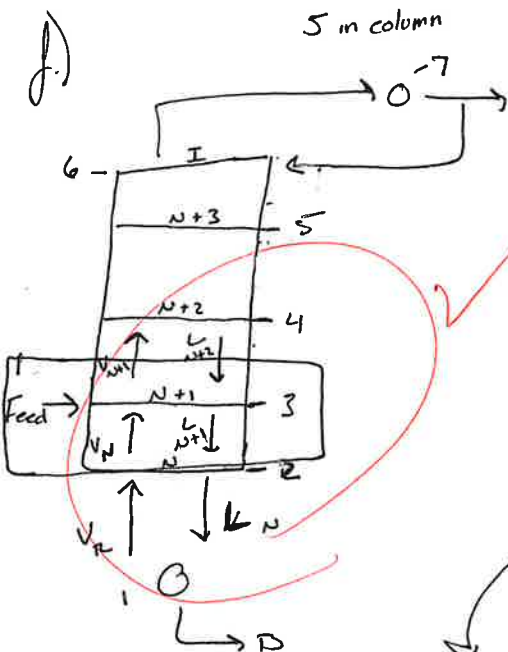
$N_{min} = 6.3667$

↳ 7

5 in column

$N_{Fmin} = 2.8796$

↳ 3



$F + L_{N+2} + V_N = V_{N+1} + L_{N+1}$

$\Delta V_{feed} = 0$ b/c sat. liquid.

$\Delta V_{feed} = 0 = \sum \frac{\alpha_{iHK} F z_i}{\alpha_{iHK} - \phi}$

$0 = \frac{26.7857(100)(0.05)}{26.7857 - \phi} + \frac{8.5714(100)(0.10)}{8.5714 - \phi} + \frac{3.0357(100)(0.3)}{3.0357 - \phi} + \frac{1(100)(0.35)}{1 - \phi} + \frac{0.3571(100)(0.2)}{0.3571 - \phi}$

$0 = \frac{133.9285}{26.7857 - \phi} + \frac{85.714}{8.5714 - \phi} + \frac{91.071}{3.0357 - \phi} + \frac{35}{1 - \phi} + \frac{7.142}{0.3571 - \phi}$

→ where $\alpha_{LHK} > \phi > \alpha_{HKL}$

$3.0357 > \phi > 1.00$

→ out of options $\phi = 1.498$

g.) $V_{min} = \frac{26.7857(5)}{26.7857 - 1.498} + \frac{8.5714(10)}{8.5714 - 1.498} + \frac{3.0357(29.4)}{3.0357 - 1.498} + \frac{1(1.4)}{1 - 1.498}$

7) 10/10

$V_{min} = 72.6437 \frac{\text{kmol}}{\text{hr}}$

$V_{min} = L_{min} + D$

$72.6437 = L_{min} + 45.8$

$L_{min} = 26.8437 \frac{\text{kmol}}{\text{hr}}$

$\left(\frac{L}{D}\right)_{min} = \frac{26.8437}{45.8} = 0.5861$

$$h.) \cdot \dot{z} = \dot{L}/D$$

$$\frac{\dot{L}/D - \dot{L}/D_{min}}{\dot{L}/D + 1} = 0.0688$$

$$\frac{\dot{L}/D}{\dot{L}/D_{min}} = 1.2$$

$$\frac{\dot{L}/D}{0.5861} = 1.2$$

$$\dot{L}/D = 0.7033$$

on graph $y = 0.6$

$$0.6 = \frac{N - N_{min}}{N + 1}$$

$$0.6 = \frac{N - 6.3667}{N + 1}$$

$$0.6N + 0.6 = N - 6.3667$$

$$-0.6N + 6.3667 - 0.6N + 6.3667$$

$$\frac{0.4N = 6.9667}{0.4}$$

$$N = \frac{17.4168}{15.91} \text{ But OK}$$

5/5

Method
is ok!

$$i.) \frac{2.8796}{6.3667} = \frac{N_F}{17.4168}$$

$$N_F = \frac{7.2}{7.8775} \checkmark$$

5/5

pretty
close.

$$j.) \frac{N_F - 1}{17.4168 - N_F} = \left[\frac{54.2}{45.8} \left(\frac{0.35}{0.30} \right) \left(\frac{0.0111}{0.0306} \right)^2 \right]^{0.206}$$

$$\frac{N_F - 1}{17.4168 - N_F} = 0.7037$$

$$\frac{N_F - 1}{+ N_F 0.7037} = \frac{12.2569}{+ 1} - \frac{N_F (0.7037)}{+ N_F 0.7037}$$

$$\frac{1.7037 N_F = 13.2569}{1.7037}$$

$$N_F = \frac{7.7812}{7.2}$$

ok!

5/5

L.) :

$$FR_{LK} = \frac{3.0357^{(6.3667)}}{\left(\frac{0.96}{0.04}\right) + 3.0357^{6.3667}}$$

$$FR_{LK} = 0.97999 \dots$$

$$= 0.98$$

→ yes. 0.98 is exactly how much FR_{LK} we want in

dist. which means we can use a relatively small amount of stages to get desired product. ✓

10/10

85/100

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

Name: Matthew Loftstrom 657010748

$q=1$

1. (100 pts) A saturated liquid feed at 100 kmol/min consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of propane is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKS do not distribute.

	Component	K (y/x) value	α (relative volatility) value	kmol/hr
LNK	C2, ethane	7.5		5
LNK	C3, propane	2.4		10
LK	C4, butane	0.85		30
HK	C5, pentane	0.28		35
HNK	C6, hexane	0.1		20
	Total in feed	-	-	100

- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

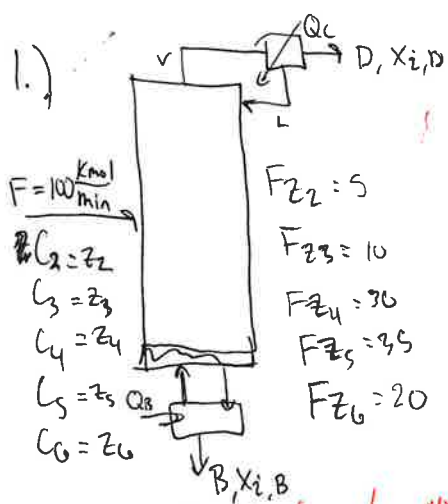
$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,Bott} = (BX_{i,Bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

$$\alpha_{HK} < \phi < \alpha_{LK}$$



$$F = B + D$$

$$Fz_i = Bx_{i,B} + Dx_{i,D}$$

$$Fz_2 = Bx_{2,B} + Dx_{2,D}$$

$$Fz_3 = Bx_{3,B} + Dx_{3,D}$$

$$Fz_4 = Bx_{4,B} + Dx_{4,D}$$

$$Fz_5 = Bx_{5,B} + Dx_{5,D}$$

$$Fz_6 = Bx_{6,B} + Dx_{6,D}$$

$$FR_{2,D} = 0.98$$

$$FR_{5,B} = 0.96$$

$$FR_{6,B} = 1$$

$$\alpha_{ij} = \frac{K_i}{K_j}$$

Distillate:

$$FR_{2,D} = 1$$

$$FR_{3,D} = 1$$

$$FR_{4,D} = 0.98$$

$$FR_{5,D} = 0.04$$

$$FR_{6,D} = 0$$

Bottoms:

$$FR_{2,B} = 0$$

$$FR_{3,B} = 0$$

$$FR_{4,B} = 0.02$$

$$FR_{5,B} = 0.96$$

$$FR_{6,B} = 1$$

2.) C_2 : LNK

C_3 : LNK $\alpha_{25} = 26.79$

C_4 : LK $\alpha_{35} = 8.571$

C_5 : HK $\alpha_{45} = 3.04$

C_6 : HNK $\alpha_{55} = 1$

$\alpha_{65} = 0.3571$

2) 5/5

3.) $FR_{i,D} = \frac{Dx_{i,D}}{Fz_i}$

$$Dx_{i,D} = FR_{i,D} \cdot Fz_i$$

$$x_{i,D} = \frac{Dx_{i,D}}{D}$$

$x_{2,D} = 0.1092$ $x_{5,D} = 0.0366$

$x_{3,D} = 0.2183$ $x_{6,D} = 0$

$x_{4,D} = 0.6419$

$Dx_{2,D} = Fz_2 = 5 \frac{\text{kmol}}{\text{min}}$

$Dx_{3,D} = Fz_3 = 10 \frac{\text{kmol}}{\text{min}}$

$Dx_{4,D} = 0.98 Fz_4 = 29.4 \frac{\text{kmol}}{\text{min}}$

$Dx_{5,D} = 0.04 Fz_5 = 1.4 \frac{\text{kmol}}{\text{min}}$

$Dx_{6,D} = 0 \frac{\text{kmol}}{\text{min}}$

$D = \sum_i Dx_{i,D}$

$D = 45.8 \frac{\text{kmol}}{\text{min}}$

14/15

Validation! $\sum x_i = 1.0$

4.) $FR_{i,B} = \frac{Bx_{i,B}}{Fz_i}$

$$x_{i,B} = \frac{Bx_{i,B}}{B}$$

$x_{2,B} = 0$

$x_{3,B} = 0$

$x_{4,B} = 0.0111$

$x_{5,B} = 0.6199$

$x_{6,B} = 0.369$

$Bx_{i,B} = FR_{i,B} \cdot Fz_i$

$Bx_{2,B} = 0$

$Bx_{3,B} = 0$

$Bx_{4,B} = 0.02 Fz_4 = 0.6 \frac{\text{kmol}}{\text{min}}$

$Bx_{5,B} = 0.96 Fz_5 = 33.6 \frac{\text{kmol}}{\text{min}}$

$Bx_{6,B} = Fz_6 = 20 \frac{\text{kmol}}{\text{min}}$

$B = \sum_i Bx_{i,B}$

$B = 54.2 \frac{\text{kmol}}{\text{min}}$

14/15

Validation! $\sum x_i = 1.0$

$$8.) \frac{L}{D} = R$$

Matthew Lofstrom
697010848

$$R = 1.2 R_{min}$$

$$X_{F,fit} = \frac{R - R_{min}}{R + 1} = \frac{0.2 R_{min}}{1.2 R_{min} + 1} \quad \left| \begin{array}{l} R_{min} = 0.586 \\ = 0.069 \end{array} \right.$$

$$Y(X_{F,fit}) \approx 0.60$$

$$Y = \frac{N - N_{min}}{N + 1} \quad \left| \begin{array}{l} Y = 0.60 \\ N_{min} = 6.37 \end{array} \right.$$

$$Y(N+1) - N + N_{min} = 0$$

$$N(Y-1) + Y + N_{min} = 0$$

$$N = 17.42 \text{ stages}$$

$$9.) N = 17.42 \quad N_{F,min} = 2.78$$

$$N_{min} = 6.37$$

$$N_F = N \frac{N_{F,min}}{N_{min}} = 17.42 \cdot \frac{2.78}{6.37} = 7.88 \text{ stages} \approx 8 \text{ stages}$$

$$10.) \frac{N_F - 1}{N - N_F} = \left[\frac{B}{D} \left(\frac{z_5}{z_4} \left[\frac{x_{4,B}}{x_{5,D}} \right]^2 \right)^{0.206} \right] \quad \left| \begin{array}{l} N_F = 7.88 \\ N = 17.42 \\ B = 54.2 \\ D = 45.8 \\ x_{4,B} = 0.0111 \\ x_{5,D} = 0.0806 \\ z_5 = 0.35 \\ z_4 = 0.36 \end{array} \right. =$$

$$\frac{N_F - 1}{17.42 - N_F} = 0.80$$

$$N_F - 1 = 0.80(17.42) - 0.80 N_F$$

$$1.8 N_F = 0.80(17.42) + 1$$

$$N_F = \frac{0.80(17.42) + 1}{1.8}$$

$$N_F = 8.30 \text{ stages}$$

Kirkbride's eqn $\Rightarrow N_{F,fit}$
better fit

In fact, they are close!

-1

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

95/100

Name: Hyungyoung Jun

1. (100 pts) A saturated liquid feed at 100 kmol/min consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of propane is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms. butane

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5	26.79	5
C3, propane	2.4	8.57	10
C4, butane	0.85 LK	3.04	30
C5, pentane	0.28 HK	1	35
C6, hexane	0.1	0.36	20
Total in feed	-	-	100

- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

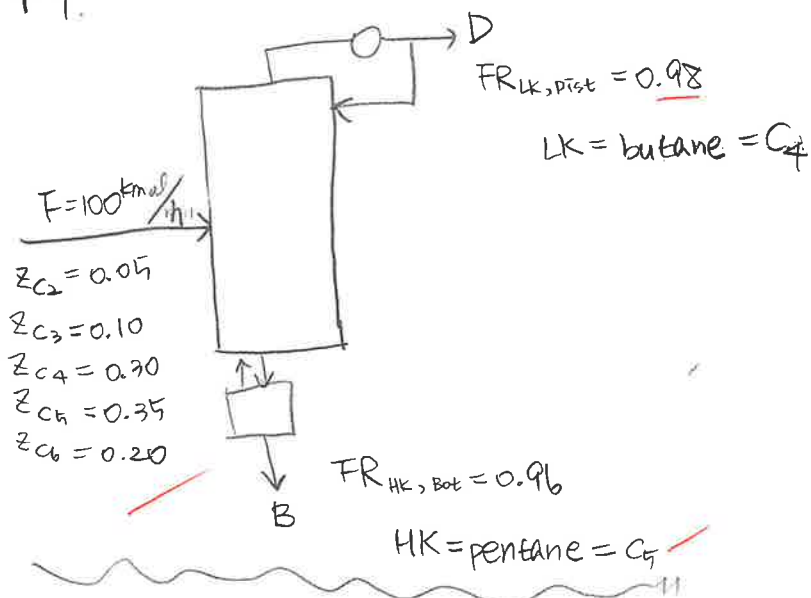
$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,bot} = (BX_{i,bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a mass balance envelope around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

1-1.



$$F = B + D \rightarrow F_i = B_i + D_i$$

$$Fz_i = Bx_{i,B} + Dx_{i,D}$$

1-2.

$$C_2 = LNK$$

$$C_3 = LNK$$

$$C_4 = LK$$

$$C_5 = HK$$

$$C_6 = HNK$$

$$\alpha_{C_2 C_5} = \frac{7.5}{0.28} = 26.785 \dots \approx 26.79$$

$$\alpha_{C_3 C_5} = \frac{2.4}{0.28} = 8.571 \dots \approx 8.57$$

$$\alpha_{C_4 C_5} = \frac{0.85}{0.28} = 3.035 \dots \approx 3.04$$

$$\alpha_{C_5 C_5} = \frac{0.28}{0.28} = 1$$

$$\alpha_{C_6 C_5} = \frac{0.1}{0.28} = 0.357 \dots \approx 0.36$$

1-3.

$$C_2: Dx_{C_2,Dist} = Fz_{C_2} = (100)(0.05) = 5 \text{ kmol/h}$$

$$C_3: Dx_{C_3,Dist} = Fz_{C_3} = (100)(0.10) = 10 \text{ kmol/h}$$

$$C_4: Dx_{C_4,Dist} = (FR_{C_4,Dist})(Fz_{C_4}) = (0.98)(100)(0.30) = 29.4 \text{ kmol/h}$$

$$C_5: Dx_{C_5,Dist} = (FR_{C_5,Dist})(Fz_{C_5}) = (1 - 0.98)(100)(0.35) = 1.4 \text{ kmol/h}$$

$$C_6: Dx_{C_6,Dist} = 0 \text{ kmol/h}$$

$$D = 5 + 10 + 29.4 + 1.4 + 0 = 45.8 \text{ kmol/h}$$

$$x_{C_2,Dist} = \frac{5}{45.8} = 0.1091 \dots \approx 0.11$$

$$x_{C_3,Dist} = \frac{10}{45.8} = 0.218 \dots \approx 0.22$$

$$x_{C_4,Dist} = \frac{29.4}{45.8} = 0.641 \dots \approx 0.64$$

$$x_{C_5,Dist} = \frac{1.4}{45.8} = 0.0305 \dots \approx 0.03$$

$$x_{C_6,Dist} = 0$$

$$0.11 + 0.22 + 0.64 + 0.03 + 0 = 1$$

⇒ correct!

1-4. $C_2: B\lambda_{C_2, \text{bot}} = 0 \text{ kmol/h}$

$C_3: B\lambda_{C_3, \text{bot}} = 0 \text{ kmol/h}$

$C_4: B\lambda_{C_4, \text{bot}} = (FR_{C_4, \text{bot}})(FZ_{C_4}) = (1-0.98)(100)(0.30) = 0.6 \text{ kmol/h}$

$C_5: B\lambda_{C_5, \text{bot}} = (FR_{C_5, \text{bot}})(FZ_{C_5}) = (0.96)(100)(0.35) = 33.6 \text{ kmol/h}$

$C_6: B\lambda_{C_6, \text{bot}} = FZ_{C_6} = (100)(0.20) = 20 \text{ kmol/h} \rightarrow FR_{C_6, \text{bot}} \approx 1$

$B = 0 + 0 + 0.6 + 33.6 + 20 = 54.2 \text{ kmol/h}$

$(F = B + D = 45.8 \text{ kmol/h} + 54.2 \text{ kmol/h} = 100 \text{ kmol/h} \Rightarrow \text{correct!})$

$\lambda_{C_2, \text{bot}} = 0$

$\lambda_{C_3, \text{bot}} = 0$

$\lambda_{C_4, \text{bot}} = \frac{0.6}{54.2} = 0.0110 \dots \approx 0.01$

$\lambda_{C_5, \text{bot}} = \frac{33.6}{54.2} = 0.619 \dots \approx 0.62$

$\lambda_{C_6, \text{bot}} = \frac{20}{54.2} = 0.369 \dots \approx 0.37$

$0 + 0 + 0.01 + 0.62 + 0.37 = 1 \Rightarrow \text{correct!}$

1-5. $N_{\min} = \frac{\ln \left\{ \frac{FR_{A, \text{dist}}}{FR_{A, \text{bot}}} \times \frac{FR_{B, \text{bot}}}{FR_{B, \text{dist}}} \right\}}{\ln \alpha_{AB}}$

$A = LK = C_4$

$B = HK = C_5$

$FR_{C_4, \text{dist}} = 0.98$

$FR_{C_5, \text{bot}} = 0.96$

$FR_{C_4, \text{bot}} = 1 - 0.98 = 0.02$

$FR_{C_5, \text{dist}} = 1 - 0.96 = 0.04$

$\alpha_{AB} = \alpha_{LK-HK} = \alpha_{C_4-C_5}$

$= \frac{K_{C_4}}{F_{C_5}} = \frac{0.85}{0.28} \approx 3.04$

$\Rightarrow N_{\min} = \frac{\ln \left\{ \left(\frac{0.98}{0.02} \right) \times \left(\frac{0.96}{0.04} \right) \right\}}{\ln(3.04)} = 6.3586 \dots \approx 6.36 \text{ (stages)}$

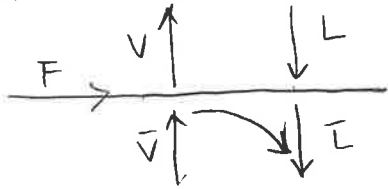
$N_{F, \min} = \frac{\ln \left\{ \frac{\lambda_{A, \text{dist}}}{\lambda_{B, \text{dist}}} \times \frac{Z_B}{Z_A} \right\}}{\ln \alpha_{AB}} = \frac{\ln \left\{ \left(\frac{0.64}{0.07} \right) \times \left(\frac{0.35}{0.30} \right) \right\}}{\ln(3.04)} = 2.891 \dots \approx 2.89 \text{ (stages)}$

$\lambda_{C_4, \text{dist}} = 0.64 \quad Z_{C_5} = 0.35$

$\lambda_{C_5, \text{dist}} = 0.07 \quad Z_{C_4} = 0.30$

5

. 1-6.



$$\Delta V_{\text{feed}} = V_{\text{min}} - \bar{V}_{\text{min}} = \sum_{i=1}^C \frac{\alpha_{i-\text{ref}} F z_i}{\alpha_{i-\text{ref}} - \phi}$$

$$\text{sat. liq.} \rightarrow q=1 \Rightarrow \text{f}$$

$$i = 2-6, \text{ ref} = \text{HK} = \text{C}_5$$

$$0 = \frac{\alpha_{\text{C}_2\text{C}_5} \cdot z_{\text{C}_2}}{\alpha_{\text{C}_2\text{C}_5} - \phi} + \frac{\alpha_{\text{C}_3\text{C}_5} \cdot z_{\text{C}_3}}{\alpha_{\text{C}_3\text{C}_5} - \phi} + \frac{\alpha_{\text{C}_4\text{C}_5} \cdot z_{\text{C}_4}}{\alpha_{\text{C}_4\text{C}_5} - \phi} + \frac{\alpha_{\text{C}_5\text{C}_5} \cdot z_{\text{C}_5}}{\alpha_{\text{C}_5\text{C}_5} - \phi} + \frac{\alpha_{\text{C}_6\text{C}_5} \cdot z_{\text{C}_6}}{\alpha_{\text{C}_6\text{C}_5} - \phi}$$

$$0 = \frac{(26.79)(0.05)}{26.79 - \phi} + \frac{(8.57)(0.10)}{8.57 - \phi} + \frac{(3.04)(0.30)}{3.04 - \phi} + \frac{1(0.35)}{1 - \phi} + \frac{(0.36)(0.20)}{0.36 - \phi}$$

$$0 = \frac{1.3395}{26.79 - \phi} + \frac{0.857}{8.57 - \phi} + \frac{0.912}{3.04 - \phi} + \frac{0.35}{1 - \phi} + \frac{0.072}{0.36 - \phi}$$

$$\alpha_{\text{HK} \cdot \text{ref}} < \phi < \alpha_{\text{LK} \cdot \text{ref}}$$

$$= \alpha_{\text{HK} \cdot \text{HK}} < \phi < \alpha_{\text{LK} \cdot \text{HK}}$$

$$= \alpha_{\text{C}_5\text{C}_5} < \phi < \alpha_{\text{C}_4\text{C}_5}$$

$$= \underline{1 < \phi < 3.04}$$

$$\Rightarrow \underline{\phi = 1.498}$$

$$1-7. \quad V_{\text{min}} = \sum_{i=1}^C \frac{\alpha_{i-\text{ref}} (D x_{i, \text{dist}})}{\alpha_{i-\text{ref}} - \phi}$$

$$= \frac{\alpha_{\text{C}_2\text{C}_5} (D x_{\text{C}_2, \text{dist}})}{\alpha_{\text{C}_2\text{C}_5} - 1.498} + \frac{\alpha_{\text{C}_3\text{C}_5} (D x_{\text{C}_3, \text{dist}})}{\alpha_{\text{C}_3\text{C}_5} - 1.498} + \frac{\alpha_{\text{C}_4\text{C}_5} (D x_{\text{C}_4, \text{dist}})}{\alpha_{\text{C}_4\text{C}_5} - 1.498}$$

$$+ \frac{\alpha_{\text{C}_5\text{C}_5} (D x_{\text{C}_5, \text{dist}})}{\alpha_{\text{C}_5\text{C}_5} - 1.498} + \frac{\alpha_{\text{C}_6\text{C}_5} (D x_{\text{C}_6, \text{dist}})}{\alpha_{\text{C}_6\text{C}_5} - 1.498}$$

$$= \frac{(26.79)(5)}{26.79 - 1.498} + \frac{(8.57)(10)}{8.57 - 1.498} + \frac{(3.04)(29.4)}{3.04 - 1.498} + \frac{(1)(1.4)}{1 - 1.498} + \frac{(0.36)(0)}{0.36 - 1.498}$$

$$= \frac{133.95}{25.292} + \frac{85.7}{7.072} + \frac{89.376}{1.542} + \left(-\frac{1.4}{0.498} \right) + 0$$

$$= 72.564 \dots \approx \underline{72.56 \text{ kmol/h}}$$

$$L_{\min} = V_{\min} - D = 12.56 \frac{\text{kmol}}{\text{h}} - 45.8 \frac{\text{kmol}}{\text{h}} = 26.76 \frac{\text{kmol}}{\text{h}}$$

$$(V_{\min} = L_{\min} + D)$$

$$\left(\frac{L}{D}\right)_{\min} = \frac{26.76}{45.8} = 0.5842... \approx 0.584$$

15/15

$$1-8. \quad X = \frac{R - R_{\min}}{R + 1} \quad Y = \frac{N - N_{\min}}{N + 1} \quad \left(N = \frac{N_{\min} + Y}{1 - Y}\right)$$

$$\frac{R}{R_{\min}} = 1.2 \quad \frac{L}{D} = R \rightarrow R = (R_{\min})(1.2)$$

$$\frac{L}{D} = (R_{\min})(1.2)$$

Need to find R!

$$R = R_{\min} \times 1.2$$

$$R_{\min} = \frac{L}{D} \times \frac{1}{1.2} = 0.584 \times \frac{1}{1.2} = 0.4866... \approx 0.487$$

$$Y \approx 0.47 \text{ (in the graph)}$$

$$Y = 0.56$$

$$N = 15.9$$

$$0.47 = \frac{N - N_{\min}}{N + 1} \quad (N_{\min} = 6.36)$$

$$0.47 = \frac{N - 6.36}{N + 1} \rightarrow (N + 1)0.47 = N - 6.36$$

$$\rightarrow 0.47N + 0.47 = N - 6.36$$

$$0.47 + 6.36 = N - 0.47N$$

$$(1 - 0.47)N = 0.47 + 6.36$$

$$N = \frac{0.47 + 6.36}{1 - 0.47} = 12.886...$$

$$\approx 12.89 \text{ (stages)}$$

Method ok!

3/5

$$1-9. \quad \frac{N_{F, \min}}{N_{\min}} = \frac{N_F}{N} \rightarrow N_F = N \left(\frac{N_{F, \min}}{N_{\min}} \right)$$

$$= (12.89) \left(\frac{2.89}{6.36} \right) = 5.857...$$

$$\approx 5.86 \text{ (stages)}$$

7/20

4/5

1-10.

$$\frac{N_F - 1}{N - N_F} = \left[\frac{B}{D} \left(\frac{z_{HK}}{z_{LK}} \right) \left[\frac{x_{LK, bot}}{x_{HK, dist}} \right]^2 \right]^{0.206}$$

$$\frac{N_F - 1}{12.89 - N_F} = \left[\left(\frac{54.2}{45.8} \right) \left(\frac{0.75}{0.30} \right) \left(\frac{0.01}{0.07} \right)^2 \right]^{0.206}$$

$$= 0.67964$$

$$\approx 0.6796$$

$$N_F - 1 = (12.89 - N_F) 0.6796$$

$$N_F = (12.89 \times 0.6796) - 0.6796 N_F + 1$$

$$(1 + 0.6796) N_F = (12.89 \times 0.6796) + 1$$

$$N_F = \frac{(12.89 \times 0.6796) + 1}{1 + 0.6796} = 5.810 \dots$$

$$\approx 5.81 \text{ (stages)}$$

Method is ok!

1-11.

$$FR_{LK, dist} = \frac{(\alpha_{LK-HK})^{N_{min}}}{\frac{FR_{HK, bot}}{1 - FR_{HK, bot}} + (\alpha_{LK-HK})^{N_{min}}}$$

$$FR_{LK, dist} = \frac{(3.04)^{6.76}}{\frac{0.96}{1 - 0.96} + (3.04)^{6.76}} = 0.9800 \dots$$

$$\approx 0.98$$

8/10

∴ This process is reasonably economical because the reflux ratio, L/D , and stages are low. Also $FR_{LK, dist} = 0.98 \approx 1$.

from $FR_{LK, dist}$ estimate
consistent with the assumption
($LWK \approx 1.0$ in distillate)!

95/100

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

Name: Christopher Baltazar

1. (100 pts) A saturated liquid feed at 100 kmol/min consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of propane is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5	<u>26.79</u>	5
C3, propane	2.4	<u>8.57</u>	10
C4, butane	0.85	<u>3.04</u>	30
C5, pentane	0.28	<u>1</u>	35
C6, hexane	0.1	<u>.357</u>	20
Total in feed	-	<u>39.76</u>	100

Not sum!

- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

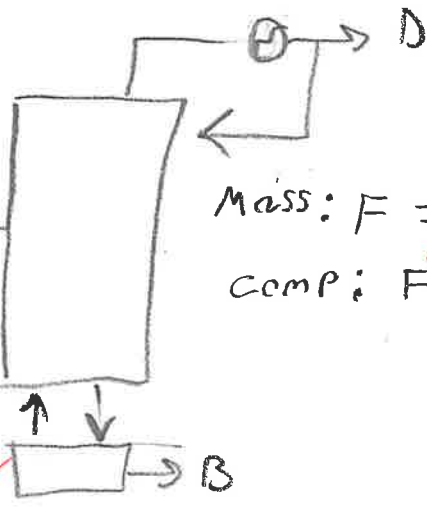
where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,Bott} = (BX_{i,Bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a mass balance envelope around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: 0.422, 1.498, and 5.969. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

Don't solve ϕ
 $L_{HK} < \phi < L_{LK}$

1)

$Q=1$
 $F = 100 \frac{\text{kmol}}{\text{min}}$
 $.05 \sim C2 \text{ LNK}$
 $.1 \sim C3 \text{ LNK}$
 $.3 \sim C4 \text{ LK}$
 $.35 \sim C5 \text{ HK}$
 $.20 \sim C6 \text{ HNK}$



$\bullet FR_{C4/D} = .98$
 $\bullet FR_{C5/B} = .96$

Mass: $F = D + B$

Comp: $Fz_i = Dx_{i,D} + Bx_{i,B}$

5/5

2) $C5 = \text{ref} = \text{HK}$

$\alpha_{C2-C5} = \frac{K_{C2}}{K_{C5}} = \frac{7.5}{.28} = 26.79$ $\alpha_{C3-C5} = \frac{2.4}{.28} = 8.57$

$\alpha_{C4-C5} = 3.04$ $\alpha_{C5-C5} = 1$ $\alpha_{C6-C5} = .357$

5/5

3) Obj: Get $Dx_{i,D}$, D

Assume: $\text{LNK} \xrightarrow{AII} D$, $Dx_{C2} = 5 \frac{\text{kmol}}{\text{min}}$ $Dx_{C3} = 10$
 $\text{HNK} \xrightarrow{AII} B$, $Bx_{C6} = 20 \frac{\text{kmol}}{\text{min}}$, $Dx_{C6} = 0$

$Dx_{C4} = (100 \cdot .3) \cdot (.98) = 29.4$

$Bx_{C4} = 30 \cdot (1 - .98) = 0.6$

$Dx_{C5} = 35 \cdot (1 - .96) = 1.4$ $Bx_{C5} = 33.6$

unit missing!

$D = D \sum x_i = 5 + 10 + 29.4 + 1.4 = 45.8 = D$

$x_{C2,D} = .11$ $x_{C3,D} = .22$

$x_{C4,D} = .64$ $x_{C5,D} = 0.031$

$154.2 = B \sum x_i = B = 100 - 45.8$

$x_{C4,B} = .011$ $x_{C5,B} = 0.62$ $x_{C6,B} = .37$

$\sum x_{i,B} = 1 \checkmark$ $\sum x_{i,D} = 1 \checkmark$

5) $N_{\min} = \frac{\ln \left(\frac{.98}{(1-.98)} \cdot \frac{.96}{(1-.96)} \right)}{\ln(\alpha_{C4-C5})} = \frac{7.07}{1.11} = 6.36 \text{ stages}$

$N_{F \min} = \frac{\ln \left(\frac{.64}{.031} \cdot \frac{.35}{.3} \right)}{\ln(3.04)} = \frac{3.181}{1.11} = 2.86 \text{ stage}$

5/5

9) $\frac{N_{Fmin}}{N_{min}} = \frac{N_F}{N} \rightarrow \frac{2.86}{6.36} = \frac{N_F}{15.36} \rightarrow \boxed{N_F = 6.94}$ ✓ C.D

5/5

10) $\frac{N_F - 1}{N - N_F} = \left[\frac{54.2}{45.8} \left(\frac{.35}{.3} \right) \left(\frac{.011}{.031} \right)^2 \right]^{.206}$

$\frac{N_F - 1}{15.36 - N_F} = [.1738]^{.206} = .697$

5/5

$N_F - 1 = 10.71 - .697 N_F \rightarrow 1.697 N_F = 11.71$

$N_F \approx 6.90$ ✓

Very close to the approx.

11) $FR_{NK,dis} = \frac{(3.04)^{6.36}}{\left(\frac{.96}{1-.96} \right) + (3.04)^{6.36}} = \frac{1177.81}{29 + 1177.81}$

$FR_{CH,D} = 0.98$ ✓

~~The reflux ratio is very small, so with 16 stages this process is economical. Assuming a 16 stage column is not very expensive.~~

$FR_{C3,D} = \frac{(8.57)^{6.36}}{\left(\frac{.96}{1-.96} \right) + (8.57)^{6.36}} = .999 \approx 1$ ✓

$CR_{CH,D} = 1$ ✓

14/15

⇒ consistent with assumption

($ENK \approx 1.0$ in distillate)

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

Name: Cody Pych

1. (100 pts) A saturated liquid feed at 100 kmol/hr consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of ~~propane~~ ^{Butane} is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5		5
C3, propane	2.4		10
C4, butane	0.85		30
C5, pentane	0.28		35
C6, hexane	0.1		20
Total in feed	-	-	100

- 1) (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- 2) (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- 3) (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- 4) (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- 5) (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,bot} = (BX_{i,bot})/Fz_i$

- 6) (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

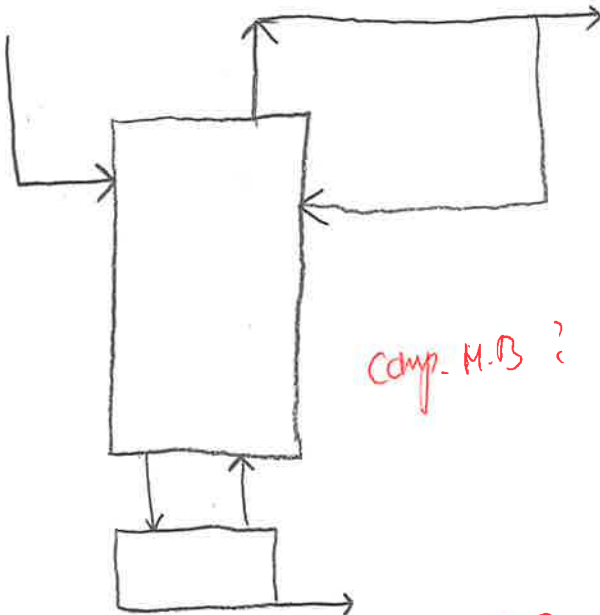
Cooly Pych
100 kmol/min

5 atm

② LNK - C2
LK - C4
NK - C3
HK - C5
HNK - C6

98%
dist ✗ C2 5
C3 10
C4 30
bot ✗ C5 35
96%
C6 20

1) 3/5



comp. M.B. ?

2) C2: LNK, α ?
C3: LNK "
C4: LK "
C5: HK "
C6: HNK "

1/5

3) 10/15

4) 10/15

$D_{C2} = \text{all } 5 \text{ mol/hr}$
 $100 \times 0.1 = 10 \text{ mol/hr}$
 $B_{C3} = 0 \text{ mol/hr}$

$D_{C3} = 10 \times 0.98 = 9.8$
 $B_{C4} = 0.06$

$D_{C4} = 30 \times 0.98 = 29.4$
 $B_{C5} = 33.6$

$D_{C5} = 35 \times 0.04 = 1.4$
 $B_{C6} = 20$

$D = 45.6$
 45.8
 $B = 54.4$
 $X_{C2} = 0$

$X_{D,C2} = \frac{5}{45.6} = .110$ ✓

$X_{B,C3} = \frac{0.2}{54.4} = .0037$

$X_{D,C3} = \frac{9.8}{45.6} = .215$ ✓

$X_{B,C4} = \frac{0.06}{54.4} = .0011$ ✓

$X_{D,C4} = \frac{29.4}{45.6} = .645$

$X_{B,C5} = \frac{33.6}{54.4} = .6176$ ✓

$X_{D,C5} = \frac{1.4}{45.6} = .031$

$X_{B,C6} = \frac{20}{54.4} = .3676$ ✓

validation!

validation!

CHE 313 Transport Phenomena III
Exam #2 (03/17/2026)

Name: Katerina Boucari

9 = 1

1. (100 pts) A saturated liquid feed at 100 kmol/min consisting of ethane (C2), propane (C3), butane (C4), pentane (C5), and n-hexane (C6) is fed to a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. K values and flow rate of each component in feed are provided in the table. A 98.0% recovery ($FR_{i,D} = X_{i,D} \cdot D / F \cdot z_i$) of propane is desired in the distillate. A 96.0% recovery of pentane is desired in the bottoms. LK

HK The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	K (y/x) value	α (relative volatility) value	kmol/hr
C2, ethane	7.5	<u>26.786</u>	5
C3, propane	2.4	<u>8.571</u>	10
C4, butane	0.85	<u>3.036</u>	30
C5, pentane	0.28	<u>1</u>	35
C6, hexane	0.1	<u>0.3571</u>	20
Total in feed	-	-	100

$F \cdot z_{C2} = 5$

- (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK. Calculate relative volatilities when HK is the reference and complete the table.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in the distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for B, D, and each component mole fraction using mass balance and stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

A: C4 $FR_{AD} = 0.98$

B: C5 $FR_{DB} = 0.96$

$\alpha_{LK,HK} = \alpha_{C4,C5} = 3.036$

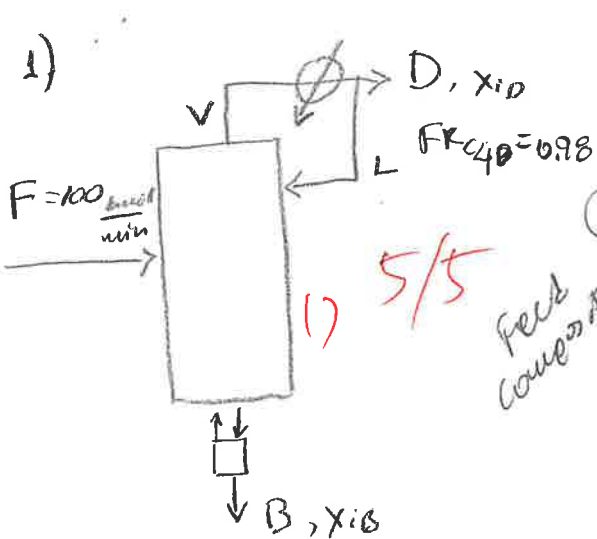
where A=LK and B=HK, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,bot} = (BX_{i,bot})/Fz_i$

- (15 pts - Underwood equation) After drawing a **mass balance envelope** around the feed stage, find ΔV_{feed} . Use the 1st Underwood (UW) equation below to set up full extended UW expression for the root ϕ . **Note: You are not required to solve the equation manually;** however, you must define all variables/values (except for ϕ) and fully expand the summation for components from $i = 2$ through 6, using the **Heavy Key (HK)** as the reference component. After solving the Underwood equation, three mathematically valid roots are obtained: **0.422, 1.498, and 5.969**. Identify the correct root and justify your selection in detail. Note that you cannot simply test these values to see if they satisfy the equation, as all three are mathematically valid roots.

$\Delta V = V - V_{min}$

$\alpha_{HK-ref} < \phi < \alpha_{LK-ref}$

$\alpha_{HK,HK} < \phi < \alpha_{LK,HK}$



2)

$$F \cdot Z_{C2} = 5 \frac{\text{kmol}}{\text{h}}$$

$$Z_{C2} = \frac{5}{100} = 0.05$$

$$Z_{C3} = \frac{10}{100} = 0.1$$

$$Z_{C4} = \frac{30}{100} = 0.3$$

$$Z_{C5} = \frac{35}{100} = 0.35$$

$$Z_{C6} = \frac{20}{100} = 0.2$$

- $C_2: \text{LNK}$
 $C_3: \text{LNK}$
 $C_4: \text{LK}$
 $C_5: \text{HK}$
 $C_6: \text{HNK}$

2) HK as reference: $K_{HK} = K_{C5} = 0.28$

Comp. M.B: $FZ_i = Bx_{iB} + Dx_{iD}$

OVMB: $F = D + B$

CMB: $FZ_i = Dx_{iD} + Bx_{iB}$

$$\alpha_{C2C5} = \frac{K_{C2}}{K_{C5}} = \frac{7.5}{0.28} = 26.786$$

$$\alpha_{C3C5} = \frac{K_{C3}}{K_{C5}} = \frac{2.4}{0.28} = 8.571$$

$$\alpha_{C4C5} = \frac{K_{C4}}{K_{C5}} = \frac{0.85}{0.28} = 3.036$$

$$\alpha_{C5C5} = \frac{K_{C5}}{K_{C5}} = 1$$

$$\alpha_{C6C5} = \frac{K_{C6}}{K_{C5}} = \frac{0.1}{0.28} = 0.3571$$

3) $D \cdot x_{iD}$, D , x_{iD} verify $\sum x_{iD} = 1$ as LNK assume $FR_{CD} = 1$

$$FR_{iD} = \frac{x_{iD} \cdot D}{F \cdot Z_i} \Rightarrow$$

$$Dx_{2D} = FR_{CD} F \cdot Z_{C2} = 1 \cdot 5 = 5 \frac{\text{kmol}}{\text{h}}$$

$$Dx_{3D} = FR_{CD} F \cdot Z_{C3} = 1 \cdot 10 = 10 \frac{\text{kmol}}{\text{h}}$$

$$Dx_{4D} = FR_{CD} \cdot F \cdot Z_{C4} = 0.98 \cdot 30 = 29.4 \frac{\text{kmol}}{\text{h}}$$

$$Dx_{5D} = FR_{CD} \cdot F \cdot Z_{C5} = (1 - 0.96) \cdot 35 = 1.4 \frac{\text{kmol}}{\text{h}}$$

$$Dx_{6D} = FR_{CD} \cdot F \cdot Z_{C6} = 0 \frac{\text{kmol}}{\text{h}}$$

$\hookrightarrow$ assume $FR_{CD} = 0$ as HNK

$$D = \sum (D \cdot x_{iD}) = 45.8 \frac{\text{kmol}}{\text{h}}$$

$$x_{iD} = \frac{D \cdot x_{iD}}{D} \Rightarrow$$

$$N_{Fmin} = \ln \left\{ \frac{x_{C4D}}{x_{C5D}} \times \frac{z_{C5}}{z_{C4}} \right\} = \ln \left\{ \frac{0.64}{0.031} \times \frac{0.35}{0.3} \right\}$$

ln dec4cs

ln(3.036)

5) 5/5

$$\frac{L_F}{F} = q$$

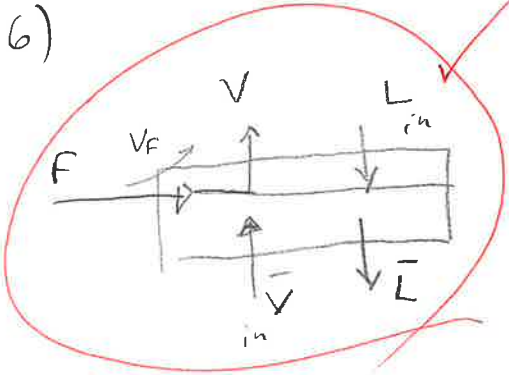
$$L_F - F$$

$$F \left(\frac{L_F}{F} - 1 \right)$$

$$N_{Fmin} = 2.86 \approx 3 \text{ stages}$$

$$F = L - \bar{L}$$

6)



$$F = \bar{V} + L - V - \bar{L} = 0 \quad V_{min} - \bar{V}_{min} = L - \bar{L} - F$$

$$= \frac{L_F - F}{F} = F(1 - q)$$

$$F - L + \bar{L} = \bar{V} - V$$

$$F - L_F = \bar{V} - V$$

$$F(1 - q) = \Delta V_{feed}$$

$$L_F = L - \bar{L}$$

15/15

$$\Delta V_{feed} = V_{min} - \bar{V}_{min} = \underbrace{F(1 - q)}_{q=1} = \sum \frac{\alpha_{i-ref} F z_i}{\alpha_{i-ref} - \phi}$$

$$\alpha_{ref} = C_5$$

$$\Rightarrow 0 = \frac{\alpha_{C2C5} \cdot F z_{C2}}{\alpha_{C2C5} - \phi} + \frac{\alpha_{C3C5} \cdot F z_{C3}}{\alpha_{C3C5} - \phi} + \frac{\alpha_{C4C5} \cdot F z_{C4}}{\alpha_{C4C5} - \phi} + \frac{\alpha_{C5C5} \cdot F z_{C5}}{\alpha_{C5C5} - \phi} + \frac{\alpha_{C6C5} \cdot F z_{C6}}{\alpha_{C6C5} - \phi}$$

$$\Rightarrow 0 = \frac{26.79(5 \frac{\text{kmol}}{\text{h}})}{26.79 - \phi} + \frac{8.57(10 \frac{\text{kmol}}{\text{h}})}{8.57 - \phi} + \frac{3.036(30)}{3.036 - \phi} + \frac{1(35)}{1 - \phi} + \frac{0.357(20)}{0.357 - \phi}$$

$$\Rightarrow \alpha_{HK-AK} < \phi < \alpha_{LK-HK}$$

$$1 < \phi < 3.036 \text{ based on possible values given}$$

$$\boxed{\phi = 1.498}$$

$$7) \quad V_{min} = D + L_{min} \quad \phi = 1.498 \quad L_{min} = V_{min} - D$$

$$V_{min} = \frac{\sum \text{direct}(D \cdot X_{ip})}{\text{direct} - \phi}$$

9/10

$$V_{min} = \frac{\text{direct}(D \cdot X_{120})}{\text{direct} - \phi} + \frac{\text{direct}(D \cdot X_{320})}{\text{direct} - \phi} + \frac{\text{direct}(D \cdot X_{420})}{\text{direct} - \phi} + \frac{\text{direct}(D \cdot X_{520})}{\text{direct} - \phi} + \frac{\text{direct}(D \cdot X_{620})}{\text{direct} - \phi}$$

$$V_{min} = \frac{26.79(5)}{26.79 - 1.498} + \frac{8.571(10)}{8.571 - 1.498} + \frac{3.076(29.4)}{3.076 - 1.498} + \frac{1(1.4)}{1 - 1.498} + 0$$

$$V_{min} = 5.296 + 12.118 + 58.035 + (-2.811)$$

$$V_{min} = 72.638 \text{ km/hr} \quad -1$$

$$R_{min} = \left(\frac{L}{D}\right)_{min} = \left(\frac{V_{min} - D}{D}\right)_{min} = \frac{(72.678 - 45.8)}{45.8}$$

$$R_{min} = 0.586$$

$$8) \quad \frac{R}{R_{min}} = 1.2 \Rightarrow R = 1.2 R_{min} = 0.703$$

$$X = \frac{0.703 - 0.586}{0.703 + 1} = 0.0687 \approx 0.07$$

5/5

$$\text{from aerve} \quad Y = 0.55$$

$$Y = \frac{N - N_{min}}{N + 1} \Rightarrow Y(N + 1) = N - N_{min} \Rightarrow N \cdot Y + Y = N - N_{min}$$

$$N - N \cdot Y = -N_{min} - Y$$

$$N(1 - Y) = -N_{min} - Y$$

$$N = \frac{-N_{min} - Y}{1 - Y} = \frac{-6.37 - 0.55}{1 - 0.55} =$$

$$Y = \frac{N - N_{min}}{N + 1}$$

$$0.55 = \frac{N - (6.37)}{N + 1}$$

$$(N + 1)0.55 = N - 6.37$$

$$0.55N + 0.55 = N - 6.37$$

$$N = 15.38 \approx 15 \text{ (stage)}$$

$$N - 0.55N = -6.37 - 0.55$$

$$N(1 - 0.55) = \frac{-6.37 - 0.55}{1 - 0.55}$$